ΕΘΝΙΚΟ ΚΑΙ ΚΑΠΟΔΙΣΤΡΙΑΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ
ΜΕΤΑΠΤΥΧΙΑΚΟ ΠΡΟΓΡΑΜΜΑ ΣΠΟΥΔΩΝ
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ΕΛΛΗΝΙΚΗ ΔΗΜΟΚΡΑΤΙΑ
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Διπλωματική Εργασία

Πρόβλεψη πρόωρου τερματισμού κλινικών δοκιμών με μοντέλα μηχανικής μάθησης vs κλασσικής παλινδρόμησης, βάση ανάλυσης των χαρακτηριστικών σχεδιασμού τους

Master's Thesis

Predicting early termination of clinical trials with machine learning vs. classical regression models, based on analysis of their design characteristics

Συγγραφέας: Ανδρέου Ευγενία
Επιβλέπων: Σπύρος Δεναξάς

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2. Περίληψη

Πρόβλημα/Υπόβαθρο:
Ο πρόωρος τερματισμός των κλινικών δοκιμών έχει ως αποτέλεσμα τη σπατάλη πόρων, ανθρώπινου δυναμικού αλλά και χρηματοδοτήσεων. Το αντίκτυπο είναι μεγαλύτερο σε προχωρημένες φάσεις, όπως οι 3 και 4, δεδομένου του μεγαλύτερου μεγέθους των παραπάνω.
Οι λόγοι μπορεί να οφείλονται μεταξύ άλλων, στα ίδια τα χαρακτηριστικά σχεδιασμού της έρευνας. Συνεπώς, ο τερματισμός μιας τόσο σύνθετης ερευνητικής διαδικασίας είναι πολύ παραγοντικός αλλά και μοναδικός για την εκάστοτε κλινική δοκιμή. Το σύμπλεγμα αιτιών καθίσταται συχνά δύσκολα ανιχνεύσιμο από τα κλασσικά μοντέλα πρόβλεψης, όταν αντίθετα τα μοντέλα μηχανικής μάθησης εκπαιδεύονται συγκεκριμένα στα εκάστοτε δεδομένα.

Σκοπός:
Το περιεχόμενο της παρούσας εργασίας είναι η ανάλυση των χαρακτηριστικών των κλινικών δοκίμων, μέσω μοντέλων πρόβλεψης της πιθανότητας πρόωρου τερματισμού τους. Η αναγνώριση των χαρακτηριστικών αυτών μπορεί να συμβάλλει στη βελτίωση του σχεδιασμού των κλινικών δοκιμών και κατ' επέκταση στην επαρκέστερη διαχείριση των ανωτέρω πόρων.

Μέθοδοι:
Για το σκοπό αυτό αναλύθηκαν οι κλινικές δοκιμές φάσης 1 έως 4 του διαστήματος 2011-2024 από το https://www.clinicaltrials.gov/ και https://aact.ctti-clinicaltrials.org/ . Για κάθε φάση ξεχωριστά, εκπαιδεύτηκαν μοντέλα μηχανικής μάθησης και κλασσικής παλινδρόμησης σε δεδομένα σχεδιασμού - αποτελέσματος και συγκρίθηκε η μεταξύ τους απόδοση.
Πιο συγκεκριμένα, μοντέλα κλασσικής στατιστικής όπως Logistic, LDA, QDA, συγκρίθηκαν με μοντέλα μηχανικής μάθησης KNN, SVC, Gradient Boosting, XGBoost μέσω ROC/PR Curves κ.α.

Αποτελέσματα:
Το μοντέλο που υπερίσχυσε σε όλες τις φάσεις ως προς τα ROC/PR AUCς αλλά και στις περισσότερες εκ των υπολοίπων μετρήσεων ήταν το XGBoost και το LightBoost.
Τα αποτελέσματα ως προς τις σημαντικότερες εκ των μεταβλητών εκτιμήθηκαν με SHAP plots και δευτερευόντως με logistic regression βάση των μεταβλητών που υπέδειξαν τα SHAP plots..
Διαπιστώθηκε ότι ο μειωμένος αριθμός συμμετεχόντων εμφανίζεται ως το κυρίαρχο χαρακτηριστικό που προβλέπει πρόωρη λήξη, ενώ ο αριθμός των δομών το αμέσως επόμενο.
Δευτερεύουσες μεταβλητές είναι, συγκεκριμένα πεδία ασθενειών όπως τα νεοπλάσματα, οι υγιείς εθελοντές, οι συμμετέχοντες μεγαλύτερης ηλικίας, τυχαιοποίηση κτλ.

Η συνεισφορά των χαρακτηριστικών φαίνεται να σχετίζεται με φάσεις, ερευνητικές φάσεις ερευνητικούς στόχους ή χαρακτηριστικά σχεδιασμού. Για παράδειγμα, η τυχαιοποίηση και το placebo εμφανίζονται στη φάση 3 όπου ο σχεδιασμός είναι πολύ σημαντικός και μάλιστα χαρακτηρίζεται και ως «συγκρίσεως θεραπείας».

Συζήτηση/Συμπεράσματα:
Τα αποτελέσματα τη παρούσας εργασίας, συμφωνούν σε μεγάλο βαθμό με την προ υπάρχουσα βιβλιογραφία. Η επιπλέον συνεισφορά της εντοπίζεται στην ευρύτερη μελέτη των παραγόντων πρόωρου τερματισμού. Αναλυτικότερα, η ανάλυση εμπεριείχε όλες τις φάσεις από 1 έως 4, και πραγματοποιήθηκε ανά φάση ξεχωριστά. Αναλύθηκαν όσο περισσότερα χαρακτηριστικά σχεδιασμού μιας κλινικής δοκιμής. Τέλος, συγκρίθηκε μεγαλύτερος αριθμός μοντέλων από ότι έχει πραγματευτεί η βιβλιογραφία.
Μετέπειτα αναλύσεις μπορούν αν βασιστούν ανά φάση, ανά έκβαση μια κλινικής δοκιμής και σε αλληλεπιδράσεις μεταξύ των παραμέτρων.
Τα στοιχεία αυτής και αντίστοιχων μελετών, μπορούν να συμβάλλουν στον προσεκτικότερο σχεδιασμό μιας κλινικής δοκιμής, στα σημεία που εμφανίζονται ως παράγοντες πρόωρου τερματισμού. Μάλιστα, η σχετική βιβλιογραφία μπορεί να αξιοποιηθεί για τις αναλύσεις των ίδιων των δημόσιων αποθετηρίων όπως το ClinicalTrials.gov, με σκοπό τον εντοπισμό αντίστοιχων ερευνών με αυτά τα χαρακτηριστικά πρόωρου τερματισμού. Αποτέλεσμα αυτού είναι η βελτίωση διαχείρισης οικονομικών και ανθρώπινων πόρων αλλά και η ασφαλέστερη έκθεση των εθελοντών σε έρευνες με ουσιαστική συνεισφορά στις επιστήμες υγείας. Βασικός παράγον για να επιτευχθούν με ακρίβεια τα παραπάνω, η ποιότητα των δεδομένων, πρέπει να βελτιωθεί και να διατηρείται.

2. Summary

Problem/Background:
The premature termination of clinical trials results in a waste of resources, such as human and/or funding. The impact is greater in advanced phases, such as 3 and 4, given the greater size of the above.
The reasons may be due, among other things, to the design characteristics of the research itself. Consequently, the termination of such a complex research process is very factorial but also unique to each clinical trial. The complex of causes often becomes difficult to detect by classical prediction models, when on the contrary, machine learning models are specifically trained on the data in question.

Purpose:
The content of this work is the analysis of the characteristics of clinical trials through prediction models of the probability of their premature termination. The identification of these characteristics can contribute to the improvement of the design of clinical trials and, by extension, to more efficient resources management.

Method:
For this purpose, clinical trials from phases 1 to 4 from 2011-2024 were analysed from https://www.clinicaltrials.gov/ and https://aact.ctti-clinicaltrials.org/ . For each phase separately, machine learning and classical regression models were trained on design-outcome data, and their performance was compared.
More specifically, classical statistical models such as Logistic, LDA, QDA, were compared with machine learning models KNN, SVC, Gradient Boosting, XGBoosting and LightBoost via ROC/PR Curves, etc. Finally, the results regarding the most important variables were estimated with SHAP plots and secondary with logistic based on features indicated by SHAP.

Results:
The model that outperformed in all phases in terms of ROC/PR AUCs and in most of the remaining metrics was XGBoost and LightBoost.
The reduced number of participants appeared as the dominant feature predicting early termination, followed by the number of facilities. Secondary variables were specific disease fields such as neoplasms, healthy volunteers, older aged participants, intervention model, allocation etc.
Features' contribution seems to be related to phases', investigational phases research objectives or design characteristics. For example, randomization and placebo occur in phase 3 where design is of much importance this 'treatment comparing' phase.

Discussion/Conclusions:
The results of this study are aligned consistently with the existing literature.
Its additional contribution is found in the broader study of factors of premature termination. More specifically, the analysis included all phases from 1 to 4 and was carried out for each phase separately. Also, as many design features as possible were analysed. Finally, a larger number of models were compared than has been mentioned in literature.
Subsequent analyses could be based on phase, and interactions between features, such as how other features affect the accruals rates.
The data from this and similar studies can contribute to a more careful design of a clinical trial, in the points that appear as factors of premature termination. In fact, the relevant literature can be used for the analyses of the public registries themselves, such as ClinicalTrials.gov, in order to identify corresponding studies with these characteristics of premature termination. This could result is improvement oof financial and human resources management, as well as safer exposure of volunteers to research with a substantial contribution to health sciences. To accurately achieve the above, data quality of input must be improved and preserved.

3. Purpose – Description

Clinical trials depend on human volunteers and cost significant investment of human, administrative and financial resources. Given the above, an early terminated trial raises financial, concerns such as cost of resources that could have supported other trials, as well as ethical issues regarding volunteers whose participation may not contribute to meaningful scientific knowledge. Furthermore, it raises scientific concerns about which design feature led to the decision of termination and, finally, safety concerns regarding the efficacy and, most importantly, the safety of the drugs provided. [1]

Clinical Trials' (CT) initiation, involve extensive planning, cost, researchers' and sponsors' deliberation etc. Funding agencies (e.g., NIH) are trying to screen only quality studies before funding, by conducting peer reviews, which however, are costly and time consuming. Even though, still many trials achieve to move on initiation but unfortunately, end up early terminated [2][3].

Early termination of clinical trials (CT) lead to great loss of funding and human (volunteers, administrators etc.) resources, especially in advanced phases were trials get more scaled. Despite the efforts of responsible monitoring committees (e.g., FDA) or public registries, there are still many phases initiated without meeting all criteria, resulting to a waste of those resources that could have been utilized elsewhere for a more meaningful scientific progress.

Many studies like this tried to provide explanations of clinical trials' early termination, in order to avoid this problem and contribute to more efficient trials' design. The purpose of this study is to analyze clinical trials from ClinicalTrials.gov registry to identify as many parameters as possible for early termination probability. Specifically, to detect trials' design characteristics as responsible for early termination and to define the best classifier for prediction of this probability.

The results of the study could then be combined with similar literature studies and provide a useful guide to responsible parties when a new trial is filing for initiation approvals.

The chapters of current project are organized as follows:

1. **Introduction**:
   Reference to ClinicalTrials.gov public database and its issues in trials registration.
   Characteristics of an CT and main purpose of each phase.
   Statistical Theory overview such as analysis of background knowledge for metrics, models, and methods used in this study.
2. **Methods**:
   Reference to methods tools and environments this study used for analysis.
   Detailed description of data sources, data engineering and data science analysis.

3. **Results**:
   Results of model comparison, feature evaluation and metrics used for comparison.
4. **Discussion**:
   Summary of key findings. Comparison of results and processes with background literature studies.
   Proposal of practical application of findings to optimization of trials' design.
5. **Appendices**:
   Additional information of code, dictionary of terms etc.
6. **References**:
   Detailed Bibliography from text citation.

4. Introduction

4.1. Clinical Trials

4.1.1. Definition

Clinical trials are designed to observe outcomes of human subjects under interventional experiment conditions controlled by the scientist. Because of this, clinical trials are also called interventional studies. They could be randomized controlled trials, where participants are allocated at random to receive one of several clinical interventions. These are the ultimate evaluations of healthcare intervention. However, CT could also be non-randomized. [4] Often, the intervention is investigational, which means it is not approved for doctors to prescribe. [5]
In current study both types were analyzed.
More Specifically from ClinicalTrials.gov randomization is assigned into three levels and is referred as:
Allocation: The method by which participants are assigned to arms in a clinical trial.

- N/A (not applicable): For a single-arm trial
- Randomized: Participants are assigned to intervention groups by chance
- Non-randomized: Participants are expressly assigned to intervention groups through a non-random method, such as physician choice [6]

4.1.2. RCT Advantages & Disadvantages

RCTs main advantage is the randomization which solves selection bias problems. Selection bias could be formed due to unknown confounders. RCT are considered the ultimate evaluation of a healthcare intervention.[7]
However, biases that may occur are 'misclassification-information bias', where outcomes or exposure are incorrectly recorded, co-interventions (e.g., additional interventions received), contamination (e.g., subjects receiving the intervention

outside the trial etc.).
Another problem 'volunteer bias' (sub-type of 'selection bias'), where the sample is not representative of the population from which it was drawn. This happens mostly due to restricted inclusion or exclusion criteria. More specifically, possible participants may be excluded due to eligibility criteria such as comorbidities, or other attributes like distance from study facility etc. Restricted criteria are defined to study design to achieve homogeneity of groups, control of confounders and thus 'efficacy' (measure of success of intervention under experimental conditions). In contrast, this leads to loss of generalizability to real world population, where comorbidities or other parameters may occur. The result is that although RCT offer very good 'efficacy' measures, they cannot promise intervention's 'effectiveness' (real world measure of success of intervention).
A well-structured RCT needs a well-defined and representative sample from population. Sample size needs to be sufficient for the study to achieve the appropriate power needed to detect a statistically significant difference. Studies' outcomes must be reliable and meaningful measures. Above all else, ensuring safety is the top priority.[5]

4.1.3. Phases

Safety is above all matters in an CT, thus, the first of 4 phases purpose is to test the safety and maximum tolerated dose (MTD) of a drug, the pharmacokinetics, pharmacodynamics and drug–drug interactions. [5]
Below are explained all stages of an CT in detail.

Pre-Clinical
Pre-Clinical stage includes animal experiments and evaluation of drug production and purity. It does not include human participants. Main concepts studied are:
- Drug safety per dose, approximately to human quantities
- Pharmacodynamics (e.g., mechanisms, dose-response)
- Pharmacokinetics (e.g., absorption, metabolism, drug interactions)

Those data are all included in IND approval submitted to FDA for further drug investigation in humans. [5]

<u>Approvals</u>
For a promising intervention in pre-clinical stage, the sponsor or/and investigator submits an investigational new drug (IND) application to FDA. In this application are included all data regarding necessary drug information and all the results from this pre-clinical phase. If approved phases I-III follow. [5]

Early Phase 1 ('Phase 0')
Their main purpose is nor therapeutic nor diagnostic, but exploratory (e.g., screening studies, microdose). They include very limited to not at all human exposure. [6] They are beyond the scope of this study and were not analysed.

Phase 1 (‘Dose-Escalation’ | ‘Human Pharmacology’)
It is the first human experiment stage.[5] In this stage are determined :
- Metabolism aspects
- Pharmacologic aspects
- Side/Toxicity effects associated with increasing doses
- Early indication of effectiveness.[6]

Thus, maximum tolerated dose (MTD) and doses before toxicity are determined. Subjects are followed closely for all dose levels for any toxic effects.
Although, the objective of this stage is dose and pharmacology testing, there is a misinterpretation from volunteers that it could be therapeutic. Some improvements regarding the consent form, could help solving more clearly this ‘therapeutic misconception’. Participants constitute a small number of healthy or/and diseased volunteers. This stage is open labeled. [5]

Phase 2 (‘Therapeutic Exploratory’)
The main purpose of this stage is :
- Mostly test safety
- Pharmacokinetics, Pharmacodynamics. [6]
- An especial role of these trials is that they are a supportive stage to proceed to phase III, as they answer crucial questions regarding doses, dose frequencies, administration routes, endpoints etc.
- They may also study preliminary evidence of efficacy by comparing the drug with other controls (e.g., drug from published trials, standard therapies etc.) randomize and examine different dose arms (e.g., control arm).

Despite the possibility of studying all the above and that they are larger that phase I trials, they are also conducted with a small number of participants only with disease/condition of interest. In combination with the safety concerns of early stages, this phase cannot serve any efficacy establishment but only support the continuity and planning of phase III.
That is the reason FDA usually requests for additional phase IV, for less common adverse events to be identified.
With the completion of the above early phases, a meeting may be conducted between sponsors, investigators and FDA to review preliminary data, IND or even manufacturing concerns, and to decide if phase III will be initialized.[5]

There is another Phase specified as Phase 1/Phase 2, Trials that are a combination of phases 1 and 2. [6]

Phase 3 (‘Therapeutic Confirmatory’ | ‘Comparative Efficacy’ | ‘Pivotal Trial’)
They are initiated after above preliminary effectiveness evidence of phase 2. They are much larger that the above with a more diverse population.
Their main purpose is to identify :
- Efficacy

- Adverse events or else benefit-risk ratio [6][5].

However, it has no more than 300-3000 participants, which gives a statistical power to establish an adverse event no less than 1 in 100 participants [8].This means that less common adverse events are not probable to be identified in this stage, and this is the reason why occasionally FDA requests for more than one phase III clinical trials or additional phase IV trial after drug approval.

These phase's trials are categorized as follows based on the purpose:

- <u>Comparative Efficacy trials ('superiority' | 'placebo-controlled trials')</u>

It is the most common type of phase III RCTs where an intervention is compared to a standard therapy or placebo. It is notable that many 'placebo effect' instances happen in this type (health improvement by placebo administration). Some believe it's due to better care under research conditions, while others believe that participants with acute symptoms would either way improve as time passes even without intervention. However, this indicates that a promising result from a study does not always establish efficacy, as placebo effect may occur to both interventions. Another problem in phase III is a debate regarding comparative design in cases like surgical placebo procedures.

- <u>Equivalency trial ('positive-control study')</u>

The objective of this type of study is to study whether the intervention of interest is similar or different from another comparator. In this type, a placebo is almost never used. The margins are prespecified by the investigator and although they are based on statistical evidence, clinical experience etc. guidance for margin specification remains poor.

  – <u>Non-inferiority trial</u>

This type is a variant of equivalency studies. Its objective is to exclude that drug of interest is less effective from comparator.

Equivalency trials often appear major issues, such as biased results towards null hypothesis. This is because they are often incorrectly designed and analyzed as if they were comparative efficacy studies. It is notable that they are most prone to false positive than any other study (which would translate into false-negative for a comparative study). Specifically, non-inferiority studies are more susceptible to false-positive results than other study designs. [5]

Study's design is very important in this phase. Some of the analysis or else balance methods used are:

- <u>Randomization</u>

The massive benefit of Phase 3 studies is randomization in treatment allocation. Its purpose is to eliminate confounders and systematic differences (bias) between groups, thus any difference is due to treatment and not confounders.

A common method used is to just randomly assign subjects per group like flipping a coin. However, this could lead to imbalances in treatment assignments or covariates' distribution.
A solution is 'block randomization' in which the number of subjects per arm is equal and balanced after a specified block size. For example, in a trial with 2 arms, a block size of 4 subjects would have 2 positions in arm A and 2 positions in arm B.
Yet, this could also result in 'unblinding' issues. For example, in a block size of 2, investigator will understand that if the first subject belongs to treatment A the next belongs to treatment B.

- <u>Stratification</u>

This method is usually used in combination with randomization. Stratification ensures that some prognostic factors of clinical importance are balanced within arm groups.
Though, in small sample sizes when block randomization and stratification had both been applied, the original intended balance may be lost. To avoid such problems, alternative randomization methods can be used such as minimization or dynamic allocation that reduce imbalances among multiple study arms. [5]

- <u>Blinding-Masking</u>

Blinding aims to reduce 'information bias' of subjects' outcomes. Blinding could be:

- 'Single: Subject unaware of the treatments provided
- 'Double: Subject and investigator unaware of the treatments provided
- 'Triple: Data analyst, subject, and investigator unaware of the treatments provided.

Still, not all types of studies can be blinded. For example, drug delivery methods or expected adverse and toxicities could indicate which drug is provided.

- <u>Intervention Model</u>

Trial's strategy for assigning interventions to participants can be customized to the above problems. The model could be:

- Single Group: Just a single arm
- Parallel: Participants are assigned to one out of two or more groups in parallel
- Crossover: Participants receive one of two (or more) interventions during the initial phase of the study and the other intervention during the second phase of the study. Thus, each subject is its own control.
- Factorial: Two or more interventions are evaluated, each alone and in combination, in parallel against a control group. Thus, more than one treatment is simultaneously evaluated.
- Sequential: Groups of participants are assigned to receive interventions based on prior milestones being reached in the study. For example, in some dose escalation and adaptive design studies [5][6]

- <u>Analysis methods</u>
- 'Intention-to-treat' | 'analyzed as randomized' rule):

Subjects are evaluated based on the arm they belong, no matter what treatment they receive. It is the primary analysis method and most used, as it has the benefits of randomization's selection bias reduction. [5]

– ‘As-treated’ | ‘per-protocol’ :
Subjects are evaluated based on the treatment received, no matter in which arm they belong. It is used mainly as complementary analysis as it diminishes the randomization benefits and is prone to selection bias. Although in comparison to the above it seems not so useful , it has much better results in the case that there are ‘adherence’ or ‘contamination’ issues where the above method fails. [5]

There is another Phase specified as Phase 2/Phase 3, Trials that are a combination of phases 2 and 3. [6]

Approvals
If phases’ I-III results are promising regarding not only efficacy but safety too, a New Drug Application (NDA) is then conducted. Except FDA an external committee is also involved with this application. After final approval, phase IV follows, in which safety and effectiveness for the indicated population is monitored.[5]

Phase 4 (‘Therapeutic Use’ | ‘Post-Marketing’)
FDA may request phase IV trial after a drug approval. Their purpose is to:
- Identify less common adverse events
- Drug effectiveness in populations and doses similar or different from the previous studies
- Evaluate cost

After post-marketing phase IV many drugs require new black box warnings post-marketing than pre-marketing phases (e.g, phase III) or many drugs even withdraw because of safety reasons.[5][6]

4.2. ClinicalTrials.gov Registry

Clinical Trials.gov is a national registry of clinical trials, created from FDA in 2000. [9] It was initiated due to a law of Food and Drug Administration Modernization Act of 1997, which required from National Library of Medicine (NLM) to create and manage a public database for investigational drugs [10][11].
In short, it consists of a website and an online database. The purpose of ClinicalTrials.gov is to provide information about clinical research studies to the public, researchers, and health care professionals. The U.S. government does not review or approve the safety and science of all studies listed on this website.[12]
Today it includes clinical trials and observational studies from over 200 countries, that are happening now, will happen, or completed/terminated, with corresponding information. [11]
Registry of studies should be done from Investigators and Sponsors [12][11].
Registration data are displayed as table containing information regarding study names, sponsors, investigators, dates, conditions, intervention types, documents, design features, eligibility criteria, study endpoints, study results, and other [10].

Especially, for FDA approved drugs results must be uploaded within one year of the study's completion, but this does not apply for all studies. [9][11]
Finally, Clinical Trials 'expanded access' provide the possibility to patients with serious illness, unable to participate in trials, to register for an unapproved treatment. [11]
Clinical trials registry dataset is a type of Electronic Health Record (EHR) data of participants, interventions, outcomes etc. EHR data, are digital records of medical information of any type. In general, healthcare datasets are complex and can be challenging to analyse due to their size and complexity.[13]

4.2.1. Importance of Accurate Registration of Clinical Trials

The importance of accurate registry of clinical trials has many benefits, while the opposite may even induce harmful consequences.
As great funding is applied to CT trials, the appropriate registration of their design characteristics (especially, those leading to early termination), could be evaluated from registries' statistical analysis. The detection of those characteristics could prevent the risk of wasted resources. Early termination is most costly on advanced phases where human and fund resources are more scaled. [14]
Clinical trials involve experiments implemented on human samples. They aim to health science evolution, and their results may have implications to future patient's care [15]. The impact of clinical trials extends to individual patients by providing alternative therapies, but also to society by increasing health care value.[5]
Especially, Randomized controlled trials (RCTs), which are fundamental for health science evolution and decision-making, should be reported transparently, regarding not only the treatment efficacy but also adverse events or safety risks that patients may occur (e.g., death risk, hospitalization or abnormalities).
As such it is harmful and thus unethical not to be reported accurately [16][5].
Furthermore, lots of studies are possibly to be published at journals with great impact on the clinical care community [10]. The reduction of literature bias will inform prescribers and patients for new treatments, and benefit researchers in systematic reviews [9], [15][9], without leading them to biased results, while improving replicability and reproducibility.[17] All in all, such registries are of public accountability [15][18].
Finally, misleading records cancel the participants' attempt to contribute to health knowledge, especially for those relying on the study results [18].
Moreover, public registration platforms, such as ClinicalTtials.gov, use those data for variety of analysis regarding. This analysis may consider sample sizes, sponsor type (e.g., industry, on- industry), number of PIs, sponsors or funders per trial, sources distribution or locations for global studies, termination reasons, results, type of intervention, national economic health burden, unmatched or missing registries within platforms etc. [15].
Due to the above, International Committee of Medical Journal Editors (ICMJE), decided in 2005 that trials, which need to qualify for publication, should first be

registered in corresponding public platforms, such as ClinicalTrials.gov, before patients recruitment [15][9]. Similar, requirements were decided from Food and Drug Administration FDA in 2007[9] which required all clinical trials, with one or more sites in the U.S, to be registered at ClinicalTrials.gov.[7]

4.2.2. Clinical Trials.gov Data issues

Variety of issues have been noticed, regarding inaccurate data registration on Clinical Trials.gov website. Most common problems are missing or misleading data (e.g., Intervention types, RP names and roles, enrolment sizes, dates, sites etc.)[15] [16] [19] Non-reporting results on time is another issue. However, most of this information are being mandatory to mention (e.g., sponsor's name).
Regarding Responsible Parties (PR)/Principal Investigator (PI) personal information, there have been noticed trials not mentioning one or both of their names. Others not matching PRs' number with number of corresponding roles. Names that were typed with various formats, within different fields of registry, or that were even presented with personal information of fake non-existent individuals. Another example is trial registries with more than one PI per trial, despite only one should normally bear the responsibility. Even for multi-location trials each location's PI registry should be distinguished from the main PI.[15] A rarer issue is multiple primary outcome registries. [19]
Further, problem is wrongly labelled studies e.g., RCTs labelled as observational and via versa or even more general as 'NA' while there not.[15] [19]
Another issue concerns the delayed publication of studies and results, although trials should register prospectively by uploading main primary outcomes before enrolment starts, and upload results within one year of the study completion. However, not all obligations are met, as there are still trials not even registered [15]
Another, serious data misleading issue is about enrolment sizes. Many studies did not mention pre-designed or actual participants' sample sizes but rather mentioned extreme outlier values which were obviously mistaken, cluster sample sizes, population proportions etc. There were also several, discrepancies between sample estimates during register and actual accruals (usually less sample achieved actually), which is noticeable if compare enrolment registries to their text reports. [19]
Another issue of ClinicalTrials.gov is the great amount of unstructured text data. Explanation summaries and descriptions are conducted through brief texts without formatting restrictions, although they include precious information e.g, termination reasons.
Finaly, the most serious problem, is the discrepancies found between registries and their published papers [16] [10] [19]. However, here ClinicalTrials.gov usually seemed to be more accurate. Specifically, published studies were found not to mention part or whole number of adverse events (usually ClinicalTrials.gov was accurate) or actual efficacy, which is harmful for the health community. [10][16]

It is also notable that on ClinicalTrials.gov documentation, is referred that some fields were only made mandatory after specific dates, while others are not even mandatory to fill, as explained through the icons below: [6]

*****	Required
***§**	Required if Study Start Date is on or after January 18, 2017
[*]	Conditionally required

For example, Study Documents ***§** column from registry, has more than 2/3 null values compared to datasets extracted.
Due to the directions and pressure of the FDA and ICMJE the situation has improved considerably over the years [15]. Although it must be noted that ClinicalTrials.gov or US government do not review the accuracy of registrations [11].

4.3. Background

4.3.1. Termination Reasons

Termination reasons seem to be difficult to obtain, even though the efforts being made for accurate registered outcomes.[2] Especially, in advanced phases the explanation for termination is even more complicated. [7] However, many background studies have attempted to identify the reasons.
The most common reasons for trials' early termination are:

- <u>Low or fully unsuccessful enrollment</u>

Especially, in specific study fields, where for example the rarity of the condition hardens the collection of sufficient sample size [20][21][1][7][3][22]. In contrast, larger sample sizes appear lower termination rates [20]. Poor accruals also affect the statistic power of a study, thus it is important to fulfil the pre-designed participant's number [22]. It is notable that 'terminated' trials in ClinicalTrials.gov registry refer to trials which no longer treat participants, while 'withdrawn' to those which did not enroll any participant. [1]

- <u>Condition/Disease</u>

Field of disease studied in a trial is another indicator, with most common the cancer/neoplasms, followed by surgical, nervous system, cardiovascular, pediatric conditions etc. Additional diseases such as nervous system, cardiovascular and many others have also been studied in relation to early termination. [2][3][23][14][21]. Moreover, disease type seems to be affecting low accruals rates (e.g., rarer diseases tend to have lower enrollment). [7][23]

- <u>Location</u>

The number of sites impacts the enrolment outcome, as it is related to limitations of participants leaving nearby or not from sites etc. [14] or even the location of facilities. [2] [21][7]

- Healthy volunteers

Trials not accepting healthy volunteers were also noticed to have greater termination rates [14] .

- Safety-Risk Ratio

Another main explanation is drug efficacy, toxicity or more general the risk-benefit ratio [2] [1] [14][24][3].

- Funding/Sponsors

Moreover, characteristics regarding funding, or type of sponsor, [2] where non-industry sponsored clinical trials were found less prone to termination in some studies. [7]
Greater number of collaborators has also been noticed to result in higher termination probability [7].

- Administrative

Administrative explanations were detected as well, regarding protocols, investigators, drug supply or withdrawal etc. [1]

Despite the above, much analysis remains to be done to uncover all the reasons why a clinical trial is led to premature termination.

4.3.2. Previous Research

While all institutions and registries try to overcome early terminations and false registries to save human and funding resources, this topic has become popular to researchers who study reasons for CTs early termination. Many technics have been used apart from classic models such as ML algorithms or even NLP text mining methods to identify early termination characteristics as soon as possible. [2]

Follett L, Geletta S, Laugerman M (2019)
They studied ‘completed’ and ‘terminated’ trials, prior to 2015 from ClinicalTrials.gov registry. They used a combination of structured data with unstructured data from ‘description’ text column, as offered from ClinicalTrials.gov. The study included transformation of ‘description’ column to a structured format by creating binary columns based on the presence of most frequently detected words per category of trials.
Data engineering processes of ‘tokenization’ followed, by removing common or unwanted characters, separating words, creating ‘tidy data frame’ of words-trials, inverse document frequencies, binarization etc.
After structuring the data, the words were assigned to one of two outcome categories of the trials. More detailed, if some pre-specified frequency threshold was exceeded in one category, then the word was assigned to that category, while words that were rarely detected were assigned the other category of trials. Many words were evenly detected in both completed and terminated categories.
Finally, all data were fitted with random forest models.

Study resulted in specific words as predictors, with the top five (based on their predictive importance) being 'treat', 'chemotherapy', 'cancer', 'patients', and 'tumor'. Study was conducted in all phases and not separate per phase using random forest model. [2]

<u>Geletta S, Follett L, Laugerman M (2019)</u>
The same researchers as above in an alternative study, used NLP and ML algorithms to predict patterns of early trials' termination on trials started prion 2015.
Specifically, Latent Dirichlet Allocation (LDA) was used on unstructured texts data from descriptions of trials (e.g., 'Brief Summary' column).
LDA uses Bayesian methods to model each document as a mixture of topics and each topic as a mixture of words, which are after corresponded to probabilities.
Thus, a document can be represented by a vector of topic probabilities while each topic can be represented by a vector of word probabilities. Number of topics choice is mostly arbitrary.
Pre-processing of LDA included 'tokenization' as above study (data transformed to 'tidy data frame' so that there is one line per word -'token' per clinical trial).
Common English words and characters were also excluded. Document-term matrix is created by calculating the frequencies per trials.
Then, LDA was applied to the document-term matrix to create topic-word probabilities (β matrix) and document-topic probabilities (γ matrix).
From all topics only 25 were used based on their probabilities. Those 25 topics were fitted with random forest classifier with 6 other structured data, to assess importance of the LDA topic probabilities. Another Random forest was fitted also for the 6 structured data only. Then a logistic regression was used for interpretation reasons. It was proven that both type of predictors gave better results, as structured alone lost in rare diseases.
For each topic, top 10 words in terms of the term-topic probabilities were evaluated and resulted in a top 5 of diseases fields that seemed to be most valuable indicators of termination (e.g., 'surgery', 'surgical' , 'pain' underlying surgical procedures. First topics were the surgical procedures, then dermatological, heart conditions, pregnancy and last HIV. [3]

<u>Kavalci E, Hartshorn A (2023)</u>
They studied only 'completed', 'terminated' or 'withdrawn' (no enrolment), clinical trials from ClinicalTrials.gov register for year 2012-2022. Before 2011 missing data were greater so those trials were excluded.
This study combined two different public datasets, ClinicalTrials.gov and CHIA dataset. CHIA contains thousands of inclusion and exclusion criteria based on phase 4 studies. The ClinicalTrials.gov dataset (CSV format) was used as source for CT data and disease categorization features, while the CHIA dataset to generate a new set of eligibility criteria. Regarding the methodology, 'text mining' technics were applied on unstructured text data, in order to be used in combination with already structured data.

Double characterized phases studies (phases 1/2, 2/3) were assigned to both phases (phase 1 and phase 2 or phase 2 and phase 3). 'Withdrawn' trials were also characterized as 'terminated' in order to create a binary a outcome of 'Completed', 'Terminated' trials.
Some examples of features used from ClinicalTrials.gov, is the number of primary and secondary outcomes, number of sites, termination by disease (e.g., neoplasm studies were most likely to fail) etc. Others regarding eligibilities such as gender, age, healthy volunteers, number of inclusion/exclusion criteria etc.
Especially, CHIA dataset was used to generate more complex eligibility criteria which were also used as search terms in text data. Those, data were transformed into a binary dataset of 12,864 features.
Missing data for numerical features were handled via Multiple Imputation by Chained Equations (MICE).
Phases were studied at separately with a dataset for each.
Train test split used was 70:30 ratio. Imbalance dataset towards completed status (85:15 ratio) was handled with random under sampling on the train set only, in order for the test set to be representative of true status distribution. It must be noted that datasets were much smaller after train-test split and random under sampling.
Feature selection was used because eligibilities created enlarged the dataset and created noise. Firstly, elbow point of features vs model error plot was used to identify k number of features. Then, ANOVA F scores determined which features to keep.
For model evaluation during training, fivefold cross validation was applied. Train set was split into 5 folds. Model was trained 5 times and each time a different fold was used as the test set. Performance of the model was the average of 5. For Hyperparameter tunning, randomness technics were used, as it is a time-consuming process.
Then features were analysed with logistic but also ensemble models like Random Forest and XGBoost to predict probability of termination. XGBoost model evaluated as the best performing. The evaluation was achieved with ROC AUC curves, F1-score, Balanced Accuracy.
After, two different types of XGBoost models were trained for all phases together. One with only study characteristics features, and the other with additionally eligibility criteria search features and disease categorisation features. McNemar Chi-squared test for paired nominal data was used to test the null hypothesis is that models perform the same.
For model interpretation (SHapley Additive exPlanations) SHAP plots for local overview were used on individual predictions. [14]

<u>Elkin M, Zhu X (2021)</u>
They also studied interventional and observational studies from 2000 onwards from (ClinicalTrials.gov) registry. Similarly to the above, features were created from the combination of structured and unstructured data.
Features were formed in three categories:

- ‘statistic features’ : (e.g., number of collaborators, industry/non-industry sponsor, study design like masking type, eligibility criteria like gender, USA/non-USA location, number of countries etc.).
- ‘Keyword features’ : resulting from unstructured data from ‘description’ field. Analysed with TF-IDF (term frequency-inverse document frequency). TF is the frequency of words in a document while IDF is the frequency of words in several documents. Based on IDF a term that occurs in many documents (e.g., ‘the’), is not a good discriminator and is given less weight.
- ‘Embedding features’ : to overcome the problem of sparse data for keywords appearing in small number, using Doc2Vec to generate vector representations of words.

Furthermore, five feature selection methods were used, including ANOVA, ReliefF, Mutual Information (MI), CIFE (Conditional Informative Feature Extraction) and ICAP (Interaction Capping). Dowdall method was implemented to combine and select features from all five methods. Dowdall favours feature with many first preferences, thus, If a feature is accidentally ranked to the bottom by a method, it will have little impact the aggregation value.
Finally, data were fitted to four classifiers, Neural Networks, Random Forest, XGBoost, and Logistic Regression. Three types of features were fitted and evaluated separately and combined. Highest metrics resulted from combination of features (all three together). Ensemble models like random forest and XGBoost outperformed logistic. Similar to the first study above, cancer words were top detected to relate with early termination but also some rarer diseases (‘Mycosis’, ‘Fungoides’, and ‘Sezary’) which may reflect the difficulty of finding sufficient sample.[7]

<u>Williams R, Tse T, [...]</u>
Studied ‘primary outcomes’ field from ClinicalTrials.gov for CT studies to detect reasons of early termination and pattern that primary outcomes were reported on registry or public literature.
Study defined 3 types of explanation for trial termination:

- Scientific reasoning was most common (e.g., risk-benefit)
- Other than scientific study (e.g, low accrual as main in this category, administrative regarding to protocol or investigators, funding, product withdrawal, insufficient drug supply, etc.)
- Unknown-not provided

The results were that approximately 60% of trials were terminated because of trial conduct-related problems. There is currently significant interest in efforts to reduce the frequency of these problems, particularly with respect to participant recruitment.[1]
<u>Gayvert K, Madhukar N, Elemento O (2016)</u>
Studied CT related to drug interventions from ClinicalTrials.gov based on their features to predict toxicity related early termination of trials. In addition, drugs that

have already been approved from FDA, but failed in toxicity trials (FFT drugs), were compared.
Features regarding drug characteristics but also tissue expression (eg., genes etc.) where considered with additional data from DrugBank registry. Examples of drug features are Hydrogen Bond Acceptor Count, Molecular Weight, Polar Surface Area etc. and examples of target tissue features are Target Liver Expression, Target Network Degree, Target LOF Mutation Frequency etc.
This study developed a new approach for predicting the odds of clinical trial outcomes using random forest (PrOCTOR). PrOCTOR integrates the two types of features mentioned above in order to distinguish FDA-approved drugs from FTT drugs.
Principal components analysis (PCA) was used due to high correlation of target expression features.
Study results showed that frequent adverse events are acceptable for drugs treating serious diseases such as cancer, which was mentioned to all above studies as well. This because many drugs may have partly failed as FFT but alternative others may have failed to all similar requirements. Analysis also indicated that both categories of features were important indicators, while combination of other although uncorrelated features provided greater discriminative power as well. [24]

<u>Chapman S, Shelton B, […] Bhangu A (2014)</u>
Studied phase III/IV clinical trials from ClinicalTrials.gov database for year 2008-2009 for surgical procedures specifically, using logistic regression. Result were that surgical procedure trials were more likely to early terminate and appear low accrual rates as well. [23]

<u>Carlisle B, Kimmelman J, […] MacKinnon N (2015)</u>
Studied trials ‘completed’ or ‘terminated’ in 2011 from ClinicalTrials.gov registry, which had unsuccessful or fulfilled less than 85% of pre-designed enrolment. Other factors which resulted in poor enrolment, were greater number of eligibility criteria, non-industry funding, early phases, smaller number of sites, use of placebo etc. Statistical methods used, where chi squared test, backward feature selection and logistic model.[22]

Some of commonly used methods in most of the studies are:
In case of multifocalities studies, the location with most facilities, is chosen as the main location of the trial.
Evaluate models on test sets or parameter tunning with validation set cross validation and k-fold cv accordingly.
Evaluation metrics always include balanced accuracy (imbalance towards ‘completed’), sensitivity, specificity and ROC curves.
Imbalance of ClinicalTrials.gov datasets towards completed was usually treated with random under sampling.

5. Methods

5.1. Hypothesis Testing

5.1.1. H_0 vs H_1

Hypothesis testing explores how likely it is that the observed difference would be seen by chance alone if the null hypothesis was true. The goal is not to accept or reject the null hypothesis but rather to evaluate how likely it is for an observed difference to happen if the null hypothesis is true [25].

- $\boldsymbol{H_0: \theta = \theta_0}$ **Null Hypothesis**

Assumes no difference between groups (e.g., in their mean or proportions). Any difference lies due to chance. [26]. It expresses a pre-specified value of an unknown parameter of population θ, such as $H_0: \theta = \theta_0$, where θ_0 is a known constant [27].

- $\boldsymbol{H_1: \theta \neq \theta_0}$ **Alternative Hypotheses**

Assumes significant differences between two groups or else that there is significant deviance of θ from θ_0. The alternative hypothesis must be based on a logical explanation based on the problem background [26].
Alternative Hypothesis can be two-sided (or two tailed) such as $H_1: \theta \neq \theta_0$, or one-sided (one tailed) such as $H_1: \theta > \theta_0$ or $H_1: \theta < \theta_0$ depending on the problem (e.g., one tailed: assuming the true time results of a drug appear equals to 10, then $H_0: \mu = 10$ and $H_1: \mu < 10$) [27].

Then a random sample of the evaluated populations is chosen. The acceptance or rejection of H_0 is based on the value of a given test statistic, which needs to have a defined distribution under the H_0. Afterwards, values of statistic test are separated into two discrete spaces. In one region H_0 is accepted while in the other rejected with second space being called '$critical\ region$' [27].

5.1.2. p value

$p\ value$ is the probability that an observed effect is simply due to chance, if H_0 is true[28] or else is the probability of seeing the observed difference, or greater, just by chance if the null hypothesis is true.
As a probability it can take any value between 0 and 1. Values close to 0 indicate that the observed difference is very unlikely to be due to chance and in contrast values near 1 indicate that observed difference is very likely to occur due to chance and random variation.
As an example, assume p value = 0.08 for a problem studied with logistic regression with given Odds ratio = 0.33. p value indicates that there is an 8% chance or else (8 out of 100 times) to observe such a difference of 66% risk reduction (or extremer) if no effect occurs, just by chance. [25]

Finally, $p\ value$ is a measure of the strength of an association but does not provide any measure of the effect size and so it must not be used as the only guide for evaluating a result.[29]. Thus, p value should not be used for clinical decisions, as they offer no evidence for effect size, adverse events, etc.
More specifically the 'significant' or 'non-significant' result based on p value is obsolete. This is because, for e.g., p = 0.05 values, researchers will wrongly accept H_1 5% of the times (5% of times the result occurs due to randomness), meaning to fall into a $Type\ I\ error$. However, wrongly rejecting the null hypothesis can induce a $Type\ II\ error$ too. The wrong evaluation is more probable in small sample sizes, where significance may be missed due to small samples.[25]
One of the limitations of the p-value in hypothesis testing, is that it assumes the null hypothesis is always determinable and exactly equal to a certain quantity (usually zero). This would mean to get results which are identical. Any minimal variability would produce a deviation from the null hypothesis.
This can lead the testing to become too sensitive to minor differences, making them appear significant although they are not. This can lead to misrecognizing natural variability or measurement error as a significant effect. This problem is likely when enlarging the sample size too much.[30]

5.1.3. Test Statistics for Logistic

For logistic regression statistic test is z statistic that divides the difference in observed and $\mathrm{H}_0's$ proportions by the null standard error of sample's p. z statistic measures the number of null standard errors that the sample proportion p falls from the null hypothesized proportion π_0.

$$z = \frac{p - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

$H_0: \pi = \pi_0$, π_0 : population proportion under H_0
n : sample size, p: sample proportion
$E(p) = \pi$: mean of p
$\sigma(p) = \sqrt{\frac{\pi(1-\pi)}{n}}$: Standard error of p
$\sigma(p) = \sqrt{\frac{\pi_{0(1-\pi_0)}}{n}}$: Null standard error of p, that holds under H_0

Note that as sample n increases the standard error of samples proportion p decreases and approximates populations proportion π and distribution of p approximates the normal. [31]

5.1.4. Confidence Intervals CL

Confidence intervals determine the range of all plausible values that a parameter can be assigned to [31] or else a range of values within which it is likely that the true population value lies.[25]

$$p \pm z_{\alpha/2}(SE)$$

The estimated standard error of p.

$$SE = \sqrt{p(1-p)/n}$$

Note that this formula substitutes unknown population parameter π with the sample proportion p in the population standard error:

$$\sigma(p) = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$z_{\alpha/2} = z_{0.025} = 1.96$: The standard normal percentile having right-tail probability equal to $\alpha_{/2}$.
$100(1-\alpha)\%$, where $\alpha = 0.05$: 95% confidence interval for π i.e., 95% of the times values of π belong in this range of values. We can be 95% confident that the population proportion is between the two values that arise from equation. [31]

It must be noted that confidence intervals regarding estimates of regression coefficients obtained from penalized methods are problematic. Penalized methods introduce bias towards the null and thus the confidence intervals obtained are not very meaningful for strongly biased estimators.[32]

5.1.5. Type I, II Error

As the rejection of H_0 is based upon one value of the statistic test, two mistakes arise to make:

- **Type I error**

Is to reject H_1 while it is actually true. This error can occur when statistic test's value belongs to the $critical\ region$ but should not.

- **Type II error**

Is to accept it when it is actually untrue. This error can occur when statistic test's value belongs outside the $critical\ regio$n but should not.[27]
[33]
It cannot be accurately specified what type of error occurs. However, the possibility of making $Type\ I\ error$ can be computed through the '$significance\ level\ \alpha$'. In contrast, possibility of Type II error cannot be computed as it is based on unknown true value of θ parameter. The unknown probability of $Type\ II\ error$ is symbolized as $\beta(\theta)$ its domain is θ values specified by the alternative hypothesis.

The probability of rejecting H_0 while actually true is called '$power$' and defined as $\gamma(\theta) = 1 - \beta(\theta)$.
The critical region is chosen based on where the probabilities of Type I and II error could be minimum. The concurrent minimization of those two is practically not achievable. So values of α, $statistic\ test$ and $critical\ region$ which lead to small values of $\beta(\theta)$, should be chosen wisely in order to minimize those two errors [27].

5.2. Statistical Tests

5.2.1. Akaike information criterion (AIC)

AIC tests a model by how close the fitted values are to the true values, as summarized by a certain distance between the two. The optimal model is the one which tends to have the closest predicted and true outcome probabilities for (e.g., for logistic regression). This model also minimizes the following:

$$AIC = -2(loglikelihood - numberofparametersinmodel)$$

AIC depends on loglikelihood and number of predictors. Increase in predictors increases AIC score and thus has a negative impact. So, smaller AIC indicated better performing model.
In case of same number of predictors compared between two compared models or even same predictors with different formatting (e.g., log, sqrt etc.), then the AIC's part of equation that is compared is $-2loglikelihood$ (i.e., the predictors part in the equation is the same for all compared models).[31]

5.2.2. Chi-Square Test

Chi square Distribution
Chi-square test was proposed from Karl Pearson back in 1900 and is used for comparing unpaired group data, for categorical/nominal variables [34] [27]. Chi square test uses frequencies (i.e., the number of observations per category).
As a non-parametric test, it doesn't make population characteristics assumptions (e.g., about distributions). However, non-parametric tests are generally less powerful to identify significant differences and reject null hypothesis [34][35] (i.e., greater Type II error rate). [35] [36]
Chi square test is used to test if observed frequencies differ significantly from the expected ones. Firstly, is it used to recognize deviations of the sample's observed frequencies from the theoretically expected ones (one-way tables) and secondly to recognize relationships between categorical variables (contingency tables), as explained below.
The closer to accept Ho (i.e., closer the observed to the expected frequencies), the smaller χ^2 value will be and via versa. Also, as the degrees of freedom (df) increase

the χ^2 value increases, as more squared differences are summarizing and the critical value for rejection of Ho increases too. [35] [37].
Chi square distribution is skewed, with skewness decreasing as the df increase. Thus, as n → ∞ it approximates $N(0,1)$ normal distribution [37].
N should be large enough for the chi square to achieve approximation (i.e., greater than sample 5).[38]. Finally, as the sample size gets greater the more Chi square test can detect smaller effects, as the statistical power increases.

Degrees of freedom

A standard normal deviate is a random sample from the standard normal distribution. The Chi Square is the distribution of the sum of squared standard normal deviates [37], thus all values are positive.[35] The degrees of freedom (df) equal the number of independent normal deviates (e.g., only 1 independent normal deviate means a $\chi^2(1)$ distribution with 1 df). [37]

Chi square applications

There are always two hypothesis H_0 and H_1. To reject Ho, it must be $\chi^2 > c$, where $c = critical\ value$, defined from literature, based on df and a sig. level. Beyond c threshold null hypothesis is being rejected. [38][35]

Most common applications of chi square test are the two below.

- Testing Goodness of Fit (One-Way Desing) (One-Way Tables)

Tests if observed data (from sample) differ significantly from an expected population frequency, that can be theoretically determined even rationally or empirically. One-Way table example:

Outcome Category	Ei	Oi
1	25	18
2	10	22
3	13	30
4	5	21
5	12	17
6	20	12

$$X^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

[27]

X = Random variable from which observations occur with mutual independent outcome.
O_i = (i.e., $O_1, O_2, \ldots, O_k$) = observed frequencies that belong to set I_i (i.e., $I_1, \ldots, I_k$).
$Ei = np_i$ = expected frequency of set I_i (or i^{th} category) based on H_0.
p_i = probability that a random observation's outcome is from set I_i (or i^{th} category).

i = 1, … , k, categories of outcomes.
df = k-1. [27]

Degrees of freedom are calculated based on the number of categories k - 1. [35] One category's frequency can be calculated from the sum of all other frequencies and total number of sample/experiments occurring.
So df = number of categories – 1 (e.g., 6 category outcomes df = 6 - 1 = 5). [39]

$$\sum_{\{i=1\}}^{k} O_i = n$$

[39]
O_i = each observed frequency of corresponding category
n = total number of independent observations or total number of independent random experiments. [37]

- <u>Independency test (Two- Way Design) (Contingency - Frequency – Cross Tabulation tables)</u>

Tests whether the distribution of frequencies in one variable is related to the distribution of frequencies to the second variable. If true (H_1 accepted) the two variables are related.

- H_0: 2 Variables A,B are independent
- H_1: 2 Variables A,B are related

[35]
df $=(R-1)(C-1)$: Degrees of freedom are calculated based on rows and columns of the table
R = number of rows, C = number of columns. [37]

	B_1	B_2	…	B_c	Sum A
A_1	O_{11}	O_{12}	…	O_{1c}	O_{1j}
A_2	O_{21}	O_{22}	…	O_{2c}	O_{2j}
…	…	…	…	…	…
A_r	O_{r1}	O_{r2}	…	O_{rc}	O_{rj}
Sum B	O_{i1}	O_{i2}	…	O_{ic}	N

The expected frequency for a cell in i[th] row and j[th] column:

$$E_{ij} = n \cdot \frac{\sum_{l=1}^{r} O_{il}}{n} \cdot \frac{\sum_{m=1}^{c} O_{mj}}{n} \quad or\ generally\ \ E_{i,j} = \frac{O_i O_j}{N}$$

[27]

O_i = Total for the i[th] row,

O_j = total for the j[th] column,
N = total number of observations. [37]

$$X^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \ \ or\ generally\ \ X^2 = \sum_{i=1}^{k}\frac{(O_i - E_i)^2}{E_i}$$

[27]

Each observation contributes to only one cell (unpaired sample). Thus, the sum of all cell frequencies is the same as the number of observations (sample) in the experiment [37]

Ai = Variables with categories: i = 1, ..., r,
Bj = Variables with categories: j = 1, ..., c.
k = categories of combination i, j of variables A,B.
Oi,j = number of observations belonging in categories of Ai and Bj variables.
pij = probability of an observation to belong in cell (i,j)
pi = probability of an observation to belong in a category of A variable.
pj = probability of an observation to belong in a category of B variable.
Under H_0 : pij = pi i.e., probability of belonging to one of i[th] categories of A variable, does not depend on probability of belonging to j[th] category of B variable.

- 2x2 Tables Contingency tables

A	B	**A+B**
C	D	**C+D**
A+C	**B+D**	**N**

$$X^2 = \frac{N(AD - BC)}{(A+B)(C+D)(A+C)(B+D)}$$

A, B, C, D = observed frequencies, O_i, in each cell
$AD = OA \times OD$
$BC = OB \times OC$
N = Total number of observations [35]

5.2.3. Mann-Whitney U

The Mann–Whitney U test is a Rank randomization test, which is commonly used as a two-sample nonparametric statistical tests. Rank Randomization tests do not depend directly on actual numbers of the variable but rather convert the numbers to ranks and compute the randomization test afterwords. [37]

It is a nonparametric test thus, it can be used for data which do not follow a normal distribution or with a small sample size N. This test is the equivalent to the parametric independent t-test. [26]
The disadvantage of rank randomization tests is that a part of information is lost when converting to ranks, thus they are less powerful than simple randomization tests. [37]
The null hypothesis (H0) states that the two samples come from the same population, indicating no difference. The alternative hypothesis (H1) states that the two independent samples come from different populations.[26]

Original Data	
Experimental	**Control**
7	0
8	2
11	5
30	9

Ranked Data	
Experimental	**Control**
4	1
5	2
7	3
8	6

The probability value is determined by computing the proportion of the possible arrangements of these ranks that result in a difference between ranks of as large or larger than those in the actual data.
Firstly, the ranks are assigned. Then the sum of all ranks of both groups is calculated (36 for numbers 1-8 in the example) and the sum of experimental ranks (24 in the example). Afterwords, it is considered how many ways the values between groups can be rearranged to achieve an experimental sum of ranks ≥ 24.
The total number of rearrangements is given by:

$$C_r^n = \frac{n!}{(n-r)!\ r!} = \frac{8!}{(8-4)!\ 4!} = 70$$

In this example from 70 total ways, 4 of them give ≥ 24 sum of ranks in the experimental group. Thus, the probability is 4/70 = 0.057.

Regarding the direction, the test ca be one-tailed or two -tailed. For example, if comparing only experimental being larger than their initial sum of ranks, then it is one-tailed. The two-tailed probability is (2)(0.057) = 0.114, equal to 8/70 ways to arrange the data so that the sum of the ranks is either as large or larger or as small or smaller than the initial sum.
In practice the above tables can be used for samples 4-10 per group. For large sample sizes, all rearrangements are computationally difficult to compute and thus the probability. In this case only critical value is used, from which the probability can be then computed. The critical value is calculated from below equation and follows approximately normal distributed.

$$Z = \frac{W_a - n_a (n_a + n_b + 1)/2}{\sqrt{n_a n_b (n_a + n_b + 1)/12}}$$

Wa: sum of the ranks for the first group
na: sample size for the first group
nb: sample size for the second group
Z: test statistic
[37]

5.3. Correlation & Collinearity

5.3.1. Categorical Correlation

For categorical correlation, the Chi-square test of independence is applied using contingency tables. Subsequently, the following measures are used to calculate the test's effect size, as explained below.

5.3.1.1. Cramer V & Phi

The effect size of this measure is calculated through Cramér's V value (or Cramér's phi). [35] [30]. It is the statistical strength test for Chi-square test. [36] Thus, Cramer's V test is used for independence testing between nominal data.
Cramer's V is used for tables bigger than 2×2, while Phi for 2x2 tables. Its values range is between 0 and 1 and are always positive. [40]

$$V = \sqrt{\frac{\chi^2}{(N)\left(df_{row/column}\right)}} = \sqrt{\frac{\chi^2}{N(m-1)}}$$

[35] [30]
Phi is an alternative of Cramer V without the correction factor $(m-1)$ for unequal ranks. [30] Thus, it is also a measure of the strength of an association between two categorical variables in a 2×2 contingency table. Generally, $\phi \leq 0{,}1$ is considered as small effect size and $\phi \geq 0.3$ are considered to have large effect size. [35]

$$\phi = \sqrt{\frac{\chi^2}{N}}$$

5.3.2. Continuous Correlation

5.3.2.1. Pearson's r

The most common test for evaluating the correlation between two quantitative variables is Pearson's correlation coefficient. Pearson's r assumes that both variables follow a normal distribution and that the relationship between them is linear. If those two, do no occur, it may give erroneous conclusions. Logarithm scale transformation may be needed for data to approach the above conditions.
The square of Pearson's value is an estimate of the percentage of variability in the values of one variable that is explained by the variability in the other e.g., r = 0.7 means that 49% of the variability of one variable can be explained by the variation in the values of the other.

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2}\sqrt{\sum(y-\bar{y})^2}}$$

[30] [41][42]

5.3.2.2. Kendall , Spearman

It is very common that samples of clinical data do not follow a normal distribution (e.g., disease severity indexes, years of study etc.). Spearman and Kendall rank correlation coefficients are the most used tests that do not assume normal distributions, [41] thus they can be used for data with extreme values or outliers. Those tests substitute the original data for their ordered ranks (each value of Continuous variable is assigned to a number with order based its value e.g., 0, 1, 2 etc.) . If there are two equal values they will be ranked with their ranks average (e.g., [1, 2, 2, 3] -> instead of [1, 2, 3, 4] -> becomes [1, 2.5, 2.5, 4] [40]. They are also used in cases in which at least one of the variables is ordinal. Another advantage those nonparametric tests is that they also do not assume linear correlations. More specifically, they detect monotonic behavior as they test any gradual relationship in the same direction (rising or falling) for the whole data. [41]
Kendall and Spearman test is more robust to outliers than Pearson's correlation coefficient.[43]
H_o: Hypothesis of conditional independence between X and Y.

Spearman's correlation coefficient:

$$r_k = \frac{\sum(u_i-\bar{u})(v_i-\bar{v})}{\sqrt{\sum(u_i-\bar{u})^2\ \sum(v_i-\bar{v})^2}}$$

u_i, (v_i) : The rank of Xi (Yi).

Thus, under H_o: $E(r_k) = 0$ (mean of all blocks) and $\mathrm{var}(\mathrm{r_k}) = (\mathrm{n_k} - 1)^{-1}$
r_k : Spearman in block *k* (subgroup), the rank analogue of the product-moment correlation coefficient [44]

Range of values for Pearson and Spearman correlation coefficients in range [-1, 1]:

+1	−1	Perfect
+0.9	−0.9	Strong
+0.8	−0.8	Strong
+0.7	−0.7	Strong
+0.6	−0.6	Moderate
+0.5	−0.5	Moderate
+0.4	−0.4	Moderate
+0.3	−0.3	Weak
+0.2	−0.2	Weak
+0.1	−0.1	Weak
0	0	Zero

[40]
It must be noted that p value of these tests does not indicate the strength of the relationship but rather gives the probability that the correlation value may or may not occur by chance. [40]

5.3.3. Continuous – Categorical Correlation

5.3.3.1. One Way Anova

In ANOVA the independent variables X are discrete/nominal, and the dependent Y Continuous. It uses the linear model to describe how Y varies when the changes in X are discrete. More specifically, it describes how the variance of a dependent Y can be explained by differences in the group means of Y (defined by levels of X) relative to the overall mean of Y. [30]
The magnitude of the effect is indicated from how much the clustering explains the variability with respect to the overall variability (scatter of all Y observations).
The general form of any ES (Effect Size) measure as:

$$ES_{variance} = \frac{Variation_{explained}}{Variation_{total}}$$

Specifically, eta test for 1 way ANOVA is defined as follows:

$$\eta^2 = \frac{SS_{factor}}{SS_{total}}$$

[30][42]
It must be noted that η^2 tends to inflate the explained variability giving quite larger ES estimates. Additionally, in models with more than one features it may

underestimate ES as the number of features increases. Due to this, for more than one features it is advisable to use the partial-η^2 instead.[30]

5.3.4. Variance Inflation Factor (VIF)

Collinearity happens when two or more variables are related to one another, as they tend to increase or decrease together. Due to that, it is difficult to determine the effect of each separate variable on the outcome. [45]
Collinearity results in reduction of the accuracy of the coefficients' estimates. More specifically, the variance of coefficients' estimates is increased, which leads to increase in their standard errors as well. Due to this confidence intervals (CI) are getting wider as well. Since the t-statistic is calculated by dividing $\widehat{\beta_J}$ by its standard error, collinearity results in its decline. A low t statistic indicates an insignificant coefficient, which in combination with the wider CI makes the model unreliable. [46]

$$z = \frac{\hat{\beta}}{SE}, \quad SE(\widehat{\beta_J}) = \sqrt{Var(\widehat{\beta_J})}$$

The power of the hypothesis test is also reduced and so, null hypothesis may not be rejected ($H_0: \beta_j = 0$). More generally, the importance of a variable is masked due to presence of another.
The simplest way to detect collinearity is the correlation matrix of predictors. However, it can be insufficient for multicollinearity problems (where three or more variables are collinear). For these cases Variance Inflation Factor (VIF) is more appropriate. VIF is the ratio of the variance of $\widehat{\beta_J}$ when fitting the full model divided by the variance of $\widehat{\beta_J}$ when fitting on its own. The smallest value VIF can take is 1, which means no correlation is present. Values greater than 5 to 10 (as a rule of thumb) indicate collinearity issues [45]. In practice the value of 10 and above is the most common choice, however, the choice depends on how strict on multicollinearity one wishes to be [47].
A solution to the problem is to drop one variable or else to combine both collinear variables to one (e.g., an average of the standardized versions). [45]

$$VIF_j = \frac{1}{1 - R_j^2}, \quad VIF_j \geq 1, \quad j = 1, \ldots, p-1$$

X_j : j-th independent variable. [47]
$R_j^2 or\ else\ R^2_{X_j|X_{-j}}: 0 \leq R_j^2 \leq 1$: R^2 from a regression of X_j onto all of the other predictors.

- $R_j^2 = 0$: Absence of multicollinearity
- $R_j^2 = 1$: Exact multicollinearity. Leads to infinite variance of coefficients.

$1 - R_j^2$: Tolerance
X_j is the dependent variable for R_j^2, while $X_i, i \neq j, i = 1, \ldots, n$ are independent variables for R_j^2

- $VIF_j \geq c$, then X_j is causing multicollinearity problem.
- $VIF_j < c$, then X_j is not causing multicollinearity problem.[47]

Although VIF can detect the presence of multicollinearity and the related variables, it cannot exactly explain which variables and how they cause multicollinearity.[46]

5.4. Missing Data Mechanisms

Definitions

- Completers : Rows of data with no missing values
- Complete data: Full dataset, including observed and unobserved entries
- Complete case analysis : Analysis on data using only rows with no missing values

5.4.1. Mechanisms

5.4.1.1. Missing Completely At Random (MCAR)

Missing data occurrence does not depend on other variables observed or unobserved but happens due to chance. $R \perp\!\!\!\perp Y$ (R , Y are independent). Thus, the probability of missingness pattern $r \perp\!\!\!\perp Y$ and $r \perp\!\!\!\perp X$ (e.g., it is unrelated to study's issues).
Complete case analysis is unbiased although inefficient. The moments (e.g., means/variances), are valid estimates of complete data and distribution also coincides with population's. MCAR mechanism can be rejected from data.
This is the least problematic missing mechanism, however, it rarely occurs.

$$P(R|Y) = P(R) = \pi(r) = constant$$

5.4.1.2. Missing At Random (MAR)

The probability of missing data depends on observed variables but not on missing data. Missingness is allowed to be also dependent on other fully observed covariates or auxiliary variables.
MAR is a middle problematic mechanism, more likely to happen and finally, many imputation methods assume MAR mechanism. MAR cannot be rejected through data.

$$P(R|Y) = P(R|Y_i^o)$$

When MAR, complete case analysis does not provide valid estimates (e.g., means, var). Also, distribution differs from the population's for same Y_i, for non-responders.

5.4.1.3. Missing Not At Random (MNAR)

Probability of missingness is a function of Y_i^m i.e., it depends on missing data itself. Observed data Y_i^o cannot be used to fill missing data, as any imputation method would give biased results.
Cannot reject or determine if the mechanism is MAR or MCAR through data.
MNAR is the most problematic mechanism.

$$P(R = r|Y) = P(R|Y) = \pi(Y) \sim Y_i^m$$

[48][49][50]

5.4.2. Imputations

Y_j for $j = (1, \ldots, p)$: One of p variables with missing data
$Y = (Y_1, \ldots, Y_p)$

- $Y^{obs} = (Y_1^{obs}, \ldots, Y_p^{obs})$
- $Y^{mis} = (Y_1^{mis}, \ldots, Y_p^{mis})$

$Y(h)$: The h^{th} imputed data sets where $h = 1, \ldots, m$
$Y_{-j} = (Y_1, \ldots, Y_{j-1}, Y_{j+1}, \ldots, Y_p)$: The collection of the $p - 1$ variables in Y except Y_j
Q : Quantity of scientific interest (e.g., a regression coefficient). It is often a multivariate vector. Q encompasses any model of scientific interest.
Q cannot be estimated from Y^{obs} without making unrealistic assumptions about the missing data
Multiple Imputations is referring to the method where to several imputed versions of the data are created by replacing the missing values by plausible data values. These values are drawn from a distribution specifically defined for each missing point. The values to be imputed are obtained from the posterior distribution by sampling iteratively from conditional distributions.
The imputed sets ($Y(h)$) are identical as it concerns the observed data but differ in their imputed values. This differences' magnitude indicates the uncertainty of imputed values.
After the imputations, estimation on each dataset's Q must be done. The estimates of $\widehat{\mathrm{Q}^{(1)}}, \ldots, \widehat{\mathrm{Q}^{(\mathrm{m})}}$ will differ due to the difference in imputed values of each produced dataset. Afterwords, the m estimates will be pooled as one $\overline{Q}$ for which its variance is going to be computed as well. For approximately normally distributed $\widehat{Q^{(\mathrm{m})}}$, $\overline{Q}$ can be calculated from their mean. Alternatively, pooling a $\overline{Q}$ based on Rubin's Rule methodology is applied. [51]
The benefits of multiple imputations is that is accounts for the uncertainty induced by missing data and provided more accurate estimates. [50]

5.5. Outliers

An outlier is a point that y_i is far from the value predicted by the model and far from the rest of datasets y_i values. An outlier can arise from various reasons such as wrong input of data. Its presence could affect the models fit, but it can also not have great effect on it. Either way it affects other values such as RSE for regression and thus p-values, CI etc. It also affects the interpretation of the model.
Residuals could be identified by plotting the residuals, however, even this does not always make clear which should be considered as residuals. Alternatively, studentized residuals could be used, which are produced by dividing the e_i by their standard error. Values greater than 3 are considered to indicate outliers.
A solution to the problem is to remove the outlier point from the data once it is identified. However, it should be considered that it is possible for the point value to be accurate, but the model fails to predict. [45]

5.6. Quality of Fit

To evaluate the quality of fit, there needs to be measured how well predictions match the true data values. The interest lies in evaluating the accuracy of predictions on unseen test data, rather than training data.
The quality of fit is evaluated through the accuracy of estimation $\hat{f}$, i.e., how well it performs when used to predict data. As for classification:

- **Training error rate**

The proportion of mistakes if estimator $\hat{f}$ is used to the training set.

$$\frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y_i})$$

$\hat{y_i}$: Predicted label of i[th] observation by using $\hat{f}$
y_i: True label of i[th] observation
$I(y_i \neq \hat{y_i})$: Indicator variable. $I(y_i \neq \hat{y_i}) = 1$ if $y_i \neq \hat{y_i}$ and $I(y_i \neq \hat{y_i}) = 0$ if $y_i = \hat{y_i}$.

- **Test error rate**

It is associated with a set of test observations (x_0, y_0) on which $\hat{f}$ has been applied:

$$Ave\big(I\,(y_0 \neq \widehat{y_0})\big)$$

$\widehat{y_0}$: Predicted label by using $\hat{f}$ classifier to the test observation.
A good classifier is one for which the test error is smallest.[45]

Overfitting & Underfitting

- Overfitting

The model appears to be a good predictor on the training set while underperforming on unseen data of test set. This is due to its low bias and high variance in which the model may adapt too strongly to the data which includes noise. When overfitting, we say that model lacks 'generalizability'. [52]

- Underfitting:

The model has high bias and lower variance. This is due to potential interrelationships between data features that may have been ignored. [52]

5.7. Bias-Variance Trade-off

Bias - Variance and their relationship is fundamental to the performance of supervised ML models. Bias variance trade-off refers to finding the best balance between bias and variance to have a more generalizable model. Increased bias translates into underfitting, while increased variance translates into overfitting [52].

- **Variance**

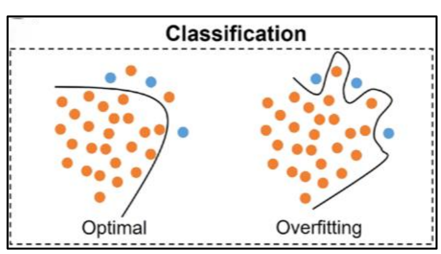
Refers to the amount by which $\hat{f}$ would change if another data set was used (e.g., different training data). Ideally, $\hat{f}$ should not change much within different sets. It gives model's flexibility. Variance induces overfitting (i.e., indicates false relationships due to noise). High variance model means that small changes of dataset (e.g., change any data point) result in great changes in $\hat{f}$ estimate. More flexible models tend to have greater variance.

- **Bias**

Refers to the *error* introduced by approximating a real-life problem, which may be extremely complicated, by a much simpler model. It gives the model's rigidity. High bias induces underfitting i.e., missing real relationships between features and the target. Low bias translates into that the model fits the data more closely. For example, a linear relationship may not fit well in real-life problems, so there will be bias in the estimation of $\hat{f}$.

- **Flexibility**

More flexible methods, have more variance and less bias. Flexibility is associated with degrees of freedom of the model, where less df means more restricted and robust model.[45]



The ultimate goal is to find the right balance between Bias-Variance (bias-variance trade-off)

[17][52]

5.8. Cross-validation (CV)

The more complex and higher dimensional the data becomes, the more the model's fit reflect the noise values in the data set. A sign of overfitting is that the model fits well to trained data but poorly to test data from the same initial sample .
k-fold has common use in which the data split into k different subsets (folds). The model is trained on every subset except one held-out subset in which the model is tested. CV needs to assure a total isolation of the test data. No test data must be part of selective (e.g., variables, parameters etc.) or training procedures. If not, then 'information leaking' is going to happen from test data's information to fitting procedure, which results to inflated accuracy scores. [53]. Also, variable selection must not be done prior to cv, however, many studies miss this and have biased results.[54]

Uses of CV
CV is commonly used to overcome overfitting by applying it for model selection, models' hyper-parameter tuning (HPO) and model assessment.[54] Another use is for estimation of test error rate, through training set, as training error rate is usually, much lower than test set's. The error curve resulting from this process from the training data is like the one resulting from true test data error. [45]

Drawbacks of CV
One pitfall cv may fall into, is that when splitting data into folds, different subsets will be produced each time, so different tunning parameters may be chosen. For this problem 'repeated grid search cv' may be used, which repeats a specified $Nexp$ times, and produces same amount of cv errors, so the minimal cv error parameter is chosen. [54]
Another problem is the assumption of CV that observations are independent. The existence of this assumption is critical to the validity of k-fold cross-validation e.g., it is violated for data from family members, which tend to have similarities. [53][55]

5.8.1. k-fold CV

Firstly, dataset is randomly divided into k groups (folds), of approximately equal size. Then the statistical method is fitted to $k - 1$ folds and tested to the remaining held out fold (validation set). This process is repeated by having different fold as validation each time. From this process k number of error scores (number of misclassifications for binary outcome), that are averaged to the final error. [45][56]
The error rate calculated from each iteration of cv, is the probability of misclassification for classification problems. The error rate is a measure of

predictive capacity of the classifier – model. Under random sampling, it can be proved that cv is almost unbiased. Bias is not too great as long as n/k is small.[56]

$$Err_i = I(y_i \neq \hat{y_i})$$

Per fold error rate: $\widehat{\varepsilon_n^{cv(k)}} = \frac{1}{n}\sum_{i=1}^{n} Err_i$

k- fold cv averaged error rate: $E\left[\widehat{\varepsilon_n^{cv(k)}}\right]$

[56][45]

e_n : true error (probability of misclassification) of a classifier designed on a sample of size n. [56]

[57]

5.8.2. Leave One Out Cross Validation (LOOCV)

The extreme version of the k-fold CV approach in which k equals the total number of individual data entries n of dataset (e.g., training dataset). Instead of splitting the dataset of interest into k-folds, here only one data point is left out as test point each time for each cross-validation step.[52] So, LOOCV can be referred as a special case of k-fold that k here equals to all data points n.

$$Err_i = I(y_i \neq \hat{y_i})$$

$$\widehat{\varepsilon_n^{l}} = \frac{1}{n}\sum_{i=1}^{n} Err_i$$

LOOCV averaged error rate: $\mathrm{E}\left[\widehat{\varepsilon_n^{l}}\right]$

[56][45]

5.8.3. Validation Set approach

Validation set is used for evaluation of a method, usually to find the performance of a model on unseen data. It includes simply dividing the initial dataset into two subsets, the 'training set' and 'test set-validation set or hold-out set'. The model fitted on the training data and then predict values of unseen test data. Resulting test error rate is like above. For n : test set data points:

$$Err_i = I(y_i \neq \hat{y_i})$$

$$\mathrm{CV}_{(\mathrm{n})} = \frac{1}{n}\sum_{i=1}^{n} Err_i$$

5.9. Scaling (Normalization, Standardization)

Scaling of features from a wide range of values into a more specific one, is crucial before applying models, especially in real world data, where even different units may be used. Scaling influences models performance and ensures no feature will over/under present than another. This ensures that no feature predominates among others e.g., in distance calculation of distance-based models like KNN. [17][58][59][52] Most common methods are:

- **Standardization**: Transforms data to zero mean and unit variance

– Z-score

The most common method. It scales values to zero mean and unit variance, resulting in negative or positive values, based on if scaled data are below or above mean value of initial data. The drawback is that in presence of outliers, they influence mean and standard deviation, which are included in scaling formula. Note that outliers' magnitudes (i.e., extreme values), vary per feature and so will their scaled range of values. Thus, some features' predominance will still occur. This is why this scaling method is not so efficient when comes to outliers.

$$x' = \frac{x - \bar{x}}{S}$$

– Robust Scaler

Useful to handle outliers, because it uses the interquartile range (IQR between 0.25-0.75 percentile). The benefit is that it is not influenced by large outliers.

$$x' = \frac{x - x_{\text{median}}}{\text{IQR}}$$

- **Normalization**: binding values between two numbers, usually [0,1]

– Min-max Normalization

It is linear transformation method. Data are scaled within range $[\min(x_{\text{new}}),\max(x_{\text{new}})]$, without losing relations among them. It results in a smaller standard deviation of scaled data. The drawback is that it is prone to outliers.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}\,(\max(x_{\text{new}}) - \min(x_{\text{new}})) + \min(x_{\text{new}})$$

  - Linear Scaling to Unit Range

    Data are scaled within range [0,1]. Results in a smaller standard deviation of scaled data. It is a special case of Min-Max scaler for $\max(x) = 1$ and $\min(x) = 0$, with similar drawbacks and benefits.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

– <u>Maximum Absolute Scaler</u>
It is a commonly used normalization method. Resulting values range approximately from -1 to 1. More specifically, if only positive values included in features, the range is [0, 1], while if only negative, the range is [-1, 0]. If both the range is [-1, 1]. The benefit is that no data of opposite sign appear after scaling, which may not be justified from dataset. For positive values of data, Maximum Absolute Scaler is also prone to outliers.

$$x' = \frac{x}{\max(|x|)}$$

– <u>Percentile transformation</u>
It is a linear method of scaling used to compare values of data or evaluate where data values stand compared to other candidates. The benefit of this method is that scaled values have simple interpretation i.e., the percentage of observations with that value or below.

$$x' = \frac{n}{N} \times 100$$

– <u>SoftMax Scaling</u>
It is a non-linear normalization method, especially useful for data not distributing evenly around their mean. Resulting values range in [0,1] interval.

–

$$\widehat{x_i} = \frac{1}{1 + e^{-y}}$$

$y = \frac{x_i - \bar{x}}{r\sigma}$ with r parameter. For small value of r, y is an approximately linear function.

There are other methods as well such as: Robust z-Score which are less sensitive to outliers, Power transformation, Tanh transformation etc.
[58][59][60][61]

5.10. Feature Format

Continuous variables could be used as is, as logarithm scale or as categorical. Categorization of continuous variables, provide easier presentation and interpretability. However, it has many drawbacks, such as the loss of information and reduction of statical power to detect feature-outcome relationships. Especially, dichotomization may increase the risk of false positive results. Also, variability may be submitted within each group, and so the variation of outcome between groups

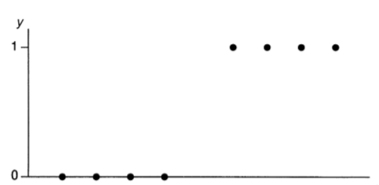
could be concealed. Similarly, non-linear relationships or the presence of confounding could be concealed as well.
The most common practice in clinical research is to dichotomize in median. An issue regarding median dichotomy is that each study uses different data samples, therefore, may use a different median cutoff point. This leads to a decrease in the comparability of their results or even makes a Meta-analysis study difficult. Dichotomization is preferred only for variables with clear boundaries. Creating ordinal variable instead of dichotomous is preferred. Finally, the continuous format is preferred above all if possible to keep.[62] [63]

5.11. Sample Size

Sample size is an important factor to avoid biased coefficients and overfit in case of small sample and many features. Models developed from data sets with too few outcome events relative to the number of candidate predictors are likely to yield biased estimates of regression coefficients.[64]
Despite, sample size must be efficient large, after a certain point, any increment in sample size will not induce any improvement in accuracy of prediction. [65] [66] The Sample sizes needed vary by dataset and method and are associated with dataset–level characteristics. Furthermore, for more features, weaker features, and more complex relationships between the predictors and the response increase expected sample sizes, thus it was preferred for some levels to be merged. [66] [65]
Regarding the models' algorithms, Logistic for example, requires smaller sample size than ML algorithms, while some studies indicate that ML models could even require a sample size of up to 200 EPV (Events per Variable) for some high dimension datasets in order to perform unbiasedly. Less informative features also increase the need in sample size in order to reduce the bias resulting from noise and overfitting. [66]

5.11.1. Sparse Data

Certain data patterns result in MLE (Max. Likelihood Estimates) to be infinite or not existing and so does the coefficient. MLE of the effect is the log odds ratio for logistic regression.
This occurs due to observing only successes or only failures over certain ranges of predictor values. It results to 'perfect discrimination' and happens for both quantitative and categorical predictors. 'Perfect discrimination' happens when it is



possible to predict the sample outcomes perfectly by knowing the predictor values (except possibly at boundary points between the two regions).
It must be noted that chi-squared approximation depends both on n and on the number of cells. It tends to improve as the average number of observations per cell increases.[31]

- <u>Quantitative Predictors</u>

The ML estimate is infinite when the x predictor values for which y = 0 are completely below or completely above those values of x for which y = 1. For example, for x = 10, 20, 30, 40 ($x \leq 40$) y = 0, while for x = 60, 70, 80, 90 ($x \geq 60$) results to y = 1. This 'perfect' fit has $\widehat{\pi} = 0$ for $x \leq 40$ and $\widehat{\pi} = 1$ $for\ x \geq 60$.) The likelihood function keeps increasing as $\hat{\beta}$ increases towards ∞.
Most software fails to recognize when $\hat{\beta} = \infty$, and also report huge standard errors.

- <u>Categorical Predictors</u>

Let's assume a contingency table between a predictor X and the dependent variable Y. When one of the cells counts is 0, that estimate is plus or minus infinity. Sparse data occur when the table has many cells (i.e., categories or variables), most cell counts are small, and many may equal 0. These types of contingency tables are also said to be 'sparse'. A cell with 0 count is said to be 'empty'. Although, in this cell's sample probability is almost always positive. Such an empty cell is called a 'sampling zero'. For each sampling zero cell the ML estimator and thus coefficient is infinite. This case can also cause bias in estimation of odds ratios.
In the example, cases 1 and 3 are sparse, as there are zero counts of success. [31]

		Response (Y)		YZ Marginal	
Center (Z)	Treatment (X)	Success	Failure	Success	Failure
1	Active drug	0	5	0	14
	Placebo	0	9		
2	Active drug	1	12	1	22
	Placebo	0	10		
3	Active drug	0	7	0	12
	Placebo	0	5		
4	Active drug	6	3	8	9
	Placebo	2	6		
5	Active drug	5	9	7	21
	Placebo	2	12		
XY	Active drug	12	36		
Marginal	Placebo	4	42		

5.11.1.1. Merge Levels & EPV

- <u>Merge Levels</u>

One solution could be to add a very small constant (e.g., 10^{-8}) in the model.
For possibly influential observations, to be deleted or merged to another category's cell. More specifically, to fit the model by excluding part of the data containing empty cells (e.g., exclude one category), or by combining that part with other parts of the data (e.g., merge one category into another), or by using fewer predictors. [31][17] By merging into other relative levels or creating general levels such as 'other', 'Unspecified' etc. it is achieved not to drop other important features of corresponding rows [17]. The decision of how many data points should be included

per cell or why a level should be considered as candidate for merging may be considered as follows.

- EPV (Events Per Variable)

For a classifier to be considered well performed, it must be accurate on training data on which it was developed, but also on unseen test data. This assumes that training phase must be implemented as good as possible with sufficient trainset sample sizes but also sample sizes per variable. [67]
Small sample sizes per variable on which the model is trained will lead to overfit, or more generally, great number of coefficients towards small number of events which induce overfit. [32]
The most common method regarding logistic regression is EPV (Events per Variable). EPV equals the number of events of the smallest outcome group, divided by the number of degrees of freedom of a model. [32][68][67][64]

$$\mathrm{EPV} = \frac{\text{Number of Events in Minority Class}}{\text{Degrees of Freedom}}$$

This indicates that the greater the number of events, the greater the df (thus the variables) can be, as the EPV can be kept to an efficient prespecified number. The 'Rule of Thumb' for regression is a count of data points per variable with $EPV \geq 10$. [32] However, studies indicate that $EPV \geq 20$ is more efficient and does not induce differences in prediction performance when alternative methods are used (eg., for evaluation of logistic performance). [67][32][64]. Finally, higher EPV value is needed when low-prevalence predictors are present in a model to eliminate bias in regression coefficients.[64] Especially, ML algorithms require greater sample sizes as mentioned above. Finally, datasets' unique characteristics alter the EPV needs as well. [64]
The drawback of this method, is that EPV should be calculated before any feature selection methods (e.g., lasso, ridge etc.) and is also a naive approach, despite the simplicity provided. [32]

5.11.2. Class Imbalance

Class imbalance means that one of classes (two if binary outcome) is much greater in size than the other. This issue is common in real-world data that need classification (e.g., spam classification of emails, where 'not spam' are obviously more than spam emails). Most of the times interest lies on the minority class, rather than majority, thus deleting such low sample categories is not a correct approach. Most ML algorithms assume balanced data classes, so the boundaries are easily biased towards the majority class, and so models' performance is decreased.
Two categories of solution are implemented: data-level and algorithm-level. Data-level solution includes resampling, which could be under-sampling or over-sampling. However, even those arise issues regarding models' performance.

5.11.2.1. Resampling

Resampling belongs to Data-level approach for class imbalance correction. The advantage of resampling methods is that the datasets need pre-processing (e.g., resampling) only once, and that I ready to be used in whichever classifier.
Due to this, resampling methods are commonly applied on training sets, with various alternatives. It is notable, that resampling should not be applied on test sets, where distribution must be representative of real-world population's distribution. The below are two data-level approaches for resampling.

5.11.2.1.1. Under-Sampling

Under-sampling is the method where a random sample from majority class is selected to be kept (or else a random sample to be dropped). This means that incidences of data are deleted from majority class. Due to that training cost is also reduced.
Under sampling drawback is that useful features/information may be erased or minimized. Another issue is that variance of classifier is increased, as sample is reduced.
Under-Sampling is optimal for low complex data and does not induce great problems regarding overgeneralization. Another advantage lies in that it does not add new data and so the risk of creating false decision boundaries is limited.
There are two types of under-sampling, Controlled under-sampling and Cleaning under-sampling.

- **Controlled under-sampling**

Controlled under-sampling allows a user-specified number of majority class to be kept. Some of the methods used are:

– <u>RUS (Random under-sampling)</u>

It is the simplest form of under-sampling. It includes randomly selecting a sample from the majority class, with or without replacement. It is the simplest method of under-sampling.
Its drawback is the information loss occurring, which is related with the level of imbalance. The greater the imbalance, the more severe the information loss.

– <u>Near-Miss</u>

Near-Miss-1 : Selects from the majority-class sample with the smallest mean distance from the minority class to the nearest N sample.
Near-Miss-2 : Selects from the majority-class sample with the smallest average distance from the minority class to the furthest N sample.
Near-Miss-3 : It is a two-stage algorithm that combines Near-Miss-1 and Near-Miss-2 and it's more robust to noise.

- **Cleaning under-sampling**

This technique does not allow for a user-determined sample size. Therefore, the resulting ratio of minority to majority class sample sizes can be different. Some of the methods used are:

- <u>TL (Tomek Links):</u>

It detects TLs in the majority class and removes them. Let d(.) be the distance between the two samples. If $x\ and\ y$ belong to different classes and there is no sample z that satisfies the conditions $d(x, y) < d(x, z)\ or\ d(x, y) < d(y, z)$, then the x, y pair is a TL. Values classified as TLs are considered noise or borderline samples.

- <u>Cluster C (Cluster Centroids):</u>

Uses k-means clustering to divide the majority class into clusters and balances the dataset using the cluster centroids.

- <u>CNN (Condensed Nearest Neighbour):</u>

It iteratively determines the sample that should be removed by using 1-nearest neighbour rule. Its purpose is to create the smallest subset of the majority class. It's drawback is the noise sensitivity. Other extensions of CNN are : ENN (Edited Nearest Neighbours), RENN (Repeated Edited Nearest Neighbours), ALL KNN , OSS (One-Sided Selection), NCR (Neighbourhood Cleaning Rule) etc

- <u>IHT:</u>

Uses a classifier algorithm to remove samples with low predictive probabilities.

5.11.2.1.2. Over-Sampling

Oversampling adds instance copies of minority classes instances to the minority class or by generating synthetic data to the minority class.
Most common problem of oversampling is the overgeneralization, which is when minority class domain is introduced into the majority-class domain, especially in complex datasets. For example, minority and majority class overlap in some areas, and oversampling interpolates samples from minority class even near boundaries, and thus a minority point may mistakenly fall into majority area. This makes training process for minority class more difficult instead of the opposite.
Moreover, oversampling complex data may result in the creation of unnecessary minority class samples (i.e., noisy samples). Similarly, complex data aggravate the problem. Two above problems, cause the training process to be more complicated, rather than simpler, with unclear boundaries and thus, unstable classifier.
Because of this, oversampling is often combined with some 'filter' methods or even in combination with under-sampling by removing detrimental samples from data.
The oversampling methods that generate new random sample without considering the majority class, induces more problems for highly complex datasets. For this, other methods like Borderline SMOTE are used.
Despite, data complexity major issue, oversample with data filtering methods is considered a better choice for more complex data is some cases, while under-sampling may be better in other. Some of the methods used are:

- <u>ROS (Random Oversampling)</u>

It randomly samples currently available instances to minority class with replacements. The advantage is that the majority class does not overlap another class during training. However, repeated sampling can induce overfitting.

– SMOTE (Synthetic Minority Over-Sampling Technique)

Most used method, which generates synthetic data to minority class.
First, it randomly selects an instance from minority class. After, the closest neighbours of this point of the same class are selected. Then a random data point from these neighbours is selected. Finally, interpolation is performed between the two data points to obtain a new minority class instance, with characteristics somewhere between the two data points.

– BSMOTE (Borderline SMOTE)

It determines the best candidates for over-sampling in the entire dataset. It most uses the points that are close to the decision boundary, as it assumes that those far from the boundary may contribute little to classification success. SVM SMOTE is a subtype of Borderline SMOTE, which uses SVM algorithms to detect the decision boundaries. KM-SMOTE employs k-means clustering before applying sampling. It groups samples together and creates new samples based on the clustering results.

– Filtering SMOTE Methods

SMOTE-TL and SMOTE-ENN are the most typical filtering methods. They help in solving the drawback of simple SMOTE of generating noisy samples. Each method adds TLs or ENNs after applying SMOTE to obtain a cleaner space.

– ADASYN (Adaptive Synthetic Sampling)

It automatically determines the sample to be over-sampled for the minority class by considering the dataset distribution. A sample that needs to be over-sampled is determined based on the learning difficulty. The learning difficulty can be quantified through the ratio of values belonging to the majority class to those belonging to the minority class, for the k nearest neighbours. [69]

5.12. Regularization

Regularization is when the coefficient estimates are constrained or else shrink towards zero, usually by adding some information to the process.
Its purpose is to minimize the overfitting of a model by reducing the effect of noise. This way, models' bias may increase slightly but the variance will be reduced. [70][52]

A case where regularization helps is when the number of predictors is large compared to the sample size. Another one is when covariates are correlated with each other. Overfitting and collinearity yield very unstable maximum likelihood estimates (MLEs) with some of the being even infinite. [70]

5.12.1. Ridge

Ridge regression has effects on coefficient estimators by shrinking them towards zero, but never equal to zero, which can be a problem for high dimensional data.

$\lambda \sum_{j=1}^{p} \beta_j^2$: Shrinkage Penalty.[45]

$\frac{||\widehat{\beta_\lambda^R}||_2}{||\widehat{\beta}||_2}$: Fraction of Ridge estimators towards initial ones.
$||\beta||_2 = \sqrt{\sum_{j=1}^{p} \beta_j^2}$ or else $||\beta||^2 = \sum_{j=1}^{p} \beta_j^2$: $l_2 norm$. [45][71]
$p(X_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j X_{ij}\right)}{1+\exp\left(\sum_{j=1}^{p} \beta_j X_{ij}\right)}$: Probability $Y_i = 1$, given the value $X_i = (X_i, ., X_n)$:

Log likelihood : $l(\beta) = \sum_i [Y_i \log p\,(X_i) + (1 - Y_i) \log\{1 - p(X_i)\}]$

$$l^\lambda(\beta) = l(\beta) - \lambda \sum \beta_j^2$$

λ : Tunning parameter. Controls the size of shrinkage
- $\lambda = 0$: No coefficients shrink
- $\lambda = \infty$: Coefficients will approach zero

As λ increases, shrinkage increases, $l_2 norm$ and $\widehat{\beta_\lambda^R}$ decrease. Its value is usually selected through cross validation
The estimate of Ridge coefficients $\hat{\beta}^\lambda$ is expected to be on average closer to the real value of β than the unrestricted MLE, i.e. $l^\lambda(\beta) < l(\beta)$ for regression.[70]
$X_j \widehat{\beta_{j,\lambda}^R}$: depends on λ value and scaling of predictors e.g., a greater scale of predictors would estimate much different on the outcome as the coefficients are also shrunk. [45]
Ridge regression shrinks correlated predictors' coefficients towards each other, allowing them to borrow strength from each other. In the extreme case of k identical predictors, they all get identical coefficients with 1/k[th] the size that any single one would get if fit alone. It is ideal if there are many predictors, and all have non-zero coefficients.[72]

5.12.2. Lasso

Lasso regularization works like Ridge, but here coefficients can get exact zero. Another difference of lasso is that $||\beta_j|_1 = l_1 norm$ is used. [45]

$||\beta_j||_1 = \sum |\beta_j|$: $l_1 norm$ [73][71]
$\lambda \sum |\beta_j|$: Shrinkage Penalty

$$l^{\lambda}(\beta) = l(\beta) - \lambda \sum |\beta_j| \quad [74]$$

It is useful for high dimensional data where variable selection is needed and also it results in and easier interpretated model. [45]
In contrast to Ridge, Lasso is indifferent to very correlated predictors and as it will pick one and ignore the rest. In the extreme case of k identical predictors however it fails. [72]

5.12.3. Elastic Net

Elastic Net combined both Ridge and Lasso. It is useful in high dimensional data and when correlated predictors occur.

$$l^{\lambda}(\beta) = l(\beta) - \lambda \, P_{\alpha}(\beta)$$

$$P_{\alpha}(\beta) = (1-\alpha)\,|\beta|_1 + \alpha\,|\beta|_2^2 = \sum_{j=1}^{p} \left[(1-\alpha)\,|\beta_j| + \alpha\,\beta_j^2\right], \qquad \alpha \in [0,1],\ \lambda > 0.$$

$$\alpha = \frac{\lambda 2}{\lambda 1 + \lambda 2} \quad , \quad 1-\alpha = \frac{\lambda 1}{\lambda 1 + \lambda 2}$$

$\lambda\, P_{\alpha}(\beta)$: Shrinkage Penalty

- $\alpha = 0$: Ridge regularization
- $\alpha = 1$: Lasso regularization
- $0 < \alpha < 1$: In between Ridge and lasso regularization [75][71]

Or else in $glmnet$ format:

$$P_{\alpha}(\beta) = (1-\alpha)|\beta|_{l_2}^2 + \alpha\,|\beta|_{l_1} = \sum_{j=1}^{p} \left[(1-\alpha)\,\beta_j^2 + \alpha\,|\beta_j|\right]$$

$$\alpha = \frac{\lambda_1}{\lambda_1 + \lambda_2} \quad , \quad 1-\alpha = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

[76][73]

5.13. Machine Learning (ML) for Healthcare

Machine Learning (ML) refers to technics, of data-driven learning from experience, without being explicitly programmed. ML is one of the most used topics of AI. They are handling unstructured (text, image etc.), complex and bigger scale data quite fast.
Due to this, they are arising in every field, one of which is healthcare that could benefit much from ML in unique challenges. ML may complement and improve diagnostic and decision-making processes of healthcare professionals in patient

care. By analysing data in real-time, recognizing patterns and thus identify diseases etc. It could contribute to providing more efficient personalized treatments, based on patient's medical history, lifestyle, diseases, genetic data, family history etc, as large volume or unstructured text or image (e.g., x-rays) data per patient can be handled with ML.
Another benefit is in reaching conclusions faster and consequently, in sooner therapeutic intervention, reduction in adverse effects etc. Similarly, disease recognitions can be achieved faster before a condition reaches greater severity level. This is crucial in case of severe conditions where early detection is crucial, especially for neoplasms that early detection is hard with conventional means. The ability of image evaluation through ML contributes further to better prognosis of neoplasm. Additionally, it leads to deeper evaluation of therapy characteristics in rarer patient groups (e.g., dosage, duration of therapy in children).
It can also be utilized is in organizational management. Specifically, in various types of data extraction from documents and Electronic Health Records (EHR) much faster than manually. This lead to identifying gaps in health care and hidden risks. The benefit of handling faster various formats of data (unstructured, images etc.) establishes ML algorithms useful for organizational purposes. Healthcare and industry could use it for automation in clinical documentation, records administration etc.
Population Health management is another field ML finds application. Health Institutions already use ML algorithms to predict pandemic outbreaks, through web or social media real-time information. This is called 'Crowdsourcing' where large volumes of real-time social-media, webpage data etc, are analysed to predict a disease outbreak etc. 'Crowdsourcing' partially solves the problem of costly and time-consuming updating and maintaining health data but still needs optimization. They also contribute to optimizing the expensive and demanding process of clinical trials designs, by computing best samples sizes, or even by being trained to recognize candidate participants through 'Crowdsourcing', detect design flaws and other criteria.
ML except supplementing healthcare decisions through analysis of complex data, also reduced human error, especially for automated processes where most errors occur.
It is notable that EHR increasing use, which include various formats of data (text, image, disease-coding etc.) has helped the integration of ML algorithms in healthcare and via versa. ML assists in reducing data errors occurring in EHR , while EHR formats of data are more efficiently handled from ML.
However, for ML algorithms to produce efficiently results, healthcare providers and researchers must be careful and precisely when they input data records. 'Garbage-in/Garbage-out' concept denotes the importance of high-quality input data or else, incomplete and erroneous values could wrongly train an algorithm. "Quality data" for ML training must be accurate, precise, complete, and generalizable (represent real-world population). Complete data is another issue arising in real-world data (e.g., due to visits missed etc.), which induces challenges for ML models too. Input

data format is a parameter for choosing the appropriate model to implement (e.g., Deep Networks are standard for image data classification).
[77][52][13]

Supervised ML
Machine learning can be categorized into two categories: supervised learning and unsupervised. For the purposes of this study only Supervised Classification algorithms were applied. Supervised algorithms are given labelled data with known outcome, which are then categorised into training and testing data.
Classification models are defined from these training data. These models can then be used to perform classification on other unlabelled data. In short, models learn the patterns from training data and apply them to test data. [13]

5.13.1. ML vs Classic Models

A huge difference between ML and classic statistical models lies in their parametric or non-parametric form. In parametric models, parameters are fixed in number and model's function has a known form-shape. In contrast non-parametric models adapt freely to data as they do not make such assumptions about the function. The most common assumptions regard distribution and shape (i.e., linear function and normal distribution of data accordingly). Common parametric models for classification are logistic regression and Naïve Bayes, while non-parametric are KNN, SVC, Trees, Random Forests etc.
ML is already starting to replace classic methods in healthcare.
Some of the common classification models' assumptions that ML algorithms overcome are:
Logistic regression may be limited due to not handling large number of features, especially if high correlated. Also, because logistic assumes uniform (constant coefficients) and linear relationship between features and outcome.
Naive Bayes uses a probabilistic approach based on Bayes theorem and thus, Bayesian logic. It also assumes that the features are independent of each other.[52]

5.14. Binomial Distribution

Distribution results from n number of Bernoulli trials which could have two possible outcomes referred as 'success' and 'failure', with 'success' usually labelled as the preferred outcome.
π: Success probability for a given trial,
Y: The number of successes out of the n trials. So, Y has the binomial distribution with index n, parameter π.
n trials are assumed to be:
- Identical: same probability of success for all
- Independent: the outcome of a trial is independent from other trials.

The binomial distribution for n trials with parameter π has mean and standard deviation:

$$E(Y) = \mu = n\pi, \qquad \sigma = \sqrt{n\pi(1-\pi)}$$

The sampling distribution of the sample proportion p has mean and standard error:

$$E(p) = \pi, \quad , \sigma(p) = \sqrt{\frac{\pi(1-\pi)}{n}}$$

The probability of outcome y for Y:

$$P(y) = \frac{n!}{y!\,(n-y)!}\,\pi^{y}(1-\pi)^{n-y}, \qquad y = 0,1,2,\ldots,n$$

[31]

5.14.1. Significance test for Binomial proportion

The z statistic divides the difference between the sample proportion p and the null hypothesis value π_0 by the null standard error of p:
For null hypothesis $H_0\colon \pi = \pi_0$:

$$z = \frac{p - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

[31]

5.14.2. Confidence Intervals for a Binomial Proportion

A significance test indicates whether a particular value for a parameter is plausible. Confidence interval to determine the range of plausible values.
With a 100(1 − α)% confidence interval for π, one can be 95% confident that the population proportion includes those values for the characteristic studied:

$$p \pm z_{\alpha/2} \cdot SE, \quad \text{with } SE = \sqrt{\frac{p(1-p)}{n}}$$

SE : estimated standard error of p.
$z_{\alpha/2}$: standard normal percentile having right-tail probability equal to $\alpha/2$.
For 95% confidence, $\alpha = 0.05$, $z_{\alpha/2} = z_{0.025} = 1.96$.

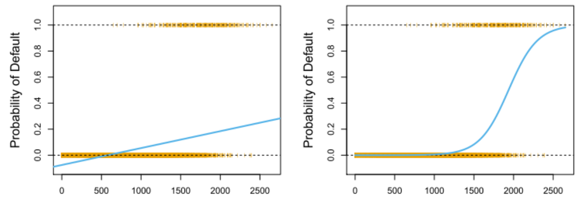
Formula needs large n to work well and is also poor for p near 0 or 1. [31]

5.14.3. Justification for Classification Models

A Linear form of the relationship of X with a binary outcome y to be interpreted: [45]
$p(X)$: Probability of 'success'.

$$p(X) = \Pr(Y = 1 \mid X)$$

Outcome Y is encoded as 0,1 for convenience.
This relation cannot be explained through linear regression, because negative probabilities or greater than 1 occur, as seen below.



[45]

5.15. Models

5.15.1. Logistic Regression

5.15.1.1. Maximum Likelihood Method

Maximum Likelihood is the method used to fit the logistic regression model. It searches for $\widehat{\beta_0}$, $\widehat{\beta_1}$ coefficients that maximize the likelihood function:

$$\mathrm{l}(\beta_0, \beta_1) = \prod_{i:y_i=1} p(x_i) \prod_{i':y_{i'}=0} \left(1 - p(x_{i'})\right)$$

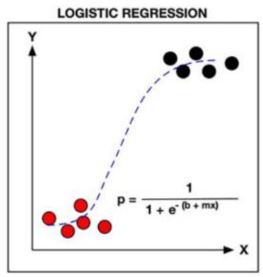
From the equation it can be concluded that maximizing likelihood means finding those coefficients, for which each data point is assigned with a predicted probability $\hat{p}(x_i)$ that gives as close as possible the actual observed value for each data point i.e., assign value close to 0 for Y = 0 and close to 1 for Y = 1, when predicted $\widehat{\beta_0}$, $\widehat{\beta_1}$ replaces coefficients in the above equation.[45]

5.15.1.2. Logistic Fit

It is a classification method that uses a logistic function for predicting probabilities to classify a binary dependent variable. The function yields a value of 0 or 1 which represents the negative (0) and the positive (1) outcome, by calculating an odds ratio probability based on the relationships between the independent variables (features) and the dependent variables (target).[52]
Logistic regression uses this transformation to interpret the binary outcome:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

From the equation is notable that p(X) cannot take values equal or less than zero and equal or greater than 1, for whichever X used. [45]



[52]

5.15.1.3. Odds & Log Odds

By transforming above equation, we receive the Odds quantity: $\frac{p(X)}{1-p(X)}$, which takes positive values [0, +∞). A unit increase in X multiplies odds by: e^{β_1}

- **Odds**

$$\frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X}$$

[45]

- **Logarithm of Odds (Log Odds/Logit)**

$$logit[p(X)] = \log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X$$

[31]

Log Odds is linear to X. Thus, a unit increase in X changes the log odds by β1.
It must be noted clearly that relation between p(X) and X is not straight line, as is in simple linear regression. (i.e., it is not $p(X) = \beta_0 + \beta_1 X$). This is understandable when plot X ~ p(X), that logistic gives an S shaped curve. Thus, a unit change in X does not correspond to change in p(X) by β1, but rather changes the log Odds by β1, as mentioned. The rate of change of p(X) per unit of X depends on the value of X.[45]

- **Odds Ratio**

The odds multiply by e^{β} for every 1-unit increase in x. Thus, the odds at level x + 1 equal the odds at x multiplied by e^{β}. Thus, for 1 unit increase of x: [31]

$$Odds(X) = \frac{p(X)}{1-p(X)} = e^{\beta_0+\beta_1 X} = e^{\beta_0}e^{\beta_1 X}$$

$$Odds(X+1) = \frac{p(X+1)}{1-p(X+1)} = e^{\beta_0+\beta_1 (X+1)} = e^{\beta_0}e^{\beta_1 (X+1)}$$

$$Odds\ Ratio = \frac{Odds(X+1)}{Odds(X)}$$

5.15.1.4. Interpreting logistic Output

	Log Likelihood				−97.2263	
		Standard	Likelihood Ratio		Wald	
Parameter	Estimate	Error	95% Conf. Limits		Chi-Sq	Pr > ChiSq
Intercept	−12.3508	2.6287	−17.8097	−7.4573	22.07	<.0001
width	0.4972	0.1017	0.3084	0.7090	23.89	<.0001

- **Interpreting Probability**

For example, $x = 33.5$ a value of feature X, results in probability of success of 0.987 or 98.7% as seen below.

$$p(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}} = \frac{e^{-12.351+0.497\cdot 33.5}}{1+e^{-12.351+0.497\cdot 33.5}} = 0.987$$

[31]

- **Interpreting Log Odds**

Log Odds increases/decreases by β for every 1-unit change in X.

$$logit[p(X)] = \log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X = -12.35 + 0.497x$$

β_1 = 0.497 change in log-odds for a 1-unit increase in X.

β : Determines the rate of increase/decrease of change in Log Odds with the corresponding change of X.

- $\beta > 0$: Increase of log odds. Ascending S curve.
- $\beta < 0$: Decrease of log odds. Descending S curve.
- $\beta = 0$: Y is then independent of X. Horizontal straight line. Then, $e^{\beta} = 1$.

$\beta\pi[1-\pi]$: The curve's slope is the actual rate of change. The slope approaches 0 as the probability approaches 1.0 or 0.

$X = -\alpha/\beta = EL_5 0$: Median Effective Level, where $p(X) = 0.50$ (outcome has a 50% chance). $p(X) = 0.50$ has slope approaches zero (straight line). Thus, $\beta(0.50)(0.50) = 0.25\beta$ slope etc.[31]

- **Interpreting Odds Ratio**

$$Odds\ Ratio = \frac{\mathrm{odds}(x+1)}{\mathrm{odds}(x)} = \frac{e^{\beta_0}\, e^{\beta_1(x+1)}}{e^{\beta_0}\, e^{\beta_1 x}} = e^{\beta_1} = e^{0.497} = 1.64$$

For each a unit increase in x there is (1+0.64) = 64% increase in Odds. [31]

5.15.1.5. Test Statistic & CL

- <u>Wald test statistic</u>

Wald test statistic (z test statistic), following approximately standard normal distribution, measures the number of standard errors that the sample proportion falls from the null hypothesized proportion.

$$z = \frac{\hat{\beta}}{SE}$$

$z^2 = \left(\frac{\hat{\beta}}{SE}\right)^2$ follows chi-squared distribution with df = 1.

The likelihood ratio test is more reliable. It compares the maximum L_0 of log-likelihood when β = 0 to the maximum L_1 of log-likelihood for β≠0. It follows chi-squared distribution with df=1.[31]

$$-2(L_0 - L_1)$$

- <u>Confidence Intervals</u>

Wald confidence interval for the parameter β coefficient and e^{β}(confidence intervals for odds) accordingly:

$$\hat{\beta} \pm z_{\alpha/2}(SE), \quad e^{\hat{\beta} \pm z_{\alpha/2}(SE)}$$

[31]

5.15.1.6. Benefits & Drawbacks

Logistic regression is easy to interpret, and a well-studied model.
However, it has limitations regarding the sensitivity to noise and high dimensional data, especially if correlated. Also, as a parametric model it includes many assumptions, such as uniform (constant coefficients and linear relationship between features and outcome.[52]

5.15.2. Generative Models

Generative models use another method to fit the classification, based on Bayes' theorem/Bayes' classifier if K classes assumed:

$$p_{k(x)} = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

The idea is that it categorizes each observation x to the class where $p_{k(x)}$ is largest.
This method is useful for cases where:

- There is a serious separation between classes, so logistic coefficients become very unstable
- X follows an approximate normal distribution in each class with small sample size.

- π_k : Prior probability that an observation x belongs to k[th] class.

- $f_k(X) = \Pr(X \mid Y = k)$: Density function of X for an observation x which belongs to k[th] class (i.e., likelihood — how likely it is to observe $x$given class k). Density function is large if probability of X ≈ x (i.e., belong to k[th] class) and small otherwise.
- $p_{k(x)} = Pr(Y = k \mid X = x)$: Posterior probability of an observation X = x (i.e., belongs to the k[th] class), given the predictor value for this observation.

The idea is that instead of computing $p_{k(x)}$, and since we do not know the real values of π_k and $f_k(X)$, we can just plugin estimations of π_k and $f_k(X)$ to the equation above. Specifically:

- π_k : Calculated by simply taking the fraction of k[th] class from a training set from a random sample of population.
- $f_k(X)$: it is more difficult to estimate, and each of the LDA and QDA models make corresponding assumption. [45]

5.15.2.1. Linear Discriminant Analysis - LDA

LDA except all the above, is also a dimensionality reduction method. Its main goal is to convert high-dimensional data into a lower-dimensional space while optimizing the distinction between different classes. This property makes LDA especially fitting for classification tasks, as well as for extracting features and visualizing multifaceted, multi-class data.[13]

- **Unique predictor, p = 1**

<u>Assumptions</u>

- $f_k(X)$ = follows normal gaussian distribution
- $\sigma_1^2 = \cdots = \sigma_K^2 = \sigma^2$ variance is equal among all classes.
- $\widehat{\mu_k} = \frac{1}{n_k}\sum_{i:y_i=k} x_i$ = is unknown and simply assumed as the average of the training sample of k[th] class.
- $\widehat{\sigma^2} = \frac{1}{n-K}\sum_{k=1}^{K}\sum_{i:y_i=k}(x_i - \widehat{\mu_k})^2$ = is unknown and simply assumed as a weighted average of the sample variances for each of the K classes.
- $\pi_k = n_k/n$ = for no other information, is simply assumed as the fraction of the number of observations in the k[th] category to the entire sample size.

$$Bayes': p_{k(x)} = \Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right)$$

By replacing $\pi_k, f_k(X)$:

$$p_k(x) = \frac{\pi_k \exp\left(-\frac{(x-\mu_k)^2}{2\sigma^2}\right)}{\sum_{l=1}^{K} \pi_l \exp\left(-\frac{(x-\mu_l)^2}{2\sigma^2}\right)}$$

Or else the log, named Discriminant function which is linear to x :

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Which replacing for mean, variance assumptions :

$$\widehat{\delta_k}(x) = x \cdot \frac{\widehat{\mu_k}}{\widehat{\sigma^2}} - \frac{\widehat{\mu_k^2}}{2\widehat{\sigma^2}} + \log(\widehat{\pi_k})$$

$\delta_k(x)$ is linear to x (linear discriminant analysis) and decision depends only on x. Decision boundaries are 'lines' themselves as well. In case $\pi 1 = \pi 2$ i.e., equal prior probability to belong to each class, then classifies observation to class for which : $2x(\mu_1 - \mu_2) > \frac{\mu_2}{1-\frac{\mu_2}{2}}$

- **Generalize for Predictors, p > 1**

<u>Assumptions</u>

- $X = (X_1, X_2, \dots, X_p)$: Predictors, follow multivariate normal distribution, $X \sim N(\mu, \Sigma)$, with some correlation between them.
- $E(X) = \mu$: Mean of X
- $Cov(X) = \Sigma_1 = \Sigma_2 =, .., = \Sigma$: Unknown p × p covariance matrix of X, common to all classes. Common covariance matrix makes LDA model linear to x.
- $X_k \sim N(\mu_k, \Sigma)$
- $\mu_1, \dots, \mu_k$: Unknown means for each class.
- $\pi_1, \dots, \pi_k$: Unknown.

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k \ (4.5)$$

Decision boundaries number is based on pair of predictors (e.g., 3 predictors mean 3 boundaries or 3 separate spaces for each class k).
Error rates are a function of posterior probability threshold, where changes can differ sensitivity for specificity and via versa. [45]

5.15.2.2. Benefits & Drawbacks

LDA in general is less flexible with its linear boundaries, with less variance. Moreover, assumption for common covariance is not actually happening to data, so LDA will perform high bias. The fact that LDA depends on all data points, considering within class means and covariance matrices, makes it very vulnerable to data changes and thus less robust to outliers or noise. However, it is recommended when there is small training sample, as variance is reduced. [45]

5.15.2.3. QDA - Quadratic discriminant analysis

Generally, the concept of QDA is like LDA, using the Bayes' theorem with plugged in values of Σ_k, μ_k, π_k, while differences lie to covariances and boundaries shape.
<u>Assumptions</u>

- $X = (X_1, X_2, \ldots, X_p)$: Predictors, follow multivariate normal distribution, $X \sim N(\mu, \Sigma)$, with some correlation between them.
- $E(X) = \mu$: Mean of X
- $Cov(X_k) = \Sigma_1 \neq \Sigma_2, \ldots, \neq \Sigma_k$: Unknown covariance matrices, different for each class k. Unequal covariance matrix makes QDA model quadratic to x.
- $X_k \sim N(\mu_k, \Sigma_k)$
- $\mu_1, \ldots, \mu_k$: Unknown means for each class.
- $\pi_1, \ldots, \pi_k$: Unknown.

$$\delta_k(x) = -\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) - \frac{1}{2}\log|\Sigma_k| + \log \pi_k \quad (5.1)$$

$$\delta_k(x) = -\frac{1}{2}x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2}\mu_k^T \Sigma_k^{-1} \mu_k - \frac{1}{2}\log|\Sigma_k| + \log \pi_k \quad (5.2)$$

[45]

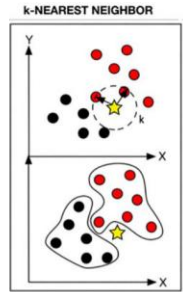
5.15.2.4. Benefits & Drawbacks

QDA model has the drawback that many covariance matrices need to be computed. However, it is more flexible and better for great sample sizes, especially when variance of classifier is not of big concern. [45]

5.15.3. k - Nearest Neighbors - KNN

KNN is a non-parametric supervised clustering ML method which was developed by Evelyn Fix and Joseph Hodges in 1951 and later optimized by Thomas Cover. KNN has various applications, especially in registering closest points, recognizing patterns, fraud detection etc.[78] It is used in data mining classification and disease detection. [13]
This model is instance based (i.e., it does not use a function defined from training set, but it rather makes predictions by analysing the data in real-time when new data points appear, without necessarily having to go through a training phase).
It works based on number of k neighbour data points, where k is equal to the

square root of the number of points and its distance (e.g., Euclidean) from a predefined point. In short, similar data points usually cluster nearby at space. A prediction for a query data point is based on the distance and thus similarity to the exciting neighbour data points. [78] [52]



Parameters

- <u>k</u>

Number of neighbour data points. Crucial tuning parameter. For large k, computational demands increase. Choice is associated with dataset's characteristics and impacts the accuracy.

- Small k values (few neighbors) increase model flexibility and noise susceptibility (i.e., overfitting).
- Large k values (lots of neighbors) can underfit, while also being computationally more demanding. Boundaries are also smoother.

- <u>Distance Metrics</u>
- Euclidean : Most used.
- Manhattan : Better for high dimension data, for different scaling or sensitivity to specific predictors.
- Minkowski : Similar uses as Manhattan. [78]

It is notable that KNN is distance-based model, so scaling of data is a very important pre-processing step, as results can vary depending on the range of features values, especially, in real world datasets, where different units are often used. [58][17]

- <u>Neighbors</u>

All data points are ordered based on their distance and the k number of closest data points of neighbors are becoming the neighbors.

- <u>Response</u>

For classification response is simply the most common occurring class of neighborhood. For regression could be similarly the mean, median etc. Note that test points will always take the same value as response, based e.g., on mean of neighborhood. [78]

5.15.3.1. Exact KNN

Exact KNN methods tend to be more accurate in finding the actual neighbor data points (i.e., actual close ones) and perform better where precision is important in the outcome (e.g., health data). Exact KNN has two main types of methos : KNN search and KNN join for selecting the k nearest neighbor's data points. [78]

- **KNN Search (Brute Force - BF or Exhaustive Search)**

Finds K nearest points of one point from the rest points. The process is repeated for all data points. However, due to calculating all distances for each data point, it is extremely computationally expensive for big datasets.
This method is better for problems which need speed and accuracy. More detailed:
D: dataset
n: data points
d: number of dimensions,
q: a query (test) point,
k: number of nearest neighbors
$dist(p, q)$: a function that calculates the distance between point p and query point q
R: a set of k neighbor points.
kNN search finds a set R of k neighbor points such that for each point p ∈ R , there is no other point $p' \in D$ where $dist(p, q) > dist(p', q)$.

- **KNN Join**

Finds K nearest points for each data point, but the process does not repeat as above per data point but happens for all data points the same time. Better when uncovering hidden relationships or patterns. It is computationally more efficient. This method is applied to k-means clustering, Outlier detection, kNN classification, Missing value computation etc. More detailed:
D_1: dataset, D_2: dataset
n: data points in D_1, D_2
d: number of dimensions
k: number of nearest neighbors
$dist(p_1, p_2)$: a function that calculates the distance between point $p_1 \in D_1$ and query point $p_2 \in D_2$
R: a set of k neighbor points.
KNN Join find pairs of points (p_1, p_2) such that for each $p_1 \in D_1$, there are exactly k points $p_2 \in D_2$ with minimal distance. Or else, for each point in D_1 , we find its k closest points in D_2.[78]

Tree Structures of Exact KNN

Tree structures are a partitioning method, with the most common to be $k - d\ tree$. Tree's root is the whole data space D which is the initial cell. Then data split at their median along the direction of a chosen coordinate. The process is repeated until split leaf cells are left with a threshold of n_0 data points. Final tree has a depth of about $log_2(n/n_o)$.
When a query point appears two methods of prediction may be chosen:

- <u>Defeatist search</u>

The query point is assigned down to the appropriate leaf and takes the value of the closest to it leaf point. Time consumption is more efficient at about $O\big(n_o +$

$log(n/n_o))$ or $O(logn)$ for constant n_0. The great negative of this is that the 'nearest point' may be giving very wrong prediction (i.e., may not be actual as similar as needed and another leaf's point may be the actual closest one to the query point). Unfortunately, failure rates may be unacceptable.

- Comprehensive search

Checks which other leaves may need to be taken into consideration for NN doners and returns the true closest neighbor point. Computationally it may take max $O(n)$ time.

A big drawback of tree-KNN is that their performance decreases a lot in high dimensions. For this problem, extra methods for dimension reduction are applied such as PCA. [79]

5.15.3.2. Alternative kNN methods

Other alternative methods help in overcoming the drawbacks of exact kNN.

Models

- Computational efficacy

Alternative KNN models like 'Approximate Nearest Neighbors' (ANN) for more complex data, are faster but with a cost of accurately choosing the neighbors, i.e., neighbors chosen might not be the actual nearest ones.

- Noise, Outlier

Advanced distance metrics, weights (i.e., far neighbor points are more influential than close ones etc.). For example, variations of model K such as Adaptive kNN, Weight adjusted KNN and Fuzzy KNN to improve precision.

- Other

k-distance join, reverse KNN join, dynamic KNN join etc.[78]

Methods

Below methods are useful for computing the actual neighbor points accurately but also as fast as possible. Especially in high dimension B or R-tree indexing methods, perform purely.

- Parallelization

Divides the concurrent process of distance computing into different units. For example, this method is regarding device's CPU way of procedure with $n_jobs$command (e.g., n_jobs = 1 work only on 1 core which is computationally slow, n_jobs = 4 work on 4 cores, n_jobs = -1 work on all available cores).

- Dimensionality Reduction

Simply, dimensionality decreases e.g., through 'principal components analysis' - PCA.

- Partitioning

Data is segmented through tree structures (e.g., R-tree, R*-tree, Ball tree, KD-tree etc.) that perform pruning on part of data.

- Space based
- Data based [78]

5.15.3.3. Benefits and Drawbacks

The highlights are that KNN makes no assumptions as a non-parametric model, and as mentioned before, it analyses data real-time, without necessarily need a training phase as it is distance based. It can also detect non-linear relationships.
Few factors are involved in its tunning, such as selecting k-value, calculating distances etc, so KNN process is easy to understand and implement. Another advantage is that requires less training time than other ML models, however, it is time consuming in classification stage of test data, especially for large datasets.
Regarding the drawbacks, the presence of unrelated features makes process more difficult for KNN.[13] Moreover, it performs better with a smaller number of features. It also requires feature scaling because it is a distance-based model and finally it is sensitive to outliers. The performance can be altered based on the distance metric and as most supervised ML it is prone to overfitting. [52]
Computational requirements: A data point's distance must be calculated from all the other data points. This escalates when it comes to bigger sample datasets [78]. Roughly, the time needed to find nearest neighbours (NN) is about $O(n)$ time, which can be a serious time passing by with large n samples. [79]

- **Noise, Outliers**

This method is based on neighbors, thus it makes it susceptible to noise and outliers, in case a non-significant predictor or an extreme value in considered as neighbor.

- **Accuracy**

By finding the efficient k number and neighbor points precisely. Especially, in high dimensions, where distances lose their meaning, it is more difficult to find the actual near data points.

- **'Curse of Dimensionality'**

Model's performance can decrease in high dimensions because distances become less meaningful.[78]

5.15.4. Support Vector Classifier - SVC

SVC has multidomain applications and is extremely popular to big data mining and pattern recognition.
The classification is based on hyperplane defining. SVC maximizes the minimum distance (margin) from either side of this hyperplane to the nearest point of the sample, as analyzed below. Finally, the largest possible margin hyperplane is chosen for the analysis. [52][13]

5.15.4.1. Hyperplane & Margins

- **Hyperplane**

A Hyperplane is a $p-1$ dimension flat subspace, which separates classes, in a p dimensional space or else a linear decision boundary. For example, for $p = 2$, it is a simple line separating two classes. For more than two classes multiple hyperplanes are used. Generalizing for p the equation of a hyperplane is simply:

$$\beta_0 + \beta_1 X_1 + \cdots \beta_p X_p = 0 \qquad (7.1)$$

Any $\left(X_1, \ldots, X_p\right)^T$ that satisfies this equation, is a point of the hyperplane and defines the hyperplane.
For training set of n training observations $x_1, \ldots, x_n$ and $y_1, \ldots, y_n \in \{-1,1\}$
$\beta_0 + \beta_1 xi_1 + \beta_2 xi_2 + \cdots + \beta pxip = 0$ X lies on the hyperplane and defines it.
$\beta_0 + \beta_1 xi_1 + \beta_2 xi_2 + \cdots + \beta pxip > 0 \quad if \quad yi = 1$ and X lies to one side of hyperplane
$\beta_0 + \beta_1 xi_1 + \beta_2 xi_2 + \cdots + \beta pxip < 0 \quad if \quad yi = -1$ and X lies to the other side of hyperplane. Or more combined, ensuring all points are on the correct side:

$$yi(\beta_0 + \beta_1 xi_1 + \beta_2 xi_2 + \cdots + \beta pxip) > 0$$

Thus, for a test data point, the categorization is achieved based on the sign (positive or negative) the equation is assigned for a x_p^* data point.

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

Thus for:

$$f(x^*) > 0\,, \qquad test\ observation = +1$$
$$f(x^*) > 0\,, \qquad test\ observation = -1$$

The farthest from zero $f(x^*)$ is, the surer for the classification, as it belongs far from the boundaries space.

- **Maximal Margin Classifier – Hyperplane Margins**

If data are perfectly separable, then there may be infinite hyperplanes, slightly rotated, that can separate them too. In order to stick with one hyperplane, the 'maximal margin hyperplane' is used among all the other hyperplane choices. More specific, two margins are defined in both sides of the hyperplane. These margins are defined by calculating the maximum distance M from each side of the hyperplane, from the closest to the hyperplane data points. So, we are looking for coefficients to satisfy the maximized value of M. The distance between margin and hyperplane is constant all along the hyperplane. Those closest to hyperplane data points that define where margins and thus hyperplane are oriented, are practically lying on the margin boundaries and are called 'support vectors'. All in all, maximal margin hyperplane depends only on the close neighbour data points, that can 'touch' the margin boundaries, rather than the whole dataset. Thus, any addition, removal or other change of support vectors or data points capable of being support vectors, will make the hyperplane orientation to change, when in contrast changes in the rest

of the dataset, will not make any difference. This, however, implies overfitting issues may occur.
Finally, another problem arises when it comes to non-perfectly separable datasets, where the violation of margins and/or hyperplane is inevitable from some data points. This problem is solved by 'soft margin classifiers' or else 'support-vector-classifiers'.

$$y_i\left(\beta_0 + \beta_1 x_{i_1} + \beta_2 x_{i_2} + \cdots + \beta_p x_{i_p}\right) \geq M (7.4)$$

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

$$\begin{aligned} f(x^*) > M \ , \quad & assign\ to\ class\ +1 \\ f(x^*) < M \ , \quad & assign\ to\ class\ -1 \end{aligned}$$

- **Support Vector – Soft margin Classifier**

Support Vector classifier solves the above problems, by allowing misclassification and boundary violation to some data points close to margins and hyperplane. Those violators can be in the wrong side not only of margins but also hyperplane itself. This makes the model more robust and overcomes the problem of overfitting.
Similarly, with above M is the margin width and needs to be as large as it can be.
Support vectors are now the data points, lying on margins or being inside them i.e. all data points which violated.
$e_1, \ldots, e_n$ are 'slap' variables, which allow observations to violate boundaries.
For the training set:

$$y_i(\beta_0 + \beta_1 x_{i_1} + \beta_2 x_{i_2} + \cdots + \beta_p x_{i_p}) \geq M(1 - e_i) \quad (7.5)$$

Classification of a test data point:

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \cdots + \beta_p x_p^*$$

$$\begin{aligned} f(x^*) > M(1 - e_i) \ , \quad & assign\ to\ class\ +1 \\ f(x^*) < M(1 - e_i) \ , \quad & assign\ to\ class\ -1 \end{aligned}$$

- $e_i = 0 \ : M(1 - e_i) \ \Rightarrow \ M \ \geq \ 0$, test point on correct side.
- $0 < e_i < 1 \ : M(1 - e_i) \ \Rightarrow \ 1 \ \geq \ M \ \geq \ 0$, test point is inside the margin.
- $e_i > 1 \ : M(1 - e_i) \ \Rightarrow \ M \ < \ 0$, point on the wrong side of hyperplane.

- **Tunning parameter C**

$$\sum_{i=1}^{n} \epsilon_i \leq C, \quad e_i \geq 0 (7.6)$$

C parameter defines the number of violations allowed from the classifier and thus the bias-variance trade-off.

- $C = 0$, each $\epsilon_i = 0$, no violations allowed.
- $C > 0$, maximum number hyperplane violations allowed are equal to C.

The higher the C value, the wider the margins and the more the support vectors.

5.15.4.2. Non-Linear Boundaries SVC

Support vector machines are a support vector classifier, which use kernels to enlarge the feature space, and produce non-linear boundaries.
The mathematical solution to above functions includes the inner products of training observations, e.g., the inner product of two observations x_i, x_i' is
$\langle x_i, x_{i'} \rangle = \sum_{j=1}^{p} x_{ij}\ x_{i'j}$.
For n observations and $\alpha_i, i = 1, \dots, n$ parameters, which are the trained coefficients per training observation and are $\alpha_i \neq 0$ only for supporting vector data points, the above function is formed as follows:

$$f(x) = \beta_0 + \sum_{i \in S} \alpha_i \langle x, x_i \rangle \ (7.7)$$

S : The collection of indices of support vectors.
β_0 and $\alpha_i, i = 1, \dots, n$ are estimated using $n(n-1)/2$ number of n data point pairs.
The classification of test point x needs to compute all the inner products of that point with all the training points.
General form of function, where K is a kernel function, which quantifies the similarity of two observations.
The following most common kernels:

- **Linear**

$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij}\ x_{i'j}$, where classifier is linear to x. It uses Pearson (standard) correlation, to quantify 2 observations similarity.

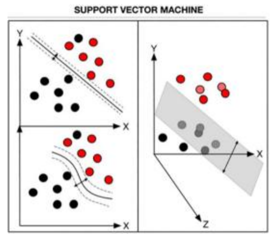
- **Polynomial**

$(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij}\ x_{i'j}\right)^d, \quad d > 0$. The greater the d the more flexible the boundaries.

- **Radial**

$K(x_i, x_{i'}) = \exp\left(-\gamma \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2\right), \quad \gamma > 0$ constant. Works very local with close to boundaries points.

The advantage of using a kernel rather than simply enlarging the feature space as we would do in a linear regression model (with polynomials, interactions etc.), is computational efficacy, as using kernels needs only to compute $K(x_i, x_{i'})$for all $n(n-1)/2\ pairs\ of\ data.$ [45]



[52]

5.15.4.3. Benefits & Drawbacks

One of the highlights of the SVC is the ability to find non-linear relationships using a kernel function. This ‘kernel trick’ allows the data to be transformed into another dimension which enhances the dividing margin between the classes. Compared with other ML algorithms, SVC achieves higher performance with large datasets, and higher accuracy. Training process is also conceptual simpler.
Despite being simpler, training is exceedingly slow and time consuming.
However, the limitation is that its prone to overfitting.[52]
It is also notable that SVC is prone to features values ranges, so scaling is a crucial pre-processing step. [17][13]

5.15.5. Classification Trees

5.15.5.1. Recursive Binary Splitting

For $X_1, X_2, .., X_p$ predictors, Trees are separating the predictor’s space into J distinct $R_1, R_2, \ldots, R_j$ regions. Regions could have any shape, but usually, ‘boxes’ are chosen for interpretation facilitation. A split can result in two or a single region or leaf.

$$R_1(j, s) = \{X \mid X_j < s\} \quad \text{and} \quad R_2(j, s) = \{X \mid X_j \geq s\}$$

The process is initiated from the top split, by considering all X_p as the first candidates for splitting, to finally choose the one that will give less Classification error rate (or RSS for regression) in both resulting regions. The first feature is the most important predictor.
The process Continuous similarly, from top to bottom. However, it is a greedy process as splits continue to be decided based on the rest of predictors and the already formed regions (i.e., next regions are divisions of the previous ones). The process ends when a cut point is reached, (e.g., a specific number of observations in leaves). [45]
Generally, a tree uses a structure that typically contains a root, internal nodes, branches, and leaves.

- <u>Internal/Decision node</u>

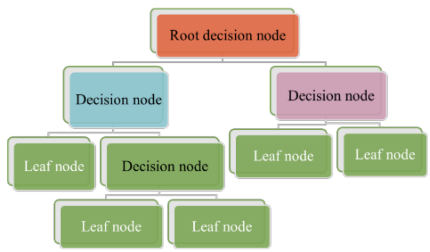
Represents a tested feature. Node is where a 'question' is tested and where the predictors split. e.g., Left: $Xj < tk$, Right: $Xj > tk$, tk the response measure for regression or classification accordingly.

- <u>Branch node</u>

It represents the test result. It is where the outcome of tested 'question' is delegated

- <u>Leaf/Terminal node</u>

Represents the class label assigned. It is where the class is finally assigned, at the bottom of the tree. Generally, it's the final decision after all results of above attributes. The end results are a set fo rules that governs the path from root towards leaves.



Trees calculate probabilities in deciding on some courses of action. The training set is recursively divided into subsets based on feature values, so the data in each subset is purer than the data in the parent set. The classifier can track the path from the root node to the leaf nodes and hold the sample's category label for new unseen test data. [45][52][13]

Practically, each split leads to a Region R_j, where the outcome values are always the same for all data points belonging in R_j. For example, for classification this value is the most 'common occurring class' in this region, while for regression it usually is the mean value of the response for points in the region. In classification the interest lies not only to the most common class but also to the proportion $\widehat{p_{m_k}}$ of this class, among others. Finally, this process runs faster for less predictors (less dimensions p). [45]

- **Separating Criteria for Classification**

– <u>Classification error rate</u>

Is the fraction of observation not belonging to kth class in a particular region and needs to be minimum. However, it is not used in practice.

$$E = 1 - \max_k(\widehat{p_{mk}})$$

$\widehat{p_{m_k}}$: $0 \leq \widehat{p_{mk}} \leq 1$ is the proportion of training observations from the kth class in the mth region.

– <u>Gini Index</u>

A measure of variance across K classes and node purity as well. For $\widehat{p_{m_k}}$ close to zero or one. i.e., great purity in node. For very pure nodes (i.e., most of observations belong to kth class).

Gini Index has a small value for pure nodes.

$$G = \sum_{k=1}^{K} \widehat{p_{mk}}(1 - \widehat{p_{mk}})$$

– <u>Entropy</u>

For $\widehat{p_{m_k}}$ near zero or one (pure node) entropy takes value near zero, i.e., entropy takes small value for pure nodes.

$$D = -\sum_{k=1}^{K} \widehat{p_{mk}} \log \widehat{p_{mk}} \geq 0$$

- **Separating Criteria for Regression**

$$\sum_{i:x_i \in R_1(j,s)} \left(y_i - \widehat{y_{R_1}}\right)^2 + \sum_{i:x_i \in R_2(j,s)} \left(y_i - \widehat{y_{R_2}}\right)^2$$

$\widehat{y_{R_1}}$ = mean response or common label for all points in R_1
$\widehat{y_{R_2}}$ = mean response or common label for all points in R_2

Note that regression criteria are oriented to both nodes, while in classification per node. [45]

5.15.5.2. Pruned Trees

Simple trees are not typically used in ML, but alternative forms are (e.g., Gradient Boosting etc.)[52]. The closest alteration on a tree is pruning. Specifically, simple trees are at risk of overfitting, as the splits and leaves increase, (i.e., bias is minimized on training data but the same isn't happening for test data). For this reason, pruned trees create less nodes and leaves in order to decrease variance with the trade of bias increase.

<u>Cost complexity pruning</u>

Initially, a T_0 very large unpruned tree is made. Then a series of T other pruned subtrees are defined by pruning the initial T_0. In order not to compute all the possible combination of T, a parameter α is defined, based on which a series of T subtrees are created (i.e., for each value of α corresponds a subtree $T \subset T_0$).
$|T|$: number of terminal nodes (leaves)

– $\alpha : \alpha \geq 0$ is a tunning parameter, that considers the bias-variance trade-off of complexity vs fitting to training set. It is usually selected through cross validation.

- $\alpha = 0$: Subtree equals the initial unpruned T_0.
- $\alpha > 0$: As α increases less leaves remain (i.e., more pruning happens).

Finally, after cross validation in the training set, the chosen subtree is the one for which the below obtains minimum value.

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} \left(y_i - \widehat{y_{R_m}}\right)^2 + \alpha |T|$$

Reason why many T subtrees are calculated, rather than one pruned based on a defined threshold (e.g., number of leaves to stop) is because tree structure is a greedy process as mentioned above. So, the pruning may have ended to a not so important split, while a more important may have followed, if continued.

5.15.5.3. Benefits & Drawbacks of Trees

Firstly, trees are fast, simple and easy to interpret, especially due to graphical representation not only regarding their 'tree' structure but also the regional spaces in cartesian planes. They can also outperform classic liner models for non-linear or complex relationships. Further, they can handle high-dimensional data. Finally, decision tree supports incremental learning, which is immutable because of the alternative functions based on each internal node.
However, in general they are less accurate than best supervised methods.
Most important, they are very prone to data changes, (i.e., they lack robustness or else suffer from high variance). For example, 2 different datasets or even 2 different random splits from the same data may result in different trees. As mentioned above, simple trees are also prone to overfitting. Lastly, it can be a lengthy process, particularly with large datasets or a high number of features. This is because the algorithm evaluates every potential split at each level of the tree, which can be computationally costly.
For more accurate optimization, below alternative tree methods have been defined, but with the trade of decreased interpretability. [45][13]

5.15.6. Ensemble Models

5.15.6.1. Benefits & Drawbacks

Ensemble methods combine several simpler models ('weak predictors') to result in a final model or prediction to achieve better performance. So, they combine the strength of other algorithms to complete training. [80][52]
One advantage of ensemble methods is that they perform in high predictive accuracy. They can also avoid overfitting problems that most unique algorithm supervised models arise. That is because, in case of a single ML algorithm can easily forecast ideally all the training data but with less accurate prediction for unseen data especially when using a small data size.
There are two standard ensemble learning algorithms: bagging and boosting algorithms. Bagging includes Random Forests, while Boosting includes Gradient Boosting Trees and XGBoost.[13]

Lastly, it is notable that Tree-based models, are better performing on wide features' values range, with scaling not being so crucial pre-processing step. [17]

5.15.6.2. Bagging Trees (Bootstrap)

Bagging means to randomly pull samples from the original data set to create a new data set of the same size. "Bagging" stands for Bootstrapping aggregation and is an ensemble method too. More specifically, initially it creates multiple models on subsets of data. Then those models' predictions are ultimately combined to make the final prediction.[52]
Because of high variance and lower accuracy scores in simple trees, there have been defined other methods to solve those problems.
Random forests use the bagging method to define more than one trees, while also using the property of reducing variance by averaging this set of trees.
More specifically, bootstrapping is taking repeatedly random samples from the training set and B separated random training samples are generated from the initial one. Thus, it is an $\hat{f}(x)$ for each of the B training sets. [45]

$$\widehat{f_1}(x),\ \widehat{f_2}(x),\ \dots,\ \widehat{f_B}(x)$$

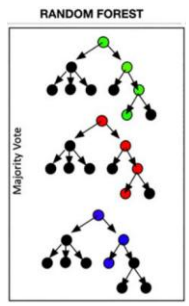
$$\widehat{f_{\text{avg}}}(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f^b}(x) \qquad (8.6)$$

A test point prediction in a classification problem, is decided by considering the class that each one of B bagging trees gave and choose the most commonly occurring. [45]
Finally, the number of trees in not so crucial. As far as an efficient number of bagging trees is chosen, a greater one will not increase the accuracy or even overfit further. Thus, one positive aspect of bagging is the lower risk of overfitting.
Moreover, bagging trees offer the capability of summarizing predictor's importances. For example, the total amount of Gini index decreased averaged over all B trees, by splits over a given predictor, can indicate and important predictor if this value is large enough. Similarly, happens with RSS for regression problems.
In contrast, due to more than one trees, they lose their interpretability efficacy compared to a single tree. [45]

5.15.6.2.1. Random Forests – (RF)

RF is another ensemble method, which is an optimized form of bagging trees regarding the 'tree decorrelation'. In contrast to other trees, RF differs to the splitting methods. Here, only a random sample of predictors are candidates (usually $m \approx \sqrt{p}$) for each split and from those, only one is chosen. This way, bagging trees are defined is not the same as one another and if for example, a very strong predictor occurs among others, he will not be present on average to (p – m)/p of the splits.

This predictor will not prevail, and trees will not give similar results as it would happen in Bagging trees. [2] Each decision tree will determine an outcome, and a majority "vote" approach is used to classify the data.[52] Averaging, uncorrelated trees lead to variance reduction, while with correlated ones this does not happen. It is called RF, since randomly generated decision trees are used to construct the final model.[52]



Benefits & Drawbacks

Using a part of predictors for fitting, is extremely useful when it comes to high dimensional datasets, or correlated predictors [45]. Random Forests are proven to be an extremely good performing model for classification, while able to detect non-linear relationships. They require less pruning than other algorithms and are also straightforward as it is simply a form of ensemble learning involving multiple classification trees.[2] Their ability of random sampling enhances generalizability and minimizes overfitting.

However, the number of trees and various internal parameters may hinder its performance. Additionally, it may be more time-consuming for great number of variables.[52]

5.15.6.3. Boosting Trees

Boosting Trees differ from the above ensemble methods, in that trees grow sequentially, (i.e., each tree grows based on the previous and that trees are fitted from the whole initial dataset, rather than bagging samples of it). [45]

Boosted Tree is an ensemble method that uses weak predictors (e.g, decision trees) that can ultimately be boosted and lead to a better performing model (i.e., the boosted tree). [52] In short, it combines a number several decision trees into a single predictor.[45] [80].

5.15.6.3.1. Benefits and Drawbacks

Boosting Trees are very useful when it comes to unbalanced datasets. However, they include limited tunning parameters that makes them more prone to overfitting compared to other ensemble methods (e.g., RF) [52]

5.15.6.3.2. Gradient Boosting Tree (GBDT) [81]

GBDT is subtype of Boosting trees. It is notable that GBDT works based not on parameters, but on functions' space and that not Y but residuals are used.[45][80]

Each new decision tree is trained based on compensating the prediction errors of the previous iteration's tree [80]. Generally, GBDT is an ensemble model of decision trees, which are trained in sequence. In each iteration, it learns the decision trees by fitting the negative gradients (residual errors). [82]
More specifically, GBDT works by approximating the function $F(x)$, of the relation of X, Y, by computing of $\hat{F}(x)$ in an additive manner. Firstly, given the initial model, a decision tree is fitted to the residuals, which equal to true y_i only for this iteration. After the first iteration, tree, each new tree is trained to minimize the following: [80]

$$\widehat{F_m} = \arg\min E\left(-\frac{\partial L(Y, P_{m-1})}{\partial P_{m-1}} - P_m\right) (8.9)$$

$-\frac{\partial L(Y,P_{m-1})}{\partial P_{m-1}}$: Residuals of previous iteration $\hat{F}(x)$.

$L(y_i, p_i)$: loss function. Measure of p_i prediction's error.
P_m : Prediction of current mth tree [80].
X : matrix of features
Y : output vector
M : Number of trees.
$m = 1, 2, .., M : M = number\ of\ trees, m = m^{th}\ tree$.
d : Number of splits for $d + 1$ terminal nodes.
$r_i = y_i$: Initializing residuals for all i in the training set.
$\sigma > 0$: Learning rate. Constant regularization parameter, that limits the influence of trees ensembled to avoid overfitting. It takes small values (e.g., 0.001, 0.01) Boosting trees lean slowly', so λ, slows the process, by determining how much trees change during the iterations. Small λ values usually need greater B. Generally, the slower the learning rate the better the performance .

$\widehat{f(x)}$: Initial function (e.g., $\widehat{f(x)} = 0$) that is going to be compensated.
$\widehat{f^m}$: Tree fitted in the b^{th} iteration.
$F(x)$: True function of relationship between X, Y
$\hat{F}(x)$: Approximately computed function of $F(x)$

Then, the tree defined is added to the initial model $\hat{f}$ to update the residuals by adding a shrunken version of the new tree, and this is repeated similarly.
Update previous tree [45][80][83]:

$$\hat{f}(x) \leftarrow \hat{f}(x) + \sigma\widehat{f_m}(x)$$

Update residuals [45]:

$$r_i \leftarrow r_i - \sigma\widehat{f^m}(x_i)$$

Final, model [80]:

$$\hat{F}(x) = \sum_{b=1}^{B} \sigma \widehat{f^{m}}(x)$$

Benefits & Drawbacks

Firstly, in GBRT trees do not need to have many splits, even one or two splits can be enough, as fitting small trees results in optimizing $\hat{f}$ to areas that do not perform well. Although, a great number of B trees can overfit. B parameter is selected, as usual, through cross validation. [45] Thus, it can avoid overfitting if properly regularized, while it is also computationally efficient. [84]

However, GBDT efficiency is decreasing in high dimensions (of features or instances). This is due to model examining all instances for all splits to find the information gain and decide the best split. Thus, the main cost lies in learning the trees, and the most time-consuming part is to find the best split. This is why algorithms such as pre-sorted, or histogram based are used in order to find the best split. It has to be noted that pre-sorted algorithms can manage sparse data, while histogram based are computationally better. [82]

The pre-sorted algorithm enumerates all possible split points on the pre-sorted feature values, so it can find the optimal split, however, it is inefficient in training speed and memory consumption. Histogram-based algorithm buckets continuous feature values into discrete bins and uses these bins to construct feature histograms during training. This algorithm is more efficient in training speed and memory consumption.[82]

5.15.6.3.3. Extreme Gradient Boosting - XGBoost

XGBoost is a regularized alternative of GMBT. The difference lies in the computation of residuals per iteration m, that is no more a loss function. [80][83] Formula used for residuals is now:

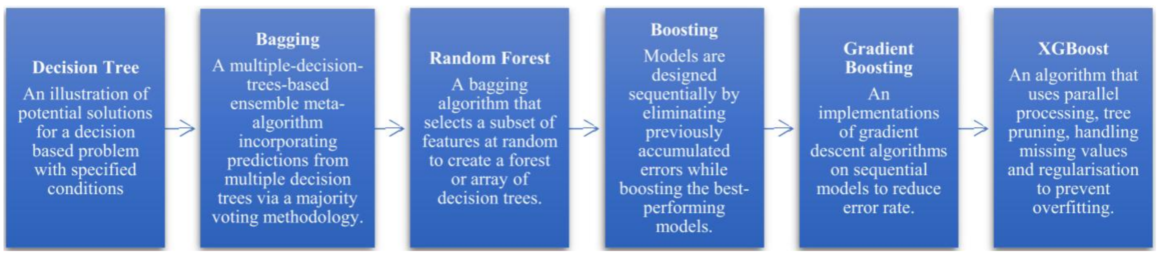
$$L_{\Phi}(y,p) = \sum_{i=1}^{I} L(y_i, p_i) + \gamma T_m + \frac{1}{2}\lambda ||w_m||^2$$

γ, λ : Regularization hyperparameters [80]
T_m : Number of leaves in the m^{th} tree
$||wm||^2$: $L2$ norm of m^{th} tree's leaf weights [80][83]

Benefits & Drawbacks

XGBoost provides high performance in many problems, outperforming most of the times the commonly used models. This happens mainly, because it a scalable method to all scenarios[85][84] [83] , while also performs faster than other models, offering a computational relief even on a common desktop, at any programming language. [83][85] It also includes features like built in cv, which is essential when evaluating algorithm performance. [85]

More specifically, it is useful and efficient for large datasets, due to its robustness against overfitting with regularization, and its flexibility through hyperparameter tunning. Thus, it has lower bias as it is more flexible but higher variance as it is more sensitive to the data than parametric methods. Although smaller datasets, could be better to be trained with simpler algorithms. Additionally, it can handle missing and sparse data in structured datasets. [85][84]
In more detail, XGBoost learns simpler trees with smoother weights, which leads to better generalization. Additionally, it employs Newton descent instead of gradient descent to optimize its trees, which leads to faster convergence. XGBoost also introduced a new feature split finding algorithm to speed up training. [80]
Finally, it provides the choice of pre-sorted and histogram-based technics. [82][84]



5.15.6.4. LightGBM

LightGBM is another Boosting algorithm, which provides faster results in high dimensional data (either on features or instances). Thus, it is used to solve the main problem of GBDT, by combining two technics, Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB).
An instance with a small gradient, means that it's training error is small and is well-trained. GOSS keeps the instances with large gradients to estimate the information gain and randomly samples the instances with small gradients, this way it excludes a significant proportion of instances.
Specifically, when computing the information gain, it multiplies data instances with small gradients with a constant. In more detail, first it sorts the instances based on the absolute value of their gradients and selects the $a \times 100\%$ of them with the largest gradient. Afterwords, it randomly samples $b \times 100\%$ instances from the rest. Then, it amplifies the randomly sampled data with small gradients by a constant $(1-a)\backslash b$ when calculating the information gain. Through this it achieved, to put focus on the under-trained instances (with large gradients) while maintaining the original data distribution. GOSS can also obtain accuracy of with a much smaller data size.
On the other hand, EFB bundles mutually exclusive features (i.e., rarely take nonzero values simultaneously), to reduce the number of features, without altering the accuracy of split.

More specifically, real world data are high dimensional, include large number of features some of which them are quite sparse (i.e., rarely take nonzero values simultaneously). EFB can bundle exclusive features into a single feature, by a carefully designed feature scanning algorithm (number of bundle << Number of features). This way feature histograms can be built from the feature bundles as those from individual features.[82]

5.16. Evaluation Metrics

5.16.1. Confusion matrix

It is a common measure used for classification problems, as it summarizes the predictions made by a classification model organized into a table by class-confusion matrix. Specifically, denoting the true positive, true negative, false-positive, and false-negative predictions. With these values various performance measures of model are calculate (e.g, accuracy, sensitivity, specificity, etc).
For binary classification sample is divided into four categories, as seen below:

Predicted Values		Actual Values	
		Positive	Negative
	Positive	True positive	False positive
	Negative	False negative	True negative

[58][57][52]

5.16.2. Classification Report

Given the below confusion matrix: [86]

		Predicted		
		Positive	**Negative**	
Ground Truth	**Positive**	True Positive (TP)	False Negative (FN) [Type II Error]	**Sensitivity (Recall)** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) [Type I Error]	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision (PPV) $\frac{TP}{(TP+FP)}$	**NPV** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

- True Positives (TP): Samples for which true and predicted labels are both 1.
- True Negatives (TN): Samples for which true and predicted labels are both 0.
- False Positives (FP): Samples for which true label is 0 and predicted label is 1.
- False Negatives (FN): Samples for which true label is 1 and predicted label is 0.

- $Predicted\ Positives: PP = TP + FP$
- $Real\ Positives: RP = TP + FN$

- $Predicted\ Negatives: PN = TN + FN$
- $Real\ Negatives: RN = TN + FP$

- $n = TP + TN + FP + FN = RP + RN = PP + PN$

- $X : True\ state\ of\ a\ subject$
- $Y = 1 = Positive\ Predicted$
- $Y = 0 = Negative\ Predicted$

- **Accuracy (ACC)**

Fraction of correct predicted outcomes to all predictions. It is one of the most used metrics for evaluating classifiers. It measures the ability of a predictor to correctly identify all samples, no matter if it is positive or negative.
Its benefits are simplicity and easy interpretation. The drawback is it cannot handle imbalanced datasets.

$$Accuracy = \frac{True\ Positive + True\ Negative}{Total} = \frac{TP+TN}{TP+FP+TN+FN} = \frac{1}{n}(TN + TP)$$

- **Sensitivity (SN) - True Positive Rate – (TPR) - Recall**

Sensitivity is the frequency/fraction of correctly predicted positive samples among all real positive samples. It measures the ability of a predictor to identify positive samples.
It is also the probability that the diagnostic test is positive, given that the subject is truly positive. It can handle imbalanced datasets.

$$Sensitivity = \frac{True\ positive}{True\ ositive + False\ egative} = \frac{TP}{TP + FN} = \frac{TP}{RP}$$

$$Sensitivity = P(Y = 1|X = 1)$$

- **Specificity (SP) – True Negative Rate (TNR) - (1 – FPR)**

The fraction of correct predicted negative to all truly negative. It measures the ability of a predictor to identify negative samples.
It is also the probability that the diagnostic test is negative, given that subject is truly negative.

$$Specificity = \frac{True\ Negative}{True\ Negative + False\ Positve} = \frac{TN}{TN + FP} = \frac{\mathrm{TN}}{\mathrm{RN}}$$

$$Specificity = P(Y = 0|X = 0)$$

- **Precision - Positive Predictive Value (PPV)**

Fraction of correct predicted positive to all predicted positive. Precision and Recall help to differentiate between error types. PPV does not take the false negatives into consideration. It can handle imbalanced datasets.

$$Precision = PPV = \frac{True\ Positive}{True\ Positive + False\ Positive} = \frac{TP}{TP + FP} = \frac{\mathrm{TP}}{\mathrm{PP}}$$

$$Precision = P(X = 1|Y = 1)$$

- **False Discovery Rate (FDR)**

The fraction of false predicted positive among all predicted positive. FDR does not take the false negatives into consideration.

$$FDR = 1 - PPV = \frac{FP}{TP + FP} = \frac{FP}{PP}$$

- **False Positive Rate (FPR)**

The horizontal axis is the FPR.

$$FPR = 1 - Specificity = \frac{FP}{FP + TN}$$

- **Negative predictive value (NPV)**

Fraction of correct predicted negative to all predicted negative.

$$NPV = \frac{TrueNegative}{TrueNegative + FalseNegative} = \frac{TN}{TN + FN}$$

$$NPV = P(X = 0|Y = 0)$$

- **Balanced accuracy (BA-BAcc)**

Accuracy metric that accounts for imbalanced samples. It solves most of the bias problems in an imbalanced dataset. It also considers the negative values.

$$BA = \frac{Sensitivity + Specificity}{2} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP + FN}} + \frac{\mathrm{TN}}{\mathrm{TN + FP}}\right)$$

- **F1 score (F1)**

Harmonic mean of PPV (precision) and sensitivity (recall). It balanced Precision and recall and is beneficial when there is a need to assess both false positives and false negatives by a single numerical representation. This measure removes the true negatives from the calculation. Useful when one may care only about positive outputs. It is especially useful for imbalanced datasets (as it is affected from Imbalanced ratio), when true positives are considered as twice important as the other samples. It does not consider the negative values as much as Bacc, so it accounts more for minority class.

$$F_{1score} = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}$$

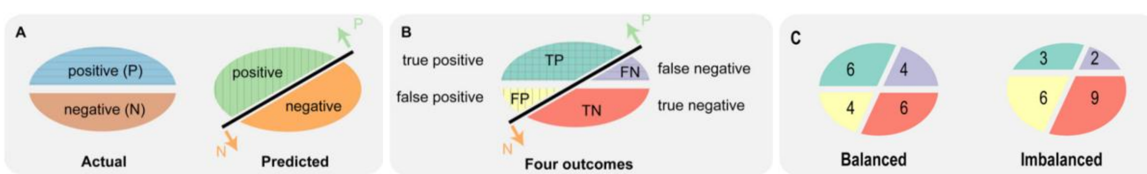
- **Matthew's Correlation Coefficient (MCC)**

MCC is generally regarded as being one of the best balanced measures for imbalanced datasets. It also considers both negative and positives predictions. MCC ranges from [–1 , 1]. Values of MCC indicate prediction accuracy as follows:

- $MCC = 0$: Random guessing
- $MCC = +1$: Perfect predictor
- $0 < MCC < +1$: Better predictor than random guessing
- $-1 < MCC < 0$: Worst predictor than random guessing
- $MCC = -1$: 'Perfect' predictor, which always predicts the opposite of true label. The output should be interpreted with the opposite meaning where Y=1 it truly is Y=0

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC is considered superior to other metrics for imbalanced datasets, such as F1 score. [87]
Usually, the scores checked are sensitivity and specificity, for which the higher they are the better the diagnostic test.
[31][88][57][58][86][89][90][91][87]



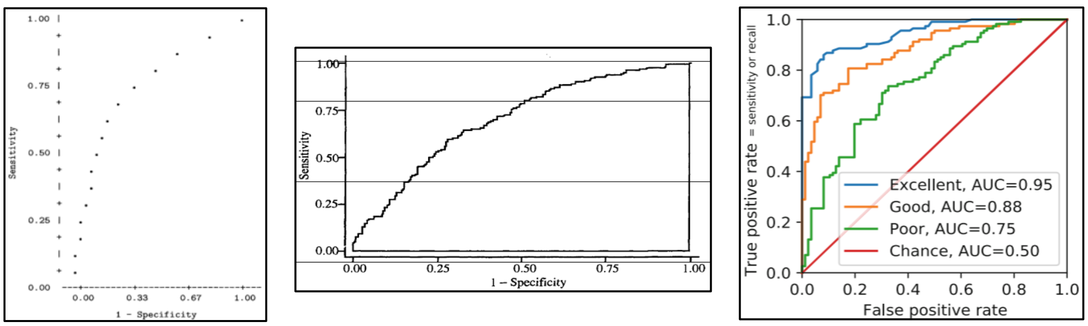
[89]

5.16.3. ROC - PR Curves & AUC

5.16.3.1. Receiver Operating Characteristic Curve (ROC Curve)

Receiver Operating Characteristic Curve (ROC) plots the probability of true signal detection (sensitivity: y-axis) vs false signal detection (1-specificity: x-axis) for a range of possible cut points. ROC curve is a completed description of a model's accuracy.

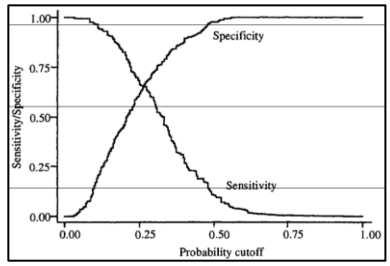
Area Under the Curve (AUC) is a measure of the ability of the model to discriminate among the individuals who would achieve the outcome (Y=1) vs those who would not (Y=0). AUC values lie from 0 to 1. [92]
The closer the ROC curve is to the upper left corner, the higher the accuracy. At the upper left corner, sensitivity = 1 (True positive rate = 1) and 1 - specificity = 0 (False positive rate = 0), where AUC = 1.0.
When sensitivity and 1 – specificity is 1:1, ROC Curve is drawn as a line of 45°, where AUC = 0.5 and indicates random guessing. [88]
The most used type is non-parametric ROC Curve (Empirical Roc Curve), as it does not consider data distribution (parametric Roc assumes normal distribution of outcome categories). By creating a 2x2 table for sensitivity and specificity on each cut-off value, creating the graph points, which are then just connected with a jagged line (rather than a smooth as in parametric). [88][31]



[57] [92][31]
The drawback of ROC curve is that it is not the best choice for high imbalanced classes. When imbalances, changes in sensitivity or specificity, may induce great practical impact but not change ROC curve much. Alternatively, Precision-Recall Curve (PR-AUC) can be used, or other metrics. [93]

Cut points
As far as it concerns cut points, the usually used in classification is $P_r(y = 1) \geq 0.5$ and $P_r(y = 0) < 0.5$ accordingly. The use of other cut points induces a trade-off between sensitivity and specificity and thus 1-specificity as the example seen below. [92]
If cut point is near 0, almost all predictions are $\hat{y} = 1$, sensitivity is near 1 and specificity is near 0, and the point for (1 – specificity, sensitivity) has coordinates near (1, 1). If cut point is near 1, almost all predictions are $\hat{y} = 0$, sensitivity is near 0 and specificity is near 1, and the point for (1 – specificity, sensitivity) has coordinates near (0, 0).
One could choose another cut point which maximizes both sensitivity and specificity, however there are statistical benefits from the usual 0.5 value. By taking the sensitivity and 1- specificity for each cut point, a dot arises, and connecting those dots, result to the ROC curve [92]. ROC curve has a concave shape connecting the points (0, 0) and (1, 1) [31].

Cutpoint	Sensitivity	Specificity	1–Specificity
0.05	99.32%	2.57%	97.43%
0.10	95.92%	15.19%	84.81%
0.15	90.48%	31.78%	68.22%
0.20	81.63%	46.26%	53.74%
0.25	65.99%	61.21%	38.79%
0.30	57.14%	72.20%	27.80%
0.35	40.14%	82.01%	17.99%
0.40	29.25%	87.38%	12.62%
0.45	18.37%	92.06%	7.94%
0.50	10.88%	97.43%	2.57%
0.55	5.44%	99.30%	0.70%
0.60	3.40%	100.00%	0.00%



5.16.3.1.1. Area Under Receiver Operating Characteristic Curve (ROC-AUC)

The AUC is the probability that a positive sample has a higher classification score (as positive) than a negative sample. [93]
The AUC is the probability that a subject achieving Y=1 outcome will have higher probability $P_r(y = 1)$ [92]. It is a measure of predictive power called 'concordance index'. More specifically, let's assume:

- n_1 : All data points for which $y_1 = 1$
- n_0 : All data points for which $y_0 = 0$
- $n_1 x n_0$: Pairs of each y_1 is paired with each y_0

AUC is the proportion of all possible positive–negative pairs for which the subject with the positive outcome ($y = 1$) also has a higher predicted probability ($\hat{\pi}$) than the negative one ($y = 0$).[31]
A perfect classification corresponds to an AUC of 1 and a random classification to an AUC of 0.5. [93] [94]

AUC must be greater than 0.5 to be meaningful, and greater than 0.8 to be considered acceptable for a method evaluation. When comparing two or more methods, this with greater AUC is considered to have the best diagnostic performance.[88]

- $AUC \geq 0.9$: Excellent. In practice it is rarely observed and if yes it may be due to perfect separation problems in data (sparse data). [92]
- $0.8 \leq AUC < 0.9$: Good. $AUC \geq 0.8$ is generally, considered acceptable.
- $0.7 \leq AUC < 0.8$: Fair
- $0.6 \leq AUC < 0.7$: Poor
- $0.5 \leq AUC < 0.6$: Fail
- $AUC = 0.5$: like random guessing [88]

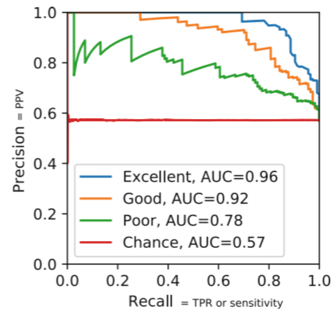
The AUC is often presented with a 95% CI because the data may be influenced by statistical errors. [88] The 95% CI provides the range of values in which the true value lies. Thus, one can be 95% sure that the CI includes the true value of AUC.[94]

5.16.3.2. Precision-Recall Curve (PR Curve)

PR-Curve focuses on the minority class so, it is a better option for imbalanced datasets[93] [95] [17] [89]. PR-Curve plots Precision (PPV) vs Recall (Sensitivity).[93] [89] The baseline for PRC is not fixed but depends on the ratio of positives to negatives such as:

$$y = P/(P+N)$$

(e.g., y = 0.09 if ratio P:N is 1:10). Also, $y = 0.5$ only for balanced datasets. In contrast for ROC curve baseline is fixed (e.g, AUC = 0.5 for random guessing). In fact, AUC (PRC) is identical to the y-position of the PRC baseline.[89]



[57]

5.16.3.2.1. Area Under Precision-Recall Curve (PR-AUC)

For AUPRC (or average precision) a perfect classification corresponds to a value of 1, in contrast to ROC AUC, a random classification does not necessarily correspond to 0.5. It depends on the prevalence ratio as mentioned before. [93]

5.16.3.3. ROC – PR Curves & Imbalanced Datasets

It is of high importance to use appropriate evaluation metrics when it comes to imbalanced datasets to realistically judge their impact. It usual for test sets to have imbalance ratio much different from real world's (e.g., due to selection bias) and thus a study's results may be evaluated more optimistically than they actually are. This is why it is useful to refer to the imbalance ratio when displaying the evaluation metrics, or else different studies of same scientific question with different ratios won't be comparable even if samples were drawn from same population.

Assume a binary classification, where $x \in X$ is an input and $y \in Y = \{-1, 1\}$ an output.

- $y = -1$: Negative class
- $y = 1$: Positive class

The positive class is the minority class and the negative class the majority.

Instead of a single real-world joint-probability distribution $p(x, y)$ we consider a parametric family:

$p(x, y; \eta) = p(x \mid y) \cdot P(y; \eta)$ where $P(y; \eta) = 1 - \eta$ if $y = -1$ & η if $y = 1$

$\eta \in [0,1]$: Parameter of the positive class prevalence
$h : X \to Y$: The classifier

As mentioned above the metrics used in ROC in PR Curves axes are:

$$True\ Positive\ Rate = TPR = Recall = Sensitivity = P(h(x) = 1 | y = 1)$$

$$False\ Postive\ Rate = FPR = P(h(x) = 1 | y = -1)$$

$$Precision = Prec(\eta) = P(y = 1 | h(x) = 1) = \frac{TPR \cdot \eta}{TPR \cdot \eta + FPR \cdot (1 - \eta)}$$

Particularly fo the test set it is sampled from $p(x, y;\ \eta test)$ where $\eta test$ may or may not correspond to a positive class prevalence of the real-world.

$$N = TP + FP + TN + FN, \qquad \text{equals the test size}$$

$$Imbalance\ Ratio = IR = \frac{TP + FN}{TN + FP}$$

p_+ : Prevalence of positive class in test set

$$p_+ = \frac{TP + FN}{N}$$

$\widehat{TPR}$: Fraction of positive samples that were classified correctly

$$\widehat{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

$\widehat{FPR}$: Fraction of negative samples that were classified incorrectly

$$\widehat{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$$

$\widehat{Prec}$: Fraction of true positives out of all the positive predictions

$$\widehat{\mathrm{Prec}}(\mathrm{p}_+) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

The values measured on test set approach the ones from true distribution $p(x, y;\ \eta)$ as the test size grows to infinity:
$p_+ \rightarrow \eta test$
$\widehat{TPR} \rightarrow TPR$
$\widehat{FPR} \rightarrow FPR$
$\widehat{Prec} \rightarrow Prec$

However, limited test sizes induce errors in the above estimations especially for imbalanced datasets.

It is notable that Precision is affected by positive class prevalence (or class imbalance ratio) while TPR and FPR are not. Considering that ROC Curve displays TPR and FPR on y and x axis accordingly it can be understood that ROC Curve is unaffected from IR, while PR Curve which displays Precision and TPR on y and x axis accordingly is affected. Thus, PR Curves can reveal poor performance of a classifier, which may not be obvious from a ROC Curve. This is why it is important to report the IR of test set for corresponding evaluation metrics.
As mentioned above, even if test set is sample from the same initial source, PR curves won't be comparable for different IR. Finally, if test set is imbalanced PR Curve is a better solution as it depends on IR and does not give too optimistic results as the unaffected from IR ROC Curve would do. Other metrics useful for imbalanced datasets are F1 and MCC as mentioned in above sections. [91]

5.16.4. SHAP Plots

Machine learning models steadily gain large applications on drug development, however their low interpretability of results, limits their implementation.
SHAP plots is a feature-based plot method that enhanced ML models to better interpretability and thereby their trustworthy. SHAP analysis is based on Shapley values, a concept from 'collaborative game theory'. SHAP provides a fair distribution of a result payout among collaborative factors (e.g., features of a drug) where factors cooperate for a common purpose, without necessarily contributing equally. For example, let's assume Feature A, Feature B and Feature C instead of 'players' and a model's outcome (e.g, outcome in percentage) instead of 'game', for which features contribute in order for the outcome to be predicted.

- Total response rate

Feature A, Feature B, Feature C: 90%

- Each drug's response rates

Feature A: 40%
Feature B: 50%
Feature C: 60%

- Paired drugs' response rates

Feature A, Feature B: 70%
Feature A, Feature C: 65%

Feature B, Feature C: 80%

Assumptions of SHAP

- <u>Efficiency</u>

The sum of each drugs' contributions equals the total response (e.g., 90%)

- <u>Symmetry</u>

If two Feature contribute always the same whichever the subset they belong (eg., pair or individually), they should receive an equal result payout.

- <u>Additivity</u>

If an intervention has multiple sub-samples and each has a separate payout result, then the contribution of each feature to the outcome is equal to the sum of contributions to each sub-sample. (e.g., two sub-populations of patients, the feature A contribution to the total population equals the sum of its contribution to each of the sub-populations).

- <u>Null player</u>

If a feature doesn't contribute to any subset (e., pair or individually), its share of the payout result is 0.

5.16.4.1. SHAP interpretation

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big(V(S \cup \{j\}) - V(S)\big)$$

- N: Set of all drugs
- V: Result payout
- j: 'Player' (e.g., Drug)
- Φ_j: SHAP value. One Φ_j is assigned to each drug and corresponds to its contribution to the total outcome. SHAP values can be described as the weighted average of a drugs contributions across all possible subsets.
- $|S|$: Size of S subset (e.g, Subset S = {Drug A, Drug B})
- $V(S \cup \{j\}) - V(S)$: Quantifies the contribution of drug j to subset S
- $\frac{|S|!(|N|-|S|-1)!}{|N|!}$: Weight of contribution
- $\frac{|S|!(|N|-|S|-1)!}{|N|!}(\mathrm{V}(S \cup \{j\}) - \mathrm{V}(S))$: Weighted contribution of subset of size $|S|$
- $\sum_{S \subseteq N \setminus \{j\}}$: Sum of all possible subset contributions without j

In practice, other approximate methods are used to calculate SHAP values, as this formula would be calculative expensive. For example, in Python SHAP packages are used (e.g., Deep Explainer, Gradient Explainer etc.)

$$f(i) = \phi_0 + \sum_{\text{features}} \phi_{i,j}$$

- i: sample i
- $f(i)$: Model prediction for sample i
- ϕ_0: Average model prediction
- $\sum_{\text{features}} \phi_{i,j}$: Sum of all SHAP values for sample i

Thus, for sample i, the SHAP values assigned to each feature describe how that feature contributed to the difference between the individual prediction of this sample from the average model prediction (i.e., Each SHAP value explains how a feature shifted the prediction of sample i away from the model's average prediction)

5.16.4.2. SHAP Plots

Reported SHAP plots refer to test dataset. Different SHAP plots can illustrate explanation of feature contributions, locally i.e., on one data point or globally i.e., whole dataset.

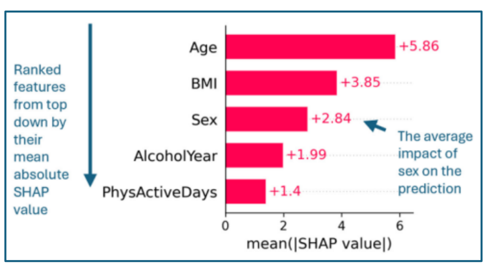
- **Bar Plot**

Displays the mean absolute SHAP value for each feature, considering all data points (i.e., global overview). Thus, it is a measure of feature importance but also ranks features top to bottom based on their mean absolute SHAP value or else impact on predictions.

Benefits & Drawbacks

Provides same units as model's predictions, and good interpretability.

Lack of direction (eg., positive, negative) and monotonicity of impact.



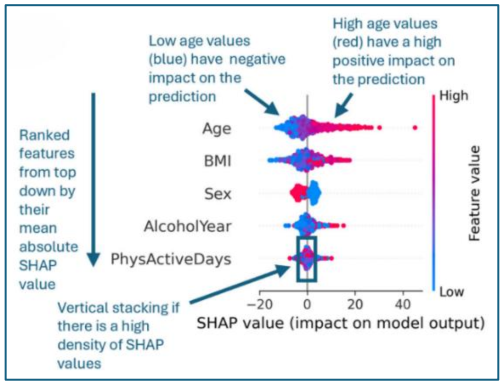
- **Bees warm Plot**

Displays SHAP value for each feature, considering all data points (i.e., global overview), ordered by the mean absolute SHAP value. Thus, it is a measure of feature importance, which ranks features top to bottom based on their mean absolute SHAP value or else impact on predictions.

The dots are the SHAP values of each data point. Color range indicates feature values. High feature values are displayed as red, while lower as blue. Missing values, if any, are displayed as grey. Each dot is placed based on its SHAP value on x-axis.

Benefits & Drawbacks

The combination of dots' color and place on x-axis indicates the direction of feature-prediction relationship, monotonicity, possible alike spreading or even outlier SHAP values.
Colour scale makes it difficult to surely explain the characteristics of relationship due to overlapping colored points, more complex relationships etc.

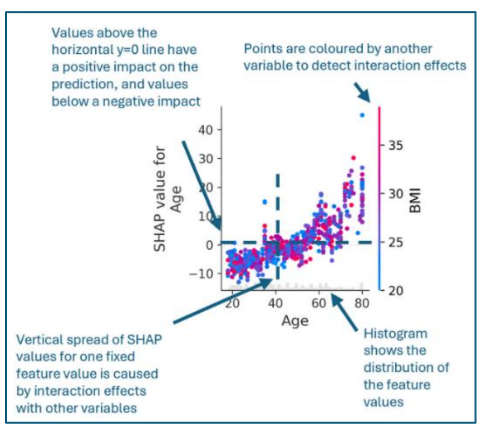


- **Scatter Plot**

Displays the relationship between one feature's values (x-axis) and SHAP values (y-axis), considering all data points (i.e., global overview) but not for all features. On x-axis it may be displayed a histogram regarding feature value distribution.

<u>Benefits & Drawbacks</u>

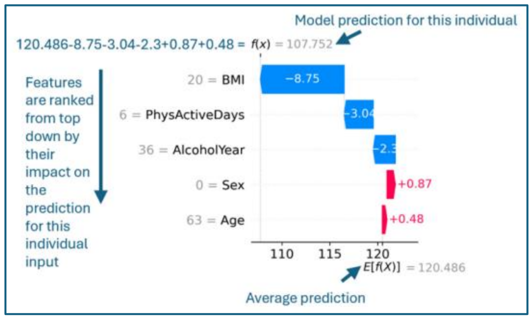
Relationship of feature-SHAP values trend indication (eg., linear/non-linear)
Interactions with other features if a vertical spread of SHAP values, or through coloring of points. The addition of trend lines makes patterns more obvious.
Can be used for k-fold validation, displaying all folds' samples, and indicating different patterns within different samples of dataset. Each fold is assigned with a specific data point coloring.
Only one feature per plot.



- **Waterfall plot**

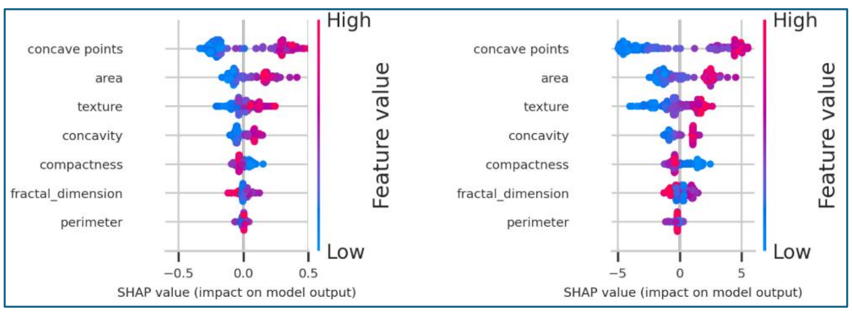
Displays SHAP values of a single data point but for all features (i.e., it is a local overview). Display features (y-axis) and SHAP values (x-axis). The value of that feature for that specific point is denoted in gray. SHAP values are displayed on the

‘arrow’ of each feature. Arrows are colored, with red color meaning that SHAP value increases the prediction $f(xi)$, and blue if it decreases.
<u>Benefits & Drawbacks</u>
Outliers, SHAP value outliers, evaluate specific features.
Display only one feature.



5.16.4.3. SHAP for Classification

In classification models, SHAP plots may interpret: 1) probabilities, or 2) log-Odds. When probabilities are displayed, it is easier to interpret predictions, however, log-odds offer easier relationship interpretation between features (e.g., interactions etc.). Best practice is to evaluate both formats.



Example left probabilities on x-axis. Right log odds on x-axis. [96]

5.17. Key Methodology Points

5.17.1. Pre-Modelling

- **Primary Purpose**

A valid scientific question must be well defined as well as the eligibility criteria analysed should be such as to reflect the actual ones from clinical practice. For example, models with too many features, may create unrealistic well performing predictors, without necessarily considering the actual useful features.[17]

- **Representative Samples**

Samples should be representative of real-world distributions. More specifically, rarer categories, should not be ignored as it could lead to ‘spectrum bias’. They could, on the contrary, be encoded to an additional level (e.g., ‘Other’, so distribution is preserved).[17]

- **Sample Size**

Especially, for high dimensional data. Sample size should be large enough to achieve stable performance, especially in training sets, in order to avoid overfitting, noise from many irrelevant features and outliers. Unfortunately, sample size determination is still mostly empirical.[17]

- **Missing Data**

Real world data commonly include missing data. Excluding all missing data could result to bias, sample reduction or even reduction in other features sample sizes. Many imputation methods could be used to replace nulls, or even models that are able to handle them (e.g., XGBoost). [17]

- **Scaling**

Real world data include wide ranges of feature values or even different units. This may result to features influencing more the model’s performance than others, especially in prone models like distance based KNN, SVC etc. Scaling is necessary for both train and test sets (whole dataset). The solution is Scaling (Standardizing, Normalizing) or utilizing a specific scale of feature (e.g., logarithm).[17]

- **Multicollinearity**

Collinearity happens when at least 2 variables are dependent one other, thus one can be linearly predicted from the other. Collinearity inflates variance of coefficients, thus, model ‘struggles’ to choose which feature coefficient to give most ‘weight’’, so changes in data lead to changes in coefficients estimations too. For example, linear models are especially prone to this problem. This way not only decreases model’s performance but also its interpretation as well. Some common solutions are bivariate analysis (pair feature correlations) and Tolerance or Variance Inflation Factor – VIF for multicollinearity detection.[17]

- **Imbalance Classes**

This happens when one classes sample is much higher than the others. This leads to bias prediction towards the majority class. This is why train sets should be resampled (over/under-resample) to restore similar distributions between classes. In contrast, test sets should not be resampled, as their distribution should be generalizable to real world’s.[17]

- **Information Leakage (Data Snooping Bias)**

It refers to transfer of data among train, test, validation sets or more generally, it happens when test set somehow influences the training process, [17] or it can be defined as the cases where information from the training set has “leaked” into the test set. [57] The problem arises due to problems in splitting or when a model is trained with features not available in unseen test data (e.g., train hospital data with discharge date, which only happens after the outcome event occurs) . Result is generalizability issues. For this reason, splitting must be always done before any transformation or process on data. Extremely high performing models should be suspicious of data leakage. [17]

The consequence is a rather falsely optimistic result. [57]
Common Causes of Data Leakage:

- Feature selection using the whole dataset.
- Dimensionality reduction using the whole dataset.
- Parameter selection using the whole dataset or the test set.
- Model or hyperparameter search using the whole dataset or the test set.
- For the same patient, if any of his data occurs in both train and test sets. For example, some of its visits are in the training set and some in the validation set [57]

- **High Dimensional Data**

When great number features occur in dataset, especially if features are more than sample size. High-dimensional data can cause overfitting and generalizability problems, because many features are 'learned' for less data points and thus, any change in data provides different results. A solution to this is dimension reduction (e.g., PCA) or feature selection technics. However, those must not be done before Cross Validation, or else positive bias is induced, especially in high dimension data.[17][97]

- **Randomness**

If randomness is not well implemented where needed (e.g., samples) then different results may occur even by using same data.[17]. Randomness helps in generalizability of method.[57]

5.17.2. Modeling

- **Hyperparameter Tunning**

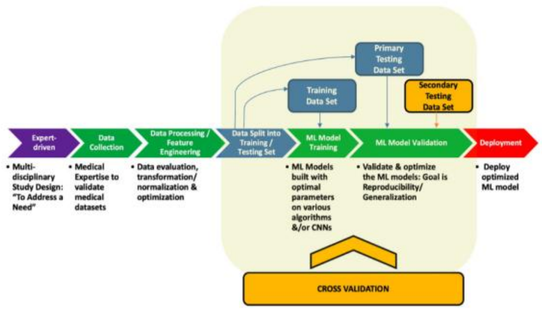
Hyperparameter optimization, lead to better performance in models. Overfitting of optimization can be avoided through Cross Validation or Regularization. [17]

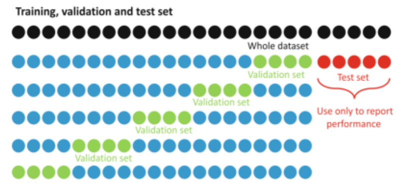
- **Overfitting (See Bias-Variance Tradeoff section)**

Overfitting happens when models are trained on limited data points and fits too 'close' to them. The problem inflates if number of features is also large, creating noise. This way model is unable to predict on unseen data. Overfitting occurs mostly on train set. It can be eliminated by using more train set samples, by implementing correctly all other concepts discussed so far. [17]

- **Generalization**

It is the ability of the model to adapt to unseen data. Thus, the use of evaluation and optimization processes should be on appropriate data sets. Specifically, hyper-parameter optimization (HPO) should be done on validation set (through cv) in training data set, while model generalization and performance evaluation should be done on out of train set data (i.e.., unseen test set). [17][57][52]





5.17.3. Post-Modeling

- **Performance Metrics**

These should be chosen based on dataset, as not all metrics are appropriate for all dataset compositions. For example, accuracy is prone to bias for imbalanced class datasets. Alternative metrics for such datasets are balanced accuracy, no information rate, the Matthews correlation coefficient (MCC), F1 measure and PR-AUC (Precision–Recall AUC). If positive class is minority, precision and recall are good choices. [17] Specifically, If class imbalance occurs, ROC-AUC is not the best choice, and PR-AUC is preferred, as it focuses on the minority class. [93]
Finally, in classification models, it is important to display the confusion matrix which provides broader insights of the models in terms of both overall and class-wise performance. [17]

- **Comparison classic vs ML algorithms**

A completed evaluation of ML models, includes comparing them to classic statistical models such as logistic regression. This way ML models impact is better reflected.[17]
Some models are easier to interpret, such as logistic regression, trees etc. However, some ML models may outperform in some problems, but their greater complexity costs interpretability. Interpretability can be done locally for one feature or one data point or globally for whole features or dataset. [96][17]
Common methods for interpretation are Shapley additive explanations (SHAP), permutation feature importance (PDP), activation atlass, local interpretable model-agnostic explanations (LIME).[17]

6. Results

6.1. Tools, Technologies & Files

- **Programming Languages**
  - Python version 3.12.10
  - Postgres SQL version 9.0

- **Environment**
- Visual Studio Code (Jupyter Notebook)
- pgAdmin4
- **Libraries**
- Pandas, numpy, sklearn, statsmodels, sqlalchemy, pickle, matplotlib, seaborn, collections, geonamescache, scipy, itertools, patsy etc.

6.2. Data Sources

Two formats of data were downloaded, one in db and another in csv format.

- **AACT Database**

AACT is a publicly available relational database (bd) that contains all information about every study registered in ClinicalTrials.gov. Content is downloaded from ClinicalTrials.gov daily and loaded into AACT.
The Clinical Trials Transformation Initiative (CTTI) enhanced AACT in October 2016. Data are accessible in cloud and can also be downloaded as a static copy. [98]
For this study, a static copy was downloaded in postgres.dmp format. The .dmp file provides SQL code for each table. Tables were be loaded by executing the corresponding SQL code for each. Specific tables were chosen to be studied, based on the features needed extraction.
After, Visual Studio Code was connected to database, and any data extracted or new tables created, were managed through this environment.[98] Thus, all processes referred below have been made with Python through Visual Studio Code environment (Thesis_SQL.ipynb jupyter file).
The db was used for data extraction, and consistency check between db and csv format. Thus, database with combination to CSV, enhanced the creation of features, especially when one source provided a more unstructured format of a feature that the other.

Filtering was applied through ‘ctgov.studies’ table of db and its columns:

- studies.study_type = ‘INTERVENTIONAL’
- studies.start_date >= ‘2011-01’
- studies.overall_status in ('COMPLETED', 'WITHDRAWN', 'TERMINATED')
- studies.phase in ('PHASE1', 'PHASE2', 'PHASE3', 'PHASE4')

Additional new tables were generated from the initial ones. These tables have the prefix ‘my_’ for convenience and to be distinguished from those pre-provided by AACT db. These new tables were defined with purpose to be used for merging with csv format into the final data frame of features.
‘nct_id’ column from ‘ctgov.studies’ table of db, was used as the unique identifier of trials in db. It was joined with any table extracted or created from db and for any merge of db tables with csv etc.

- **CSV**

Data downloaded in CSV format from Clinical Trials.gov [11]. CSV was mainly used for comparison with db's data and extraction of structured data that may not be provided in database source. Also, as the main source of trials according to their unique NCT numbers. All features' changes and addition columns added from db were merged to csv into the final data frame.

Filters applied during download from ClinicalTrial.gov website:

- Study Status
- – No longer looking for participants: Completed, Terminated
- – Other: Withdrawn
- Study Type
- – Study Type: Interventional
- Date Range
- – Study starts: From: 0/01/2011 , To: 12/31/2024
- – Completion Date: From: 0/01/2011 , To: 12/31/2024

Studies after 2011 were chosen due to many missing data occurring in ClinicalTrials.gov dataset prior to that year. [14]

6.3. Files

Jupyter files were organised based on the order of data engineering and analysis processes. Two files were created for each process of data engineering and descriptives statistics: one for analysis on full dataset and one for analysis on train set. This separation was implemented to prevent information leakage, therefore, procedures such as feature selection, collinearity analysis, and similar preprocessing steps were applied only to the training set and its corresponding files. For model fitting and results presentation, a single final file was used.
The files created are as follows:

Data Engineering - Exploratory analysis files (on full dataset)
These files are the main pipeline files that lead to final models' fit file. All data engineering processes were implemented in these files. It must be noted that transformations and feature selection decisions were not made based on these files results to avoid information leakage, as they regard full datasets per Phase (train and test sets not yet split)

- Thesis_SQL.ipynb
  Data extraction, features creation from AACT db.
- Thesis_Data.ipynb
  Data engineering. Merging db tables with csv format data frame. Different scale formats of features creation. Merging rarer levels of categorical features. Regular expression text mining for filling missing values.

- Thesis_Dummies.ipynb
  One hot Encoding of categorical features. Interaction terms creation.
- Thesis_Models.ipynb
  Models fit on training data. Models' evaluation and comparison.
- Thesis_Results.ipynb
  Summary of top performance models and top predictors per phase. ROC AUC Curves of models' comparison per phase.

Exploratory analysis - Feature selection files (on train set)

These files purpose was exploratory analysis on train set only. The purpose of this analysis was feature selection, dimension elimination, scale transformation choices, sparsity and collinearity identification etc.
However, all decision resulting from these files, were implemented directly to above pipeline files, as those are the ones leading to final models' fit file (e., dropping columns).

- Thesis_Data_Unmerged.ipynb
  Data engineering same as the above corresponding file. Feature levels were unmerged. Data provided in following files.
- Thesis_Dummies_Unmerged.ipynb
  Data from Thesis_Data_Unmerged.ipynb file One hot Encoding of categorical features. No Interaction terms creation. This file provided data to Sparse and Collinearity files.
- Thesis_Sparse.ipynb
  Data from above file were analysed on the actual training sets used in models. Sparsity risk feature levels were identified to be merged back in the above Thesis_Data.ipynb file.
- Thesis_Collinearity.ipynb
  Unmerged Data were analysed for correlation and AIC test for feature selection. Features dropped in Thesis_Visual.ipynb file or merged back in Thesis_Data.ipynb file

Exploratory Analysis on Full and Train Set – Common working files

- Thesis_Visual.ipynb
  Visualizing-Distribution-Descriptives exploratory analysis on full and train datasets. It must be noted that only train set analysis was the criteria for feature selection methods etc., although there is a common file due to convenience and similarity of descriptives for both samples of data (full and train). The train set analysis was used in combination with Thesis_Collinearity.ipynb and Thesis_Sparse.ipynb files.
- Thesis_Metrics.ipynb
  Combined descriptive data frames of pivot tables, continues distributions and p values generated from above files of both full and train data.

6.4. Data Engineering

6.4.1. Missing values

(Thesis_Data.ipynb file)
Firstly, csv and db format were compared, so records missing from csv columns were filled from db corresponding tables, where possible.
Especially, for categorical features, text mining using regular expression was used to extract words from unstructured text data (e.g., str.contains() command etc.). For example, 'Brief Summary' column of csv was widely used for text mining and null values filling. Similarly, happened for database text mining through different columns of tables.
Moreover, missing data were numerous for some categorical features. Dropping them would greatly decrease the sample size of other valuable characteristics. Because of this, missing values which could not be filled with text mining technic, were transformed into additional or pre-existing levels of these categorical features such as 'Other', 'None' etc. (e.g., adverse events nulls meant no adverse occurred). [17]
For Continuous features, missing values were far too less than in categorical and especially compared to whole dataset size, thus they were simply dropped. Imputation methods were not used, as most missing data appeared on categorical features rather than Continuous.
An exception to that is regarding locations count features (e.g., City_Counts, Country_Counts etc) where missing values were far more (about 2500). Imputation with IterativeImputer class from sklearn.impute library was chosen. In practice one dataset was produced by using 10 iterations, [99] [100] by using only location counts columns for the imputation. Simpler methods (e.g., mean impute) were not chosen, as data were highly skewed, and this would induce bias in imputed values. More complex methods (e.g., Multiple Imputation) were also not chosen, as they mostly serve interpretability, from producing multiple datasets and pooling parameters to fit a final model. The purpose here was just value imputation to be used in many models, which moreover could not be executed for multiple datasets due to time consumption. [50] [99]
Lastly, some features appeared more than half the dataset's size in null values, so it was not possible to fill with any method. These feature's columns were totally dropped, (e.g., Study Documents, Locations, Outcomes, Enrollment etc).

6.4.2. Duplicates

(Thesis_Data.ipynb file)
Some dozens of studies occurred in dataset, which had same Study Status, Titles, Dates etc. and differed only in some of their characteristics. Thus, these were considered duplicated and were dropped.

6.4.3. Features

6.4.3.1. Format & Scale

(Thesis_Data.ipynb file)
Three data types and scales of features were analysed (categorical, binary, continues). Their names were based on that format:

- **Binary**

Suffix '_Bin' : Recognition of binary format. They refer to data of 'Yes/No' levels.

- **Categorical**

Suffix '_List' : Recognition of categorical format with more than 2 levels.
These data were initially unstructured text data. They were transformed into categorical features of list element rows (e.g., ['CHILD', 'ADULT', 'OLDER_ADULT']). This way it was achieved that one trial record could belong to more than one levels of the categorical feature, while secondly data frame's number of rows remained the same without inflating the sample number.

- **Continuous**:

Suffixes such of '_Counts', '_Log', '_Counts_Log' : Recognition of Continuous features.
They were evaluated in different scales (e.g., initial, logarithmic, categorical):

– <u>Initial scale</u>

'_Counts' : Was not chosen for analysis as real world data were highly skewed. This induced problems in parametric models, distance-based models etc. [17][58][59][52]

– <u>Logarithmic</u>

'_Log' scale : Eventually, log scale was chosen for all Continuous features, due to helping with skewness of real-world raw data. Skewness and outliers' effects were decreased (e.g., Enrollment), which was necessary for parametric models. [17][58][59][52] Specifically, due to having many variables with distribution around zero until median values, log1p() was used, in order to avoid inf values.

– <u>Categorical (Binary)</u>

'_Categ' : Had only two interval type levels and were produced. They were named with this suffix to be distinguished from binary features of Yes/No. However, finally, no '_Categ' format was chosen due to inducing sparsity risk.[31]

Continues variables were preferred to be used as logarithmic scales due to skewness as mentioned above. Dichotomization was avoided due to information and relationships shapes loss where possible. [63] [62]

6.4.3.2. Outcome Feature (Study Status)

There are three types of 'closed' clinical trials in ClinicalTrials.gov Registry:

- Completed: Study has concluded normally (i.e., last participant has occurred).

- Terminated: Study stopped prematurely and will not resume. Participants are no longer receiving intervention.
- Withdrawn: Study stopped prematurely, before enrolling any participant. [6]

Below table displays the ratios between different study statuses before any data engineering (e.g, drop rows etc.) was done.

('Enrollment', 'Study Stat...	# count
(0.0, 'WITHDRAWN')	3837
(30.0, 'WITHDRAWN')	1
(34.0, 'WITHDRAWN')	1
(240.0, 'WITHDRAWN')	1
(430.0, 'WITHDRAWN')	1
(3000.0, 'WITHDRAWN')	1

Study_Status_Bin	# count
COMPLETED	57584
TERMINATED	9664
WITHDRAWN	3842

Study_Status_Bin	# count
COMPLETED	57584
TERMINATED	13506

Finally, the outcome feature is defined as a binary variable with levels encoded as:
- 'Completed' : 0
- 'Terminated' : 1 / 'Withdrawn' : 1

Terminated encoded as the 'positive' label, as this study's purpose is to evaluate the probability of early termination, rather than completion. So, the encoding is interpretively aligned with the purpose.
Withdrawn studies were analysed separately.

6.4.3.3. Predictors

As mentioned above all the features resulted from data extracted from db or combination of csv with db tables or even directly from columns provided from csv that were already in structured format. For the features provided directly from csv no particular comments for the source of extraction are noted below. Features, used directly from csv, were compared with corresponding db columns of same feature, to fill any null values, or check any possible differences. Features extracted mainly from db in cases where csv did not provide them in structured format or did not provide them at all. Were possible, data was preferred to be used directly from csv. All transformations or new columns and data frames (df) created were conducted or merged on initial data frame of csv.
Some features were only used for the basic analysis of 'Completed', 'Terminated' outcome, while others were only used for the 'Withdrawn' analysis. Some of the features were completely dropped for the 'Withdrawn' analysis, such as Enrollment, which is always zero for these trials or adverse even counts, which could not yet occur as these trials did not initiate any intervention.

- **Placebo_Bin**, **Standard_Care_Bin**, **Healthy_Bin, Global** : ['Yes', 'No']

Binary features for placebo intervention, standard care intervention, healthy volunteers' acceptance and trials regarding covid accordingly. Global refers to trials being held to more than one country.

Extracted from db. Specifically, new db tables were created and then merged to csv as new columns:

my_placebo:

From db tables: design_groups (Columns: group_type)

Separated trials to binary of containing placebo and not. Evaluation of termination relation to placebo intervention.

my_soc:

From db tables: interventions (Columns: description), design_groups (Columns: description, group_type)

Separates trials to including standard care (SOC) treatment and not. Evaluation of termination to standard care.

my_covid:

From db tables: browse_conditions (Columns: mesh_term)

Separated trials to binary covid and non-covid. Evaluation of termination relation to covid.

my_eligibilities:

From db tables: eligibilities (Columns: gender, healthy_volunteers)

Containing features of gender and trials accepting healthy volunteers or not.

- **Funder_Industry_Bin**: [FUNDER_OTHER, INDUSTRY]

The initial levels were: INDIV, UNKNOWN, AMBIG, INDUSTRY, OTHER_GOV, FED, NIH, OTHER, NETWORK. Some of those levels had small sample size which produced sparse data in visualization and models. So, these levels were merged to greater ones. [17] This way no data were omitted. Also, industry funding that analysed was of the main interest level of this feature.

- **Allocation_Bin**: [RANDOMIZED, NON_RANDOMIZED]

Initial level NA_RANDOMIZED is referring to one arm trials, thus is not randomized and merged into corresponding level. [6].

Created from db data extraction. Specifically, new db tables were created and then merged to csv as new columns:

my_designs:

From db tables: designs (Columns: allocation, intervention_model, primary_purpose, masking, subject_masked, caregiver_masked, investigator_masked, outcomes_assessor_masked)

- **Age_List**: [CHILD, ADULT, OLDER_ADULT]
- **Sex_List**: [FEMALE, MALE].

the initial levels of [ALL, FEMALE, MALE] were merged to [‘MALE’, ‘FEMALE’]

- **Intervention_Type_List**: [DRUG, INTERV_OTHER, BIOLOGICAL, DEVICE, PROCEDURE, INTERV_UNSPES]

Extracted from db. Specifically, new db tables were created and then merged to csv as new columns:

my_interventions_types:

From db table: interventions (Columns: intervention_type)

Some of the initial levels such as [COMBINATION_PRODUCT, DIAGNOSTIC_TEST,

GENETIC, RADIATION, , BEHAVIORAL, DIETARY_SUPPLEMENT] were merged to conduct a new INTERV_UNSPES level, due to sparsity risk.[17]

- **Intervention_Route_List**: ['Injection', 'Oral', 'Surgical', 'Topical'].

Created by combining db with csv data. Specifically, new db tables were created and then merged to csv as new columns:

<u>my_intervention_methods:</u>

From db tables: interventions (Columns: intervention_type, description).

Then, several words were searched with regular expression, both in db and csv, to categorise the intervention method to the above four categories.

- **Intervention_Model_List**: [PARALLEL, SINGLE_GROUP, SEQUENCIAL-FACTORIAL-CROSSOVER].

Created from db data extraction. Specifically, new db tables were created and then merged to csv as new columns:

<u>my_designs:</u>

From db tables: designs (Columns: allocation, intervention_model, primary_purpose, masking, subject_masked, caregiver_masked, investigator_masked, outcomes_assessor_masked)

Some of the initial levels such as [SEQUENCIAL, FACTORIAL, CROSSOVER] were merged into one due to sparsity.

- **Conditions_Detail_List**: (used only for the 'Completed-Terminated' analysis)

Created by combining db with csv data. Specifically, new db tables were created:

<u>my_conditions:</u>

From db tables: mesh_headings (Columns: qualifier, heading), conditions (Columns: name)

Conditions' categories from <u>MeSH tree view</u> [101] was also used for this table. Two levels of conditions from MeSH tree view were candidates for analysis, the general (e.g., C = Diseases) and the more detailed (e.g., C4 = Neoplasms). Almost all the background studies refer to more detailed conditions when associating conditions with early termination. Thus, the more detailed level was chosen to be analysed. [2][7][3]

Similarly, text format csv 'Conditions' column was used for conditions extraction by regular expression search.

Finally, both db table and csv column were merged to one.

Levels of conditions with small sample size, where imputed into others due to sparsity risk. [17]

<u>MeSH Terminologies Tree View</u>

Mesh terminologies were used to create conditions features for Conditions categories. Terminologies and their codes were used in combination with corresponding tables of db. [101] ClinicalTrials.gov registry purposes the use of NLM's Medical Subject Heading (MeSH)-vocabulary terms, if possible, as the most appropriate.[6]

- **Adverse_List**: [Death, Serious, Other]

Refers to severity level of adverse events. Missing data were considered as no adverse events occurring.

Created by db data extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_adverse:</u>
From db tables: reported_event_totals (Columns: event_type, subjects_affected)

- **Masking_List**: [MASK_NONE, SINGLE, DOUBLE, TRIPLE, QUADRUPLE].

Created from db data extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_designs:</u>
From db tables: designs (Columns: allocation, intervention_model, primary_purpose, masking, subject_masked, caregiver_masked, investigator_masked, outcomes_assessor_masked)

- **Primary_Purpose_List:** [TREATMENT, PREVENTION, DIAGNOSTIC, PRIM_PURP_OTHER]

Created from db data extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_designs:</u>
From db tables: designs (Columns: allocation, intervention_model, primary_purpose, masking, subject_masked, caregiver_masked, investigator_masked, outcomes_assessor_masked)
Some of the initial levels of [DEVICE_FEASIBILITY, SCREENING, HEALTH_SERVICES_RESEARCH, BASIC_SCIENCE, SUPPORTIVE_CARE, ss INTERV_OTHER] were merged into new PRIM_PURP_OTHER level due to sparsity.

- **Founder_Counts_Log**:

Count of Sponsors and Collaborators.

- **Adverse_Counts_Log**: (used only for the 'Completed-Terminated' analysis)

Number of adverse events. This feature was excluded for withdrawn analysis, as no adverse occurred to these trials which did not manage to initiate any intervention and thus adverse events.
Created by db data extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_adverse:</u>
From db tables: reported_event_totals (Columns: event_type, subjects_affected)

- **Arm_Counts_Log**

Created by db data extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_studies:</u>
From db tables: studies (Columns: nct_id, brief_title, official_title, source, phase, number_of_arms, enrolment). My_studies table was used for comparison to csv, number of arm groups, titles etc.

- **City_Counts_Log**

Created by database extraction. Specifically, new db tables were created and then merged to csv as new columns:
<u>my_locations:</u>
From tables: countries (Columns: name), facilities (Columns: country, city)

Produced table of facilities sites locations and number of them. Geonamescache library was used for this table too. Facilities information are available at ClinicalTrials.gov [6]
Countries_List and Continents_List columns were also defined only for descriptive analysis.

- **Enrollment_Counts_Log**: (used only for the 'Completed-Terminated' analysis)

Regarding number of participants.
Enrollment ClinicalTrials.gov definition: The estimated total number of participants to be enrolled (target number) or the actual total number of participants that are enrolled in the clinical study. [6] Meaning that 'Enrollment' column as displayed in csv/db referring to the total number or participants for all arm groups.
This feature was excluded for withdrawn analysis, as no participants were managed to be enrolled, thus, the values are always zero for withdrawn studies.

6.4.3.3.1. Interaction terms

- **DRUG_x_Funder_Industry_Bin** (used only for the 'Completed-Terminated' analysis)

Created as the product of Intervention_type and Funder_Industry_Bin features. Some of the initial levels were dropped as they were in risk of sparsity. Also, as mentioned in literature, funder type is associated with trial termination, especially if industry (most frequent of dataset are pharmaceutical industries). So, DRUG_x_Funder was in the greatest interest to be studied rather than the dropped levels. This feature was dropped in withdrawn analysis because the withdrawn class was too rare to support a stable interaction term.

6.4.3.3.2. Predictors not Used

Some of the following predictors were provided in structured format from ClinicalTrials.gov, while others were created from db or other features. However, some of them were checked for analysis (e.g., Enrollment_Log_x_City_Counts_Log)

- **Enrollment_Log_x_City_Counts_Log:**

Created as the product of the corresponding features.

- Facilities count is related to enrolment rates. [2] [21][7] [14] However, this feature was not used, as it appeared high VIF value, due to presence of Enrollment_Counts_Log feature. Although this feature was not used, the analysis could be implemented with that in Thesis_Models.ipynb file if needed.
- **Enrollment_Log_x_Conditions_Detail_List:** (used only for the 'Completed-Terminated' analysis)

Created as the product of the corresponding features. Rarer conditions or more severe ones like cancer tend to appear lower accrual rates and thus, earlier termination. [20][21][1][7][3][22]. Although this feature was not used, the analysis could be implemented with that in Thesis_Models.ipynb file if needed.

- **nct_id, Other IDs, Acronym, Study URL:**

Their purpose is to be unique identifiers of trials' records, for registry organization reasons. They are given from administrators or in case of nct_id from the public registry.

- **Study Title, official_title, Brief Summary, Conditions:**

They are text data columns. These were used only for filling missing values with regular expression. Especially, 'Brief Summary'.

- **Documents:**

From db tables: provided_documents (Columns: has_protocol, has_icf, has_sap) Documents that are uploaded to ClinicalTrials.gov are the [Analysis Plan, Protocol, Consent Form] etc. Unfortunately, not all trials fill this field properly and there even were up to 2/3 of the dataset missing values, so this feature was not used. This field was made mandatory to fill only after 2017 (***§**). As those documents (especially protocol) are necessary for the study design, the null values appearing were indeed neglected to be filled.

- **Outcomes (Primary Outcome Measures, Secondary Outcome Measures, Other Outcome Measures)**: [PRIMARY, SECONDARY, OTHER, POST_HOC]

From db tables: outcomes (Columns: outcome_type). Outcomes do not conclude a pre-designed feature (except the main outcomes a trial is designing to produce). So, it could not indicate a prognostic factor of termination. Also, this column has so many missing values that could not be used.

- **Covid_19_Bin:**

The conditions was already displayed as one of the Conditions levels (i.e., Respiratory diseases)

- **Study Results:**

They are not predesigned trial's characteristics but occurred after the completion. So, they cannot be part of predictor variables, or else they could lead to 'Data Snooping Bias'. [17]

- **Adverse_Detail_List:**

Categorical feature regarding organ system that adverse was located. Created by database table.
<u>my_adverse_system:</u>
From db tables: reported_events (Columns: organ_system)
Produced table regarding organs affected from adverse events e.g., [General, Nervous System, Skin]. This feature was not used, as other features like Adverse_List, were correlated by providing similar information.

- **Start Date, Completion Date, Primary Completion Date, First Posted, Results First Posted, Last Update Posted, Completion_Gap_Counts (Start-Completion Date)**

Dates were not used for analysis, but only for descriptive statistics and plots. This because, they are not casual factor, but events happened during these periods are e.g., for year 2019 when covid studies started, covid trials were analysed. Especially, Completion date, could lead to 'Snooping Bias'.
Datetimes of appeared into two formats: YYYY-MM-DD or YYYY-MM. All datetime data were converted to format : YYYY-MM.

- **Country_Counts**, **Continent_Counts, Cities_List, Countries_List, Continents_List:**

These features were created and applied exploratory analysis but not used for model fit. This is due to Cities_Counts_Log was giving similar information regarding the multi-facility or even international studies and was chosen instead. However, they were also, used in combination with Geography libraries fo missing values in the chosen location columns.

- **Masking_Detail_List**: [CARE_PROVIDER, INVESTIGATOR, PARTICIPANT, OUTCOMES_ASSESSOR]. This feature was not used, as other features like Masking_List were correlated with this one, by providing similar information.

6.4.4. Data Frames (df) – Phase Split

(Thesis_Data.ipynb)

Phases_List

Only, Phases 1, 2, 3, 4 were selected. They are most associated to the purpose of current study, as they include human participants, requiring great funding in later phases etc. Thus, early Phase 1, Phase N/A were excluded. The special case of combined Phases 1/2 and 2/3 were also excluded, as they have characteristics from both stages.

- <u>Phase N/A or NA</u>

Trials without phases (e.g., regarding devices, behavioural etc.).[6] Those Phases were excluded, as they are out of the study's scope.

- <u>Early Phase 1 - Phase 0</u>

They mostly do not include human experiments and thus, they are beyond of the study's scope.[6]

- <u>Phase 1/2 and Phase 2/3</u>

They are a special case of phase, designed as a combination of both stages. They were excluded because, as a hybrid stage, their characteristics represent a unique combination of both phases.[6] Including them would have distorted the analysis of each phase by duplicating overlapping features.

df1, df2, df3, df4 : Finally, four data frames were produced from the initial one (df), based on phases, and analysed separately. Specifically, the initial df that was produced from db and csv combination, was separated into 4 other dfi based on its 'Phases_List' column, by including only the corresponding phase. Each data frame's name was based on the phase (e.g., df1 for phase 1). This way Phases were analysed separately for the possibility of termination. [14] In most part of the code a phase's df is presented as dfi for convenience. Each time a specific phase/dfi is needed for analysis, dfi parameter was assigned to that specific df of phase (e.g., dfi = df1).

6.5. Exploratory Analysis (Full Dataset)

6.5.1. Visualization & Descriptives

(Thesis_Visual.ipynb file)
Coding Note:
All procedures were implemented through creation of functions.
For achieving analysis per phase of trial, a dfi input cell was placed on top of the file, so one can input the data frame of the phases needed for analysis (full or train datasets).
In this section univariable analysis was implemented. In more detail, logistic univariable analysis, plots and descriptive statistics were applied per dfi/phase.
It must be noted that no decision for feature selection was not done through whole dataset's exploratory analysis, to avoid information leakage.[102][17][103]
The analysis was done as explained below:

- **Binary features**

2x2 cross table, chi-square test (a = 0.05), logistic univariable model and bar plot are displayed.

- **Multilevel categorical features**

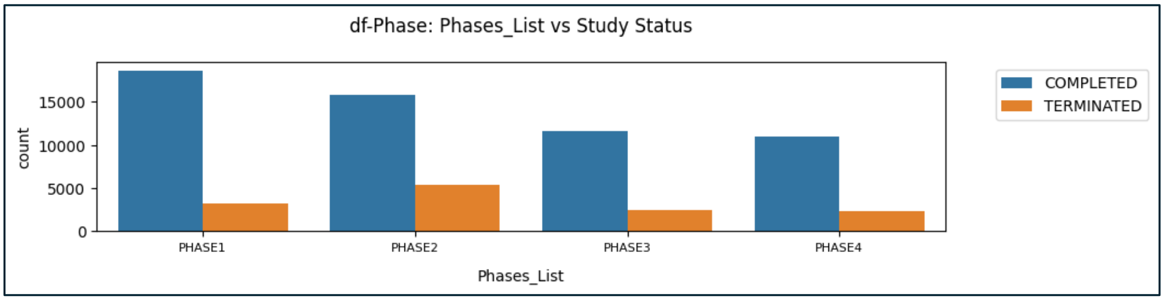
Pivot table, univariable logistic model and bar plot for visualization were displayed.

- **Continuous features**

Mann-Whitney t tests (a=0.05) were applied due to skewed data and logistic univariable model was applied as well. Plotting was done with through boxplots and kde-plots. The persistence in visualizing continuous features was done because real world health data are highly skewed, so distributions visualization helped choosing a log scale format for continuous features. It was observed that logarithm transformation of continuous features was more appropriate for limiting outliers influence and skewness (e.g., in Enrollment). More detailed, almost all continuous features were analysed in their initial, logarithmic and binarized into two ranges format. Continuous features in log scale appeared a much better distribution as far as it concerns skewness and outliers, while their categorical form was giving highly imbalanced samples per outcome level. The distribution was taken into consideration for the scale chosen, as log scaled feature helps in skewness reduction and thus the distance-based models (KNN, SVC etc) [17][58][59][52] and ensures no feature outperforms another due to scale.
From univariable logistic models, a table of p values and Confidence Intervals was also produced for all types of features but were not taken into consideration from any feature selection as they regard full datasets.
As mentioned above, all decisions for feature selection, formats or scales were not done while executing full datasets, in order to avoid data leakage. The purpose of full dataset analysis was simply for descriptive- exploratory reasons per Phase, regarding population's distributions and skewness.

It must be noted that any AIC tests, displayed in this file were only taken into consideration for train sets analysis.
Lastly, combined data frames for all phases, regarding descriptives such as frequency tables and continuous features distribution quantile tables, applied on were displayed for full dataset in Thesis_Metrics.ipynb file.

Phases_List	# COMPLETED	# TERMINATED
PHASE1	18646	3183
PHASE2	15804	5424
PHASE3	11557	2409
PHASE4	10937	2309



A pivot table of whole dataset (of all phases and records) regarding the reasons for early termination, as produced with RegEx text mining from database data:

Why_Stopped_Full_Df	# PHASE1	# PHASE2	# PHASE3	# PHASE4
Enrollment	1331	2950	1127	1479
Funding	1037	1288	574	356
Administration	384	514	198	230
Efficacy	258	455	268	50
Covid	35	91	30	71

6.5.2. Encoding

(Thesis_Dummies.ipynb file)
Encoding Selected Features, with Merged levels of categorical features.
In this section, suffixes of features' names helped to dummy and binary encoding through functions. Generally, list element data were transformed into dummies, binary data were encoded with mapping (0,1) and lastly continuous data were kept as is.

- **'_Bin'**

Binary data were encoded simple by Label Encoding with 2 levels with integer labeling displaying the presence/absence or else the yes/no levels (i.e., Integer encoding).

- **'_List'**

List element data, were One-Hot Encoded through dummy creation, as they cannot be used directly from models. List-element rows were exploded to one row per element of list for each record. However, this inflates sample size, as rows increase and duplicate for same nct_id record if more than one element occurs in a record's list-row. For this reason, these data were then grouped by nct_id, so number of rows remained the same as initial datasets (df1, df2, df3, df4). The levels that were dropped as 'Drop First' were mostly the general ones such as 'other', 'none', 'unspecified' etc.

- **'_Log'**

Continuous data were used as is and scaled. All Continuous features were chosen on a logarithmic scale.

It must be noted than in Thesis_Dummies.ipynb file Interaction term features were also created and encoded (e.g., Enrollment x Condition, RUG x Funder Industry).

Finally, any feature selection made was implemented by dropping corresponding features in this file.

6.6. Exploratory Analysis (Train Set)

Every analysis in this section was applied only in train set. Any feature selection choice was based on train set, to avoid information leakage. [102][17][103]

6.6.1. Data Engineering

(Thesis_Data_Unmerged.ipynb file)
Practically, it is a copy of the Thesis_Dummies.ipynb file. The difference is that the levels of categorical features remain unmerged.
This was done to provide raw data to below files, for exploratory analysis and feature selection purposes. These analyses were applied after selecting only train set's data, identical to those used for model fitting as explained below.

6.6.2. Visualization & Descriptives

(Thesis_Visual.ipynb file)
<u>Coding Note</u>:
All procedures were implemented through creation of functions.
For achieving analysis per phase of trial, a dfi input cell was placed on top of the file, so one can input the dfi of (full or train datasets) corresponding to the phase needed for analysis.
The process is similar to the one done in full dataset, with the difference that train sets were chosen to be executed. More detailed, in this section logistic univariable analysis, plots and descriptive statistics were applied per dfi/phase. The analysis was made on same train set data as used for model fit. However, no random under

sampling was applied here. The use of same data as the ones trained the models was achieved through same random_state of train_test_split() function.
From the train set oriented file, decisions regarding feature scales were based on their distributions shape and outliers' influence. It must be noted that no decision for feature selection was done through whole dataset's exploratory analysis, to avoid information leakage.[102][17][103] Any decision for feature selection or comparing metrics like AIC are considering train sets only.
The analysis was done as explained below:

- **Binary features**

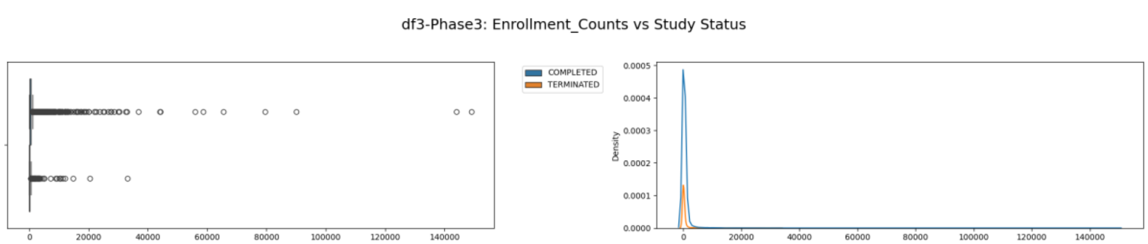
2x2 cross table, a chi-square test (a = 0.05), logistic univariable model and bar plot are displayed.

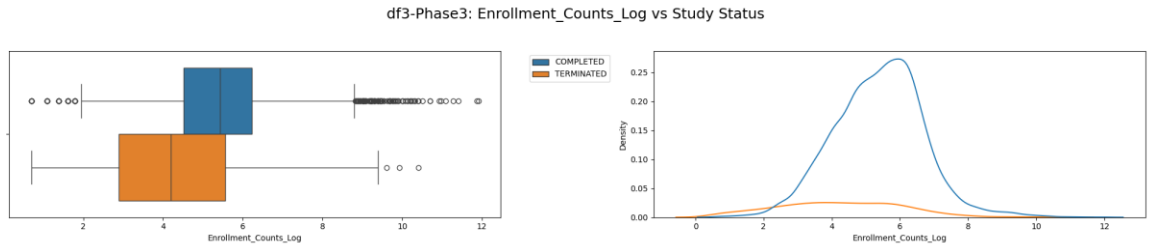
- **Multilevel categorical features**

Pivot table, univariable logistic model and bar plot for visualization were displayed.

- **Continuous features**

Mann-Whitney t tests (a=0.05) were applied due to skewed data and logistic univariable model was applied as well. Plotting was done with through, boxplots and kde-plots. The persistence in visualizing continuous features was done because real world health data are highly skewed. Considering their distribution through plotting, it was decided that logarithm transformation of continuous features was more appropriate for limiting outliers influence and skewness (e.g., in Enrollment). More specifically, choices for features format were done in train set visual file. The choice of features' formats was applied mostly on continuous features and was based on distribution skewness and sparsity risk. Almost all continuous features were analysed in their initial, logarithmic and binarized into two ranges format. Continuous features in log scale appeared a much better distribution as far as it concerns skewness and outliers, while their categorical form was giving highly imbalanced samples per outcome level even before any train test split. Thus, it was decided to use only logarithmic scaled data (log1p to avoid inf results), while no categorical formats were used due to sparsity induction, and no initial scale data were used as they were highly skewed. The distribution was taken into consideration for the scale chosen, as log scaled feature helps in skewness reduction and thus the distance-based models (KNN, SVC etc) [17][58][59][52] and ensures no feature outperforms another due to scale.
Below figure shows example of distribution for Enrollment feature for Train set of Phase 3:





From univariable logistic models, a table of p values and Confidence Intervals was also produced for all types of features.
In this step, the first correction for sparse data was made. Specifically, through crosstabs, pivot tables, and failure of logistic univariable models to convert for some highly risk sparse data. However, in this file, no random under sampling was applied, thus the sparsity analysis was done in another file as explained below. These data were merged into other levels of categorical features or more general ones like 'Other', 'None' etc. in order not to drop useful information due to rareness. (Levels were merged directly to Thesis_Data file).
Additionally, AIC tests were applied for similar information features or different scales of same feature. However, due to AIC test applied only on logistic model, while this study compares various other algorithms, AIC was not the only criteria for feature elimination, as this would induce bias towards classic statistical models. The combination of AIC tests (Thesis_Sparse.ipynb and Thesis_Visual.ipynb), and mostly the interest in format/scale of features providing the best distribution were the criteria. For example, continuous features, that their bar plots for two outcome categories did not overlap, would probably induce sparse data, if sample too small in one outcome category. Also, regarding the feature format, the magnitude was considered as well. For example, one may be more interested in analyzing how the termination is associated with the increased number of adverse events, rather than the existence of adverse events.
Finally, pivots with Chi2 for binary and Mann-Whitney p values were created and provided to Thesis_Metrics.ipynb file.
After this exploratory analysis on train sets, the excluded features were dropped back into this file on full datasets data frames, as those were used to provide data to next pipeline files leading to final models' fit file.
Lastly, as mentioned above, combined data frames for each phase, regarding descriptives such as frequency pivots, univariable analysis p values and Confidence Intervals, as well as Chi-square and Mann Whitney test tables were displayed for train data in Thesis_Metrics.ipynb file.

Outliers

In the same file outliers were also displayed as a simple scatterplot of y_i values of train set. Some of the data points seemed to have greater y_i values compared to the rest of dataset. Such outlier points were noticed in continuous features such as

Enrollment, Adverse, Arms, Locations etc. However, the log scaling and MinMax scaling of continuous features decreased their effect. The analysis was tested without those data points, however, no particular changes were noticed in models output (e.g., coefficients), metrics or performance, so it was decided to be left as they are, as they include real world data entries.

6.6.3. Encoding

(Thesis_Dummies_Unmerged.ipynb file) This file is actually a copy of Thesis_Dummies.ipynb with the difference that levels of categorical features remained unmerged and no interaction terms were produced.
This file was used as a source for other files such as the ones regarding sparsity and collinearity risk as explained below.

In this section, suffixes of features' names helped to dummy and binary encoding through functions. Generally, list element data were transformed into dummies, binary data were encoded with mapping (0,1) and lastly continuous data were kept as is.

- **'_Bin'**

Binary data were encoded simple by Label Encoding with 2 levels with integer labeling displaying the presence/absence or else the yes/no levels (i.e., Integer encoding).

- **'_List'**

List element data, were One-Hot Encoded through dummy creation, as they cannot be used directly from models. List-element rows were exploded to one row per element of list for each record. However, this inflates sample size, as rows increase and duplicate for same nct_id record if more than one element occurs in a record's list-row. For this reason, these data were then grouped by nct_id, so number of rows remained the same as initial datasets (df1, df2, df3, df4). The levels that were dropped as 'Drop First' were mostly the general ones such as 'other', 'none', 'unspecified' etc.

- **'_Log'**

Continuous data were used as is and scaled. All Continuous features were chosen on a logarithmic scale.

It must be noted than in Thesis_Dummies_Unmerged.ipynb file Interaction term features were not created or encoded (e.g., Enrollment with Condition).

6.6.4. Sparsity

(Thesis_Sparse.ipynb)
Exploratory analysis for sparse risk was applied per dfi/phase on the same data as actual training sets.

Sparse data analysis was done mostly for categorical features which produce zero cell counts for one of two binary outcomes. This leads to infinite coefficients and models cannot complete the fit or report biases. [31] Thus, without addressing this issue, many models like logistic, would not convert appropriately their coefficients. The analysis was applied only on train set to avoid information leakage and to indicate which levels of categorical features were creating sparse data on the actual data set used for model fit. [17][31]
Sparsity risk data were analyzed initially by train test splitting the data frames of each phase. For this purpose, the same random state as in model's fit was used, to produce the same train data points when splitting each dfi into train and test sets [103]. Secondly, a random under sampling was applied on train set, similarly by using the same random_state as the one used in model's fit.
Pivot tables regarding the frequencies per level of a categorical feature and per outcome were displayed.
The levels having less than 25 data points per cell in the smallest outcome category (EPV) were merged to other relevant levels of same feature or combined into more general levels like 'Other', 'None' etc. Relevant levels were identified from the correlation heatmaps analysis, as explained below.
This way no rows were dropped, thus no other valuable features information was lost, while sample size was kept the same and efficiently large. [17] [31]
Furthermore, more features, weaker features, and more complex relationships between the predictors and the response increase expected sample sizes, thus it was preferred for some levels to be merged. [66] Thus, dimensional reduction and merging weaker levels were achieved.
Any merge of levels was applied directly back to Thesis_Data.ipynb file, as this was the one providing data to the pipeline files leading to final models fit file.
The decision of sample size per cell was made based on EPV rule [67][32] with respect to feature's levels of interest. More specifically, EPV = 25 value was decided to be used based on logistic literature. However, logistic regression requires less sample size than ML algorithms, so no less than 25 EPV was used, as the sample per cell would be inefficient for ML models and also because not all levels could be merged for a larger EPV. It must be noted that EPV rule was mostly used for parametric models, such as logistic, as ML models must be evaluated through more empirical ways like cross validation, AUC etc.[66] Also, some ML algorithms such as boosting models, can handle sparse data. [85][84]

6.6.5. Collinearity

(Thesis_Collinearity.ipynb file)
The analysis was applied on the same train set as the one used for models' fit. This was achieved by using the same random_state value in train test split as this used in models' fit. Also, randomly under sample was not used unlike to sparse file above.
Collinearity was studied mostly for features giving similar information. The analysis indicated that different forms of similar features were highly correlated as expected.

For example, different formats for locations of sites such as Cities, Countries or Continents all indicated the multisite trials or even the global character of them. Also, the different formats of Adverse events, with severity feature and the organ system were the adverse detected (features of 'Adverse_List' with 'Adverse_System_List'). Masking type and Masking individual was another example. This file's results were considered in combination with Thesis_Visual_Train.ipynb for feature selection and scale selection due to distribution and collinearity and also for some levels to be merged based on their relevance with other levels if sparsity risk was high.
More specifically the process followed:

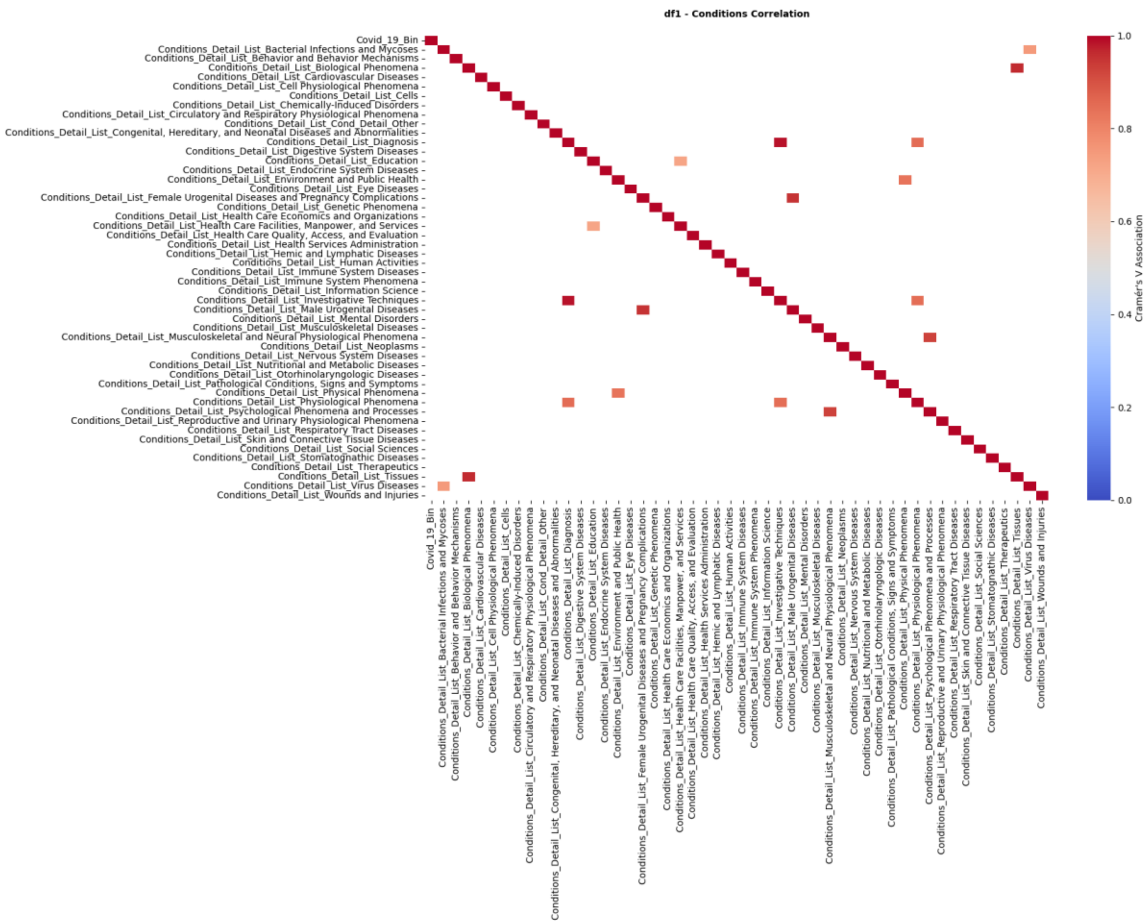
- **Continuous_x_Continuous Heatmap**

For Continuous features a normality test based on D'Agostino and Pearson's was first applied, to evaluate the method of collinearity test. Even, in log scale Continuous features were not approximating normal distribution, so Spearman correlation test was applied for Continuous with Continuous variables correlation heatmap [44].
The variables correlation was occurring in locations counts, where the count features of Cities, Countries and Continents facilities location were candidates for analyses. Finally, only Cities number of facilities were used, as it includes the most information of all others regarding facilities number, and indirectly global occurrence of studies.

- **Categorical_x_Categorical Heatmap**

Chi-square test with Cramer's V effect size were used. Examples of variables that seemed to be correlated were Adverse events severity vs Adverse of organ systems, masking type vs masking individual (e.g., participant, investigator etc.).
In the first feature, the severity was chosen because it was of greater interest for the purpose of this study and due to the Adverse_System feature increases the dimensionality more. Regarding masking, similarly, masking type (e.g., single, double etc.) was of greater interest and it was the one kept.

Also, categorical features’ levels swere analyzed for any within levels correlation, for those levels to be merged into one. Similarly, other variables were analyzed such as conditions, where correlation of their levels helped to merge those with sparsity risk as seen in the picture below for Phase 1 train set:



- **Continuous_x_Categorical Heatmap**

For continuous with categorical features one way anova test with eta-squared (η^2)σ effect size were used. The heatmap indicated, mostly features correlated due to similar information provides. For example, Adverse events with Adverse counts feature etc.

Finally, AIC tests were applied for similar information features or different scales of same feature. However, due to AIC test applied only on logistic model, while this study compares various other algorithms, AIC was not the only criteria for feature elimination, as this would induce bias towards classic statistical models. The combination of AIC tests (Thesis_Sparse.ipynb and Thesis_Visual_Train.ipynb), and mostly the interest and format/scale of features providing the best distribution (Thesis_Visual.ipynb) was the criteria.

After this exploratory analysis on train sets, the excluded features were dropped back into Thesis_Visual.ipynb file. This is due to this file is part of the pipeline leading to final models' fit file.

6.7. Model Pre-Processing

(Thesis_Models.ipynb)
Coding Note:
All procedures were implemented through creation of functions.
For achieving analysis per phase of trial, a dfi input cell was placed on top of the file, so one can input the dfi corresponding to phase needed for analysis.

In this section, each phase was analyzed for early termination probability based on its characteristics.
The data frames for each phase, after all above steps and before train test split of VIF feature elimination had the following shapes:
df1: (21012, 73)
df2: (19747, 73)
df3: (13317, 73)
df4: (12420, 73)

6.7.1. Train-Test Split

Initially, each dataset was split into train and test sets. The train-test set proportion was decided based on background studies that proposed 70:30 [14]. However, the high imbalance, the shape of current dfis and the random underdamping applied, which minimized sample sizes a lot, lead to finally deciding a test/train ratio = 80:20 for 'Complete'-'Terminated' analysis.

6.7.2. Random Under Sampling

Secondly, only the train set was Randomly Under Sampled, due to great imbalance towards 'Completed' outcome, which would produce bias towards this majority class[17]. The final ratio of binary outcome was 1:1. Test set was maintained as is, as it is supposed to represent the true distribution of data [17].
Random Under sampling was chosen as the simplest method [69]. In contrast, random over-sampling, which was not used, includes resampling methods to create greater 'termination' category (e.g., resampling minority class). This decision was made based on literature background research [7] [14] and because oversample extremely increased the computational time. Also, random oversampling induces bias by creating copies of records, especially in such imbalance and large dataset. Randomness ensures the quality of the sampling and replicability of studies [17]. In under-sampling methods, train set is minimized, as many rows are dropped from

majority category. Combining the great sample sizes of data frames and the appropriate train-test split analogy per data frame, train samples remain efficient, with predictors by far less in size ($p \ll n$).

6.7.3. Scaling

Scaling with Standardization method was used on train and test set. More specifically, Min-max Normalization was used [59][61]:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \left(\max(x_{\text{new}}) - \min(x_{\text{new}})\right) + \min(x_{\text{new}})$$

The decision was made due to dataset did not logically support negative values that standard scaler induces.

6.7.4. Collinearity

For collinearity reduction VIF ≥ 10 was used based on literature. [47] After, the eliminations of high correlated features only 54 were left. The same features were excluded for all phases, due to interpretability reasons.
Below figure displays features excluded provided similar information, as seen below for phase 1 train set:

# df1	Feature	# VIF
72	Enrollment Count x City Count	19.56423384030251
66	Enrollment Count x Phenomena Processes	15.669197836515105
35	Conditions Phenomena Processes	15.281293571179315
67	Enrollment Count x Psychiatry Psychology Anthropology Sociology	14.281478575625236
10	City Count	14.233827630698036
36	Conditions Psychiatry Psychology Anthropology Sociology	13.867346895221408
58	Enrollment Count x Endocrine System	13.754224712959173
27	Conditions Endocrine System	13.265662502762833
63	Enrollment Count x Neoplasms	12.63013906325982
55	Enrollment Count x Bacterial Mycoses Virus	12.54532715154548
62	Enrollment Count x Musculoskeletal Neural	12.2530889620802
57	Enrollment Count x Digestive System Nutritional Metabolic	12.15056133754928
31	Conditions Musculoskeletal Neural	12.01611276449148
32	Conditions Neoplasms	11.85533011096292
24	Conditions Bacterial Mycoses Virus	11.847919727096984
39	Conditions Stomatognathic Otorhinolaryngologic	11.538587843302377
70	Enrollment Count x Stomatognathic Otorhinolaryngologic	11.509199644327923
26	Conditions Digestive System Nutritional Metabolic	11.363503799767253
68	Enrollment Count x Respiratory	10.546302695880769
64	Enrollment Count x Nervous System	10.08030602761456
65	Enrollment Count x Pathological Conditions Signs Symptoms	10.043691819402783
37	Conditions Respiratory	10.03242188847304
29	Conditions Hemic Lymphatic Immune System	10.020079911066729

6.8. Models – Features

In this section classic models were compared to ML algorithms. This way the real world clinical impact of ML outperformance is achieved. [17] A simple logistic regression could be compared to Random Forests and XGBoost, as done in previous research. Although, the comparison was chosen to be more complete and wider, including different types of models. Models that were not reviewed from literature appeared at the top performers in some phases. For example, SVC is a model not extensively studied, however it was a good classifier for some phases.

6.8.1. Compared Models

- **Classic Statistical Models**
  - Logistic Regression with Lasso (L1 penalty)
  - Linear Discriminant Analysis (LDA)
  - Quadratic Discriminant Analysis (QDA)

Note: LDA and QDA are considered ML models, however, they are based on matrices and classic statistics thus, they were recorded with classic models for this study.

- **Machine Learning Models**
  - K-Nearest Neighbors (KNN)
  - Support Vector Classifier (SVC)
  - Random Forest
  - Gradient Boosting
  - Extreme Gradient Boosting (XGBoost)
  - LightBoost

6.8.2. Hyperparameter Tunning (HPO)

Parameter tuning for models was managed through GridSearchCV through k-fold cross validation with k=5 folds. [104][105][106]

6.8.3. Evaluation Metrics

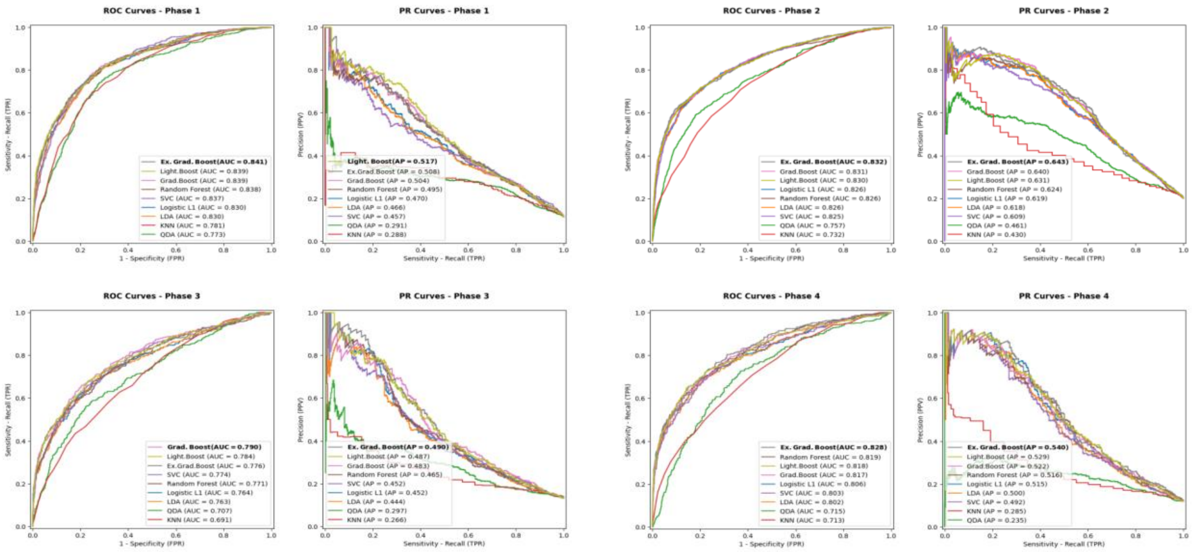
The final model evaluation of each model was done on unseen data of test set. Classification Report Metrics used were the accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), Balanced Accuracy (BA). More specifically, F1-score, MCC and BA scores were basically taken into consideration from all the scores of the classification reports matrices. This is because the test datasets were imbalanced, and thus scores like accuracy are highly biased [31][88][57][58][86][89][90].

Plots like ROC and PR curves were also displayed. For the same reason as in metrics above, PR Curves were used as they are more informative when it comes to imbalanced datasets than ROC Curves. [93] [95] [17] [89] Indeed, the results of current study indicated that a model could display worse performance in PR curve than in ROC.

Furthermore, predictors contribution for each model, was evaluated through SHAP plots.[14] [96] Thus, achieved to see the contribution of each top predictor feature in all algorithms, especially for those lacking in interpretability like most ML models. Finally, logistic model was also implemented for the to 8 top features of the best performing model for interpretability of results. This was done as SHAP plots provide only the features' contribution to the outcome based on the model and thus its mechanism of fitting the data, but does not provide the statistical inference such as significance etc.

6.8.4. Selected Model

Model selection was mostly based on PR Curves and MCC, as they are considered more accurate for imbalanced datasets. [107][89][91]
Results indicated that Boosting models, such as XGBoost and LightBoost outperformed at all phases.
More specifically, in phases 2,3,4, XGBoost appeared as the top performing model in PR Curves, while LightBoost only in Phase 1 (LightBoost is well performing for large datasets such as Phase's 1 [82]).
Regarding MCC metric, LightBoost appeared as the top performing model in phases 3,4, while XGBoost only in Phase 1. Random Forest appeared as top perming model in MCC in phase 2 as well.
The differences between top models are small, even to the third decimal place for the above measurements. However, Boosting algorithms, were always at top 3 in PR Curves, and top 1-2 for ROC Curves. Random Forest as another ensemble method consistently ranked 4th in PR Curves. Ensemble tree methods in whole had greater difference than the other models.
Model performance is in perfect alignment with background studies'. Additionally, here GBoost and LightBoost were also examined, while in literature mostly XGBoost, Random Forests and Logistic are compared, with Random Forests and XGBoost the most frequently compared. Here RF seemed to induce more unbalanced metrices to as seen for example in ROC Curves, where it held 2nd – 5th place.
Moreover, Boosting methods tend to be consistently in top performing algorithms in any of metrices used, when alternative dataset used (e.g., different scaler, Grid Search, merged levels of data, different dimension data etc). As mentioned before, Boosting Trees are also very useful when it comes to unbalanced datasets. However, they include limited tunning parameters that makes them more prone to overfitting compared to other ensemble methods (e.g., RF) [52].
It must be noted that, the more data engineering or normalization/standardization methods applied, the better classical static models perform and models' performance gap decreases. This indicates that Boosting algorithms re better for real world or low precision data, as the dataset used in this study, with missing, false or optionally input values. [11][10][16][15] [6] [19]
All the above indicates, ML algorithms like XGBoost may be more useful in clinical practice, as they can handle missing, high dimensional data, in variety of methods, which makes them more generalizable [52] and easier when statistical analysis is not able to be done in clinical practice when a health decision must be supported.
Low predictive value models were always KNN and QDA with great difference from the other algorithms. It is also notable that SVC was extremely computationally expensive, taking a long time to be completely executed. In combination with low predictive performance, SVC is not appropriate for this analysis. Thus, distance-based models like KNN and SVC also did not perform as good classifiers.



# df1	Models	BACC	MCC	F1
0	**Ex_Grad_Boost**	**76.280%**	**0.368**	**0.809**
1	Grad_Boost	76.445%	0.367	0.804
2	LightGBM	75.808%	0.364	0.811
3	Logistic	76.343%	0.361	0.797
4	Random_Forest	75.608%	0.361	0.810
5	LDA	76.130%	0.358	0.795
6	SVC	75.797%	0.352	0.792
7	QDA	72.045%	0.303	0.785
8	KNN	71.821%	0.292	0.764

# df2	Models	BACC	MCC	F1
0	**Random_Forest**	**75.664%**	**0.457**	**0.808**
1	LightGBM	75.712%	0.451	0.802
2	Grad_Boost	75.331%	0.450	0.805
3	Ex_Grad_Boost	75.491%	0.446	0.799
4	SVC	75.254%	0.438	0.793
5	LDA	75.129%	0.428	0.783
6	Logistic	75.098%	0.425	0.779
7	QDA	67.650%	0.283	0.663
8	KNN	66.487%	0.270	0.699

# df3	Models	BACC	MCC	F1
0	**LightGBM**	**72.070%**	**0.331**	**0.792**
1	Grad_Boost	71.819%	0.321	0.780
2	Random_Forest	69.789%	0.305	0.795
3	Ex_Grad_Boost	70.093%	0.296	0.775
4	LDA	70.435%	0.290	0.750
5	Logistic	70.185%	0.286	0.748
6	SVC	70.129%	0.286	0.749
7	QDA	64.092%	0.192	0.666
8	KNN	63.204%	0.183	0.694

# df4	Models	BACC	MCC	F1
0	**LightGBM**	**74.872%**	**0.391**	**0.840**
1	Random_Forest	74.066%	0.389	0.846
2	Ex_Grad_Boost	74.720%	0.384	0.836
3	Grad_Boost	73.719%	0.368	0.832
4	SVC	73.060%	0.342	0.812
5	Logistic	73.717%	0.339	0.797
6	LDA	73.532%	0.339	0.800
7	QDA	67.328%	0.231	0.718
8	KNN	64.360%	0.195	0.730

6.8.5. Features Contribution

Features' contribution interpretation was managed through SHAP plots and logistic regression for the top 8 features of the best performing model. The logistic analysis of top 8 features was decided, based on SHAP plots' pattern and EPV. As seen below feature importances are related to each phases objectives, however, most features, appear in more than one phases. In contrast, below the 8^{th} SHAP position, feature patterns seem random, while maintaining very low ϕ values. Adding more features to logistic would just increase the noise. Secondary, EPV = 25 is decided for logistic regression throughout the study. Thus, for EPV = 25, and phases sample size of y test for outcome 'Terminated' , indicates that the number of variables that could be analyzed with logistic ranges between: (y test == 1)/EPV = 11 - 31. [108] Combining

these two considerations, the number of 8 was chosen to be the same for all phases, for interpretability reasons.
In more detail, SHAP plots helped in interpreting ML models, but could not give the other statistical metrics such as p values, confidence intervals etc.
It must be noted that the features were renamed into simpler forms (e.g., Enrollment_Counts_Log -> Enrollment Counts) for display reasons.
Below the icon (+) means positive contribution towards 'Termination' and (–) means negative contribution.
The features indicated as important contributors based the combination of SHAP and logistic interpretation were:

- **Enrollment Count (+)**

Top feature contributor for all phases, with the greater ϕ value given from SHAP plots. This is aligned with literature; were all papers refer to the number of participants as the main factor leading to a trials early termination. However, this feature needs more investigation, as there are other factors leading to low accrual rates. [7][23] [20][21][1][3][22] [14]

- **City Counts (+)**

Number of city sites is appearing to all phases. Greater number of facilities is related to greater probability of termination, especially the distance of participants to the facility. [2] [21][7] [14]
Here locations number has a positive impact on early termination. Due to that, this feature may be related to the complexity or scale of a trial rather than offering to participants more locations/distances choices.

- **Global (+)**

Similarly to City Counts. Appears as top contributor mostly in phase 3, which is large scaled, and for which design characteristics is very important. Thus, global feature (1 or more than 1 country included), indicates the complexity of a trials. It appears non-significant in logistic regression.

- **Condition Neoplasms (+)**

Appears as top contributor in early phases 1 and 2. This may be due to worst health state of patients, or greater caution in new interventions that may be proven insufficient.

- **Age_OLDER_ADULT (+)**

Seems to be a good predictor for phase 1 trials. The earlier phase and thus the one providing the newest treatment. This may again indicate hesitation when it comes to new, innovative treatments, regarding safety concerns of more vulnerable patients.

- **Healthy (-)**

In contrast to older adults, healthy volunteers seem to be an important negative contributor especially for phase 1. This may be due to the opposite reason compared to older aged participants.

- **Intervention Model PARALLEL (+) / CROSSOVER (-)**

Intervention model PARALLEL (different groups receiving different interventions in the same period) appears in phases 2 and 4. This may indicate the that participants hesitate when receiving unknown intervention, due to their wish of receiving specific type of treatment i.e., new treatment vs placebo. It is notable, that crossover model (groups receive all treatments), indicates the opposite results than parallel.

- **Allocation (+)**

Randomization seems to be a good contributor on phase 3, for which design characteristics are very important. A trial being randomized has positive effect on trials termination. This may indicate similar beliefs of participants as in PARALLEL model.

- **Placebo (+)**

The existence of placebo intervention with its positive impact n termination, may be indicating similar information to the other design characteristics (e.g., Allocation). Thus, the ignorance of intervention type received is a deterrent factor.

- **Arms (+)**

Appearing in phases 1 and 2. Greater number of arms is related to greater probability of termination, indicating similar reasoning with allocation and model parallel.

- **Adverse Counts (-)**

Number of adverse events even if appearing in SHAP plots is indicated as non-significant in logistic regression. This may be due to adverse being the result of other underlying factors i.e., interactions (2 predictors work together) or multicollinearity (predictors move together). VIF indicated Adverse events value above 5, which indicates weak such relations. [108]

An alternative explanation is a non-linear relationship of Adverse and Termination, which is weakly visible from SHAP plot, where adverse counts ϕ values overlap. In contrast, logistic regression assumes linear and monotonic (one direction) relationships. [108]

However, this feature's contribution seems to be related with study phase, as Adverse Counts appear in SHAP plots as top contributor in phases 2-3 in which adverse events are mostly investigated.

It is notable that Adverse events seem to have negative impact in SHAP plots, although positive in logistic regression. Any of counts or severity of adverse appear as important, indicating that adverse as a major part of phases' investigation do not constitute reason for termination, but rather an expected outcome.

- **Funder Industry (+)/(-)**

Funder type industry seems to have positive impact on termination compared to non-industry, which aligns with literature information a well. [7] However, logistic regression does not indicate funder type as significant.

The funder appearing in phase 1-2, indicates the importance of this phases results, for the next ones to be initiated. However, for phase 1 funder has negative impact while for phase 2 positive towards termination outcome.

- **Intervention Route Injection (-)**

Injection interventions seems to contribute positively to a study's completion in SHAP plots.

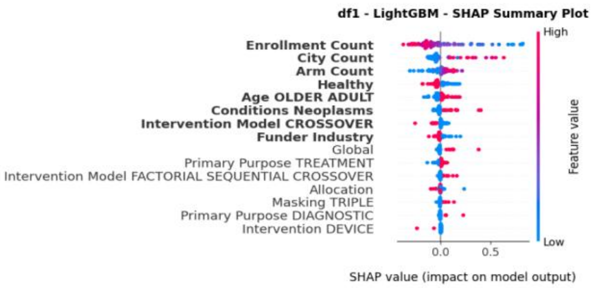
- **Conditions Respiratory (+)**

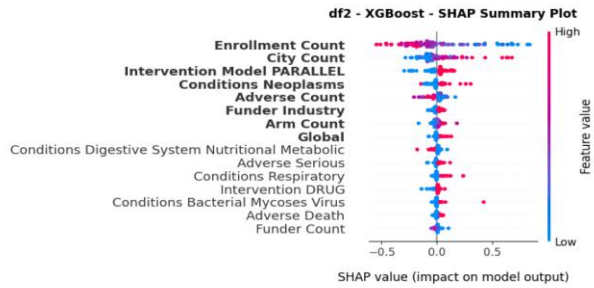
Respiratory conditions contributes positively to study's termination. This feature may be related to trials terminated due to covid, as seen in pivot table above of termination reasons above. In logistic this feature is not significant.

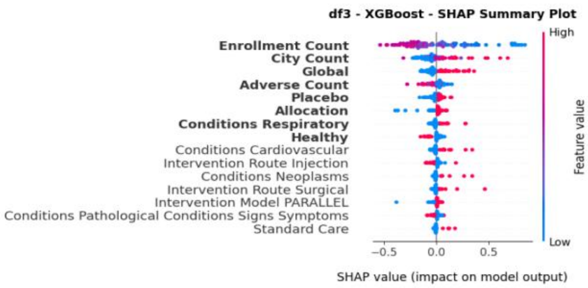
Result of this study is in alignment with background results of previous similar studies in most of the outcomes, as far as it concerns feature contribution.
As far as it concerns Enrollment and neoplasms, they are in full alignment with background studies. Finally, intervention model is not much mentioned in literature, as well as allocation method. Differences in feature contribution, is indicated to Funder type, and adverse events due to interpreting with SHAP and logistic.
From logistic regression above, Enrollment appears extremely high values of coefficient at all phases. This is due to sparsity effect of the continuous feature, where low values are mostly grouped to terminated outcome, while almost all high values to competed. Coefficients increase, tending to inf when the sparse effect occurs (great x values completely predict/separate the two outcome levels).
Similar happens to City Counts feature, with smaller coefficients, thus, weaker sparse effect. Other reasons, such as interactions may be affecting the coefficients as well. It is known from literature that other features are related to accrual rates as well, which may be inflating coefficients.
Dividing these continuous features to categorical (e.g., binary) could possibly fix the problem but is not a good practice as too much information regarding the variance of the relationship between feature and outcome is lost. [108] [62] [63] Thus, it was decided to be left as is, to highlight the problem.
Finally, features' importance seems to be related with phases. For example, design characteristics (e.g., allocation, placebo) appear in phase 3 , where study design is of high importance. Similarly, adverse events, appear as contributor after phase 2, and mostly for phases 2-3, when safety and adverse events are being investigated.
Additionally, health status of participants (e.g., healthy, older adults), seem to be related to phase 1 where a new, unknown is provided. The most vulnerable group of participants seem to lead to termination.
All these features, need to be evaluated as interaction with others, such as Enrollment, which seems to be the most affected from its logistic coefficient.

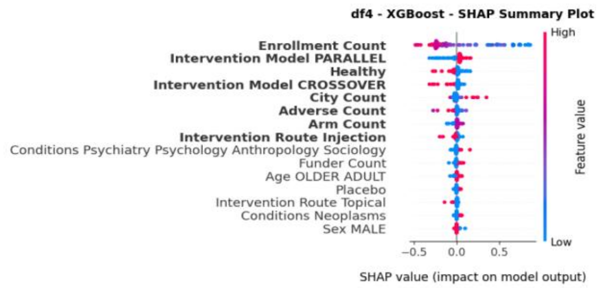
To sum up we could say that features related to phases like below:

- Phase 1: New treatment ~ Health status
- Phase 2: Safety, Pharmacokinetics ~ Adverse, Therapeutic, Sponsor
- Phase 3: Design, Treatment comparison ~ Design, Treatment characteristics
- Phase 4: Approved Treatment ~ Intervention characteristics









# df1	Feature	# df1_LightGBM_Shap_Mean
0	Enrollment Count	0.189
1	City Count	0.099
2	Arm Count	0.058
3	Healthy	0.055
4	Age OLDER ADULT	0.045
5	Conditions Neoplasms	0.031
6	Intervention Model CROSSOVER	0.026
7	Funder Industry	0.025

# df2	Feature	# df2_XGBoost_Shap_Mean
0	Enrollment Count	0.236
1	City Count	0.114
2	Intervention Model PARALLEL	0.056
3	Conditions Neoplasms	0.043
4	Adverse Count	0.039
5	Arm Count	0.021
6	Conditions Digestive System Nutritic	0.02
7	Funder Industry	0.02

# df3	Feature	# df3_XGBoost_Shap_Mean
0	Enrollment Count	0.261
1	City Count	0.109
2	Global	0.081
3	Adverse Count	0.057
4	Placebo	0.039
5	Allocation	0.036
6	Healthy	0.032
7	Conditions Respiratory	0.029

# df4	Feature	# df4_XGBoost_Shap_Mean
0	Enrollment Count	0.294
1	Intervention Model PARALLEL	0.063
2	Intervention Model CROSSOVER	0.035
3	Healthy	0.032
4	City Count	0.029
5	Arm Count	0.016
6	Adverse Count	0.016
7	Intervention Route Injection	0.011

	coef	std err	z	P>|z|	[0.025	0.975]
const	-0.0889	0.206	-0.432	0.666	-0.492	0.314
Enrollment Count	-8.4681	0.515	-16.452	0.000	-9.477	-7.459
City Count	2.1904	0.323	6.784	0.000	1.558	2.823
Arm Count	2.7608	0.353	7.816	0.000	2.069	3.453
Healthy	-0.4305	0.166	-2.600	0.009	-0.755	-0.106
Age OLDER ADULT	0.5300	0.153	3.469	0.001	0.231	0.829
Conditions Neoplasms	0.7002	0.132	5.289	0.000	0.441	0.960
Intervention Model CROSSOVER	-0.6076	0.208	-2.918	0.004	-1.016	-0.200
Funder Industry	-0.1844	0.132	-1.393	0.164	-0.444	0.075

df1

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.2048	0.141	8.533	0.000	0.928	1.482
Enrollment Count	-12.1648	0.500	-24.354	0.000	-13.144	-11.186
City Count	2.3351	0.305	7.649	0.000	1.737	2.933
Intervention Model PARALLEL	0.7715	0.125	6.184	0.000	0.527	1.016
Conditions Neoplasms	0.7426	0.107	6.948	0.000	0.533	0.952
Adverse Count	0.4324	0.226	1.912	0.056	-0.011	0.876
Arm Count	1.6907	0.563	3.003	0.003	0.587	2.794
Conditions Digestive System Nutritional Metabolic	-0.0139	0.121	-0.115	0.908	-0.250	0.222
Funder Industry	0.0499	0.116	0.430	0.667	-0.178	0.277

df2

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.1437	0.211	0.680	0.496	-0.270	0.558
Enrollment Count	-8.5247	0.606	-14.074	0.000	-9.712	-7.338
City Count	2.4520	0.472	5.196	0.000	1.527	3.377
Global	0.0141	0.195	0.073	0.942	-0.367	0.396
Adverse Count	-0.5853	0.350	-1.674	0.094	-1.271	0.100
Placebo	0.4207	0.135	3.115	0.002	0.156	0.685
Allocation	0.7569	0.195	3.884	0.000	0.375	1.139
Healthy	-0.5444	0.251	-2.165	0.030	-1.037	-0.051
Conditions Respiratory	0.3377	0.178	1.894	0.058	-0.012	0.687

df3

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.1653	0.228	5.100	0.000	0.717	1.613
Enrollment Count	-14.4001	0.871	-16.538	0.000	-16.107	-12.693
Intervention Model PARALLEL	0.9974	0.235	4.246	0.000	0.537	1.458
Intervention Model CROSSOVER	-0.5148	0.345	-1.490	0.136	-1.192	0.162
Healthy	-0.7174	0.195	-3.676	0.000	-1.100	-0.335
City Count	3.3975	0.634	5.356	0.000	2.154	4.641
Arm Count	0.7218	0.815	0.885	0.376	-0.876	2.320
Adverse Count	-0.4569	0.455	-1.004	0.315	-1.349	0.435
Intervention Route Injection	-0.2827	0.144	-1.968	0.049	-0.564	-0.001

df4

6.8.6. Withdrawn Analysis

Analysis on 'Completed-Withdrawn' study status was implemented in the following files.

6.8.6.1. Files

- Thesis_SQL.ipynb
  Data extraction, features creation from AACT db.
- Thesis_Data_Withdrawn.ipynb
  Data engineering. Merging db tables with csv format data frame. Different scale formats of features creation. Merging rarer levels of categorical features. Regular expression text mining for filling missing values.
- Thesis_Dummies.ipynb
  One hot Encoding of categorical features. Interaction terms creation.
- Thesis_Models_Withdrawan.ipynb
  Models fit on training data. Models' evaluation and comparison.

6.8.6.2. Data Engineering

All steps mentioned above for Completed-Terminated data engineering were implemented for this analysis as well.
The difference lies in that only Completed-Withdrawn studies were selected. From this a data frame named df_with with shape (60717, 73) was created before train test split or VIF feature elimination. After train test split and VIF implementation, only 49 features were left for analysis. The smaller number of features compared to terminated analysis is due to some extra features being dropped, such as Enrollment and Adverse as they were always zero for these trials, which did not enroll any participants and thus did not provide any intervention for adverse events to occur. Moreover DRUG x Funder Industry feature was not used, due to the rare withdrawn event could not support an interaction term.
Also, the data is much more imbalanced than above, with ratio of almost 7:93. Due to this a train test split of 90:10 was selected.
Finally, no exploratory analysis (visualization, sparsity, collinearity etc.) was implemented on this dataset, as the same features/feature formats as in the basic analysis above were used.

6.8.6.3. Selected Model & Features

As seen in the diagrams below, best performing model is XGBoost for ROC and RR curves, while all three boosting algorithms for the other metrices. The next best performing is Random Forest. Thus, ensemble methods outperformed in Withdrawn analysis, like Terminated analysis.
The features indicated as more important are:

- **City Counts (-)**

In contrast to Terminated analysis, a larger number of facilities has negative impact on withdrawal. This difference possibly lies in the fact that participants did not experience distance difficulties, as any enrollment happened, but rather the number of facilities provided more location choices during enrollment stage. It is notable that SHAP plot displays a non-linear relationship, where less facilities have positive impact, more have negative, but a great number of facilities has also positive impact. This nonlinear relationship is not identified from logistic model.

- **Funder Industry (-)**

Industry sponsors seem to have negative impact on withdrawal i.e., non-industry sponsors are more prone to end a trial. [2] Again the direction of contribution is the opposite of terminated studies.

- **Healthy (-)**

Trials accepting healthy volunteers, are less prone to withdrawal. This feature's direction of impact is aligned with terminated analysis.

- **Arm Counts (-)**

Arm counts seem non-significant in logistic regression. As seen in SHAP plots, ϕ values overlap, which may indicate a non-linear relationship.

- **Global (+)**

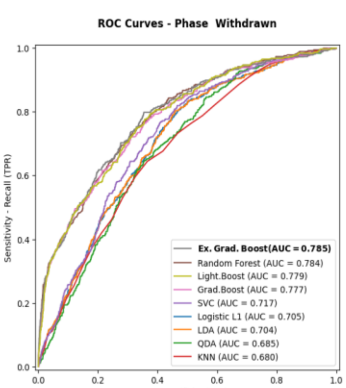
Global trials (including more than one country) seem to have a positive impact on withdrawal.

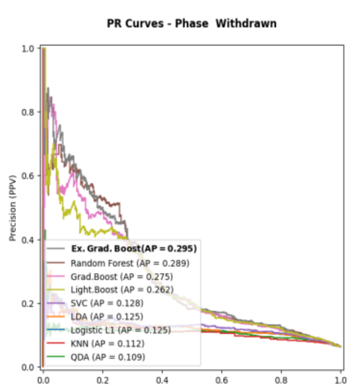
- **Neoplasms (+)**

Similarly, with terminated analysis, neoplasms have a positive impact. In contrast to the expected outcome, neoplasm conditions tend to induce less complete trials, not only due to termination but withdrawal (study did not initiate) too. The underlying factors leading to failure of such important scientific research (neoplasm treatments) need to be evaluated. Logistic regression does not indicate neoplasms as significant.

- **Older Adults (+)**

Similarly to terminated analysis and phase 1, here older aged participants, lead to very early termination of a trial. Logistic regression does not indicate age as significant.



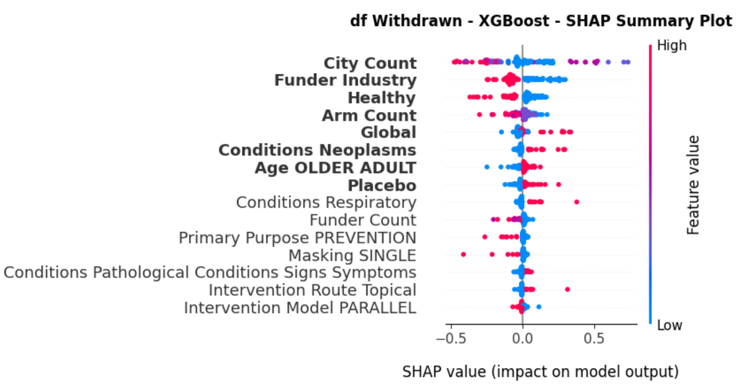


# df_with ···	Models	BACC	MCC	F1
0	**Ex_Grad_Boost**	**71.005%**	**0.214**	**0.767**
1	LightGBM	70.537%	0.209	0.766
2	Grad_Boost	70.034%	0.203	0.764
3	Random_Forest	70.577%	0.202	0.724
4	SVC	67.420%	0.169	0.687
5	LDA	66.013%	0.157	0.706
6	Logistic	65.976%	0.156	0.709
7	QDA	64.262%	0.143	0.739
8	KNN	63.546%	0.138	0.755

7. Discussion

7.1. Comparison to Literature

Although, most of the results are aligned with background's similar studies, differences occur either in which features and models were analysed or in data engineering methods used.
Initially, missing data were not dropped where possible, but rather merged into levels of categorical features, or filled mostly with text mining technics. This way no valuable information was lost from other features of same trials records.
Regarding features, enrollment is indicated as the top contributor, followed by the number of facilities, neoplasms, healthy volunteers etc. which are all mentioned in literature. However, This study analysed more factors and evaluated more models, for all trial phases 1-4 and separately per phase and thus, provides a more general approach, which is not limited to specific topics (e.g., conditions) or stages of CT. In detail, features not directly provided from CinicalTrials.gov were analysed, such as counts of occurrences, standard care treatment, interaction terms etc. Additionally, more models were compared, which indicated the LightBoost algorithm a top performing classifier in some phases, however, it was not mentioned in literature. Finally, all phases from 1 to 4 were analysed, which indicated that feature contribution is related to each phase.



# df_with ⋯	Feature	# df Withdrawn_XGBoost_...
0	City Count	0.189
1	Funder Industry	0.109
2	Healthy	0.079
3	Arm Count	0.05
4	Global	0.048
5	Conditions Neoplasms	0.034
6	Age OLDER ADULT	0.03
7	Placebo	0.03

	coef	std err	z	P>|z|	[0.025	0.975]
const	-2.4436	0.161	-15.209	0.000	-2.759	-2.129
City Count	-3.0255	0.520	-5.814	0.000	-4.045	-2.006
Funder Industry	-0.7264	0.129	-5.616	0.000	-0.980	-0.473
Healthy	-0.6063	0.166	-3.655	0.000	-0.931	-0.281
Arm Count	-0.8339	0.465	-1.793	0.073	-1.745	0.077
Global	1.7618	0.192	9.184	0.000	1.386	2.138
Conditions Neoplasms	0.2727	0.133	2.050	0.040	0.012	0.533
Age OLDER ADULT	0.1992	0.145	1.378	0.168	-0.084	0.483
Placebo	0.1141	0.131	0.873	0.383	-0.142	0.370

df_with

7.2. Strengths

As mentioned above, current study considered alternative features, not mentioned in previous studies, such as interactions of funder type with intervention type or the route of intervention (e.g., oral, topical, injection, surgical).
Additionally, current study analysed different formats of features, (e.g., adverse events regarding severity type, number of adverse events, organ system of adverse event etc). This analysis also checked for different scales of features (e.g., enrolment, enrolment_log etc.).
More models were evaluated as well as alternative algorithms (LightBoost) of the best performing model indicated from literature (XGBoost).
ML models were also compared to non-ML (i.e., ML vs logistic), as evaluating them without considering the traditional methods would not reflect their actual impact [17] .In contrast background studies mostly compare Random Forests and XGBoost.
Furthermore, current study had multiple purpose. First to compare classic statistical models with ML models, secondly to evaluate which model outperformed in termination probability prediction and finally to indicate the features contribution per phase. From this comparison not only XGBoost but also LightBoost were indicated as top performers.
Models evaluation was based not only on ROC curves but also on imbalance data sensitive metrices, such as PR Curves, MCC etc. This way results are not over optimistic.
Features contribution was analysed through SHAP plots, serving ML models, but also with logistic to obtain statistic metrics.
Moreover, all phases from 1 - 4 were studied, with each phase being analysed separately for its unique characteristics that lead to early termination. In contrast, literature is mentioning phase specific evaluation only in a few papers [14], while most studies, include part of those phases (e.g., phases 2,3)
Regarding data pre-processing, EPV value was considered for sample size per cell to be sufficiently large, while in background studies this is not mentioned. A small EPV eliminates models performance, especially in ML algorithms. [66] Finally, missing data were filled when possible and not dropped. This was achieved through comparing of records from csv and db format, or filling columns from other unstructured text data columns, through regular expression search methods, and through merging rare or missing levels of categorical data to other related levels. This way other features were not dropped, and the sample size for important features remained efficiently large.

7.3. Limitations

The presence of missing values decreased sample size for many features and features defined from the initial ones, while excluded the use of others (e.g., Documents, Geographical location etc.)

Regarding missing data, they were replaced by just regular expression (RegEx) text mining search methods, rather that alternative which are mentioned in literature and could be more appropriate (e.g., NLP). More specifically, related to feature's words were searched from unstructured text data. The drawback of regular expression text mining is that it is not possible to search for all appropriate words and expressions leading to the desired result.

Additionally, no other methods such as Neural Network analysis were made, while it was mentioned in some papers. Also, no alternative methods of classic statistical models (e.g., splines).

The use of Random Under sample induced loss of information but was chosen due to computational efficiency.

The choice of EPV ≥ 25, is not sufficiently large for ML models, but only for logistic. Furthermore, AIC tests were executed based on logistic regression, despite more algorithms were compared.

Special case of phases 0, 1/2, 2/3 and NA were not analyzed, as they were beyond the purpose of the study. However, they belong to ClinicalTrials database and could be original future research. Similarly, study status like suspended etc. were excluded.

Finally, datasets data quality needs improvement, something highly mentioned in literature for missing or wrongly recorded data. For example, enrollment and adverse event features have been recorded to display misleading values which affect the analysis, especially when sparse effect is observed. [16] [10] [19]

7.4. Recommendations for Future Research

Literature seems to provide common outcomes, regarding the early termination reasons of CTs.

What needs to be done is further investigation those specific reasons, including interactions with other features for better evaluation. Specifically, low accrual is a common top contributor for trial termination, but the factors which lead to low accruals rates are not deeply investigated. For example, analysis of enrollment as interaction to disease field, location etc. to conclude why low enrolment occurs. Similarly, for the other top contributors such as neoplasm or cities, where one could expect to have greater enrollment and completion rates.

Furthermore, analysis per study status of trials such as 'Withdrawn' , 'Terminated', 'Suspended' etc. separately and combined to evaluate the per status features contribution.

Moreover, features must be evaluated per Phase more precisely, as each phase's investigational objective is related to the features leading to termination.

Best performing models, must be evaluated in alternative methods, such as LightBoost, CatBoost or even completely different algorithms like Neural Networks etc, as the scientific problem included large scale and complex datasets.

All the above needs further investigation to detect deeper reasons leading to early termination, thus improving trials' design or even communicating consent forms in a more efficient way to attract more volunteers, especially for crucial disease fields.

8. Appendices

8.1. Code Repository

https://github.com/AndreouEugenia/Thesis

8.2. Glossary

- **Responsible Party (RP) :** Is either "Principal Investigator" or "Sponsor-Investigator"
- **Primary Purpose :** The main objective of the intervention(s) being evaluated.
  - Treatment: One or more interventions for treating a disease/condition.
  - Prevention: Preventing the development of a specific disease/condition.
  - Diagnostic: Identifying a disease/condition.
  - Supportive Care: For maximizing comfort, minimizing side effects, or mitigating against a decline in the participant's health.
  - Screening: Identifying a condition, or risk factors for a condition, in people who are not yet known to have the condition/risk factor.
  - Health Services Research: Evaluating the delivery, processes, management, organization, or financing of healthcare.
  - Basic Science: Examining the basic mechanism of action (for example, physiology or biomechanics of an intervention).
  - Device Feasibility: Device product evaluated in a small clinical trial (generally fewer than 10 participants) to determine the feasibility of the product; or a clinical trial to test a prototype device for feasibility and not health outcomes. Such studies are conducted to confirm the design and operating specifications of a device before beginning a full clinical trial.
  - Other: None of the other options applies.
- **Interventional Study Model :** The strategy for assigning interventions to participants.
  - Single Group: Clinical trials with a single arm
  - Parallel: Participants assigned to one of two or more groups in parallel.
  - Crossover: Participants receive one of two (or more) interventions during the initial phase of the study and receive the other intervention during the second phase of the study
  - Factorial: Two or more interventions, each alone and in combination, are evaluated in parallel against a control group

- Sequential: Groups of participants are assigned to receive interventions based on prior milestones being reached in the study, such as in some dose escalation and adaptive design studies

- **Arm Type ***

Definition: The role of each arm in the clinical trial.

- Experimental
- Active Comparator
- Placebo Comparator
- Sham Comparator
- No Intervention
- Other

- **Outcomes**

- Primary Outcome Measure Information
- Secondary Outcome Measure Information
- Other Pre-specified Outcome Measures

[6]

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